

Lecture no.1

Topics (1-7)

Statistical Inference and its importance

The basic objective of the statistical inference is to learn the characteristics of the population by using sample information. Generally, we call population characteristics as the parameter(s) and sample characteristics as the statistic(s). For illustration purpose, two examples are quoted to estimate the characteristics of a population: the average height of the people of Pakistan or the percentage of the voters in Punjab who think the government is doing well etc.

In real it is not possible to get data from every member of the population. So what should we do? We obtain the data from a sample and on the basis of it we make inferences about the population. According to the International Encyclopedia of Education (3rd edition), statistical inference is used to draw conclusions about some unknown aspect of a population based on a random sample from that population. Some preliminary conclusions may be drawn by the use of explanatory data analysis (EDA) or by the computation of summary statistics as well, but formal statistical inference uses calculations based on probability theory to validate those conclusions.

Probability theory is the foundation of the statistical inference. In probability theory, we have a random variable (r.v) X having a probability density function (pdf) or probability mass function (pmf) of the form $f(x)$ or $p(x)$; where θ is a parameter space), e.g. $X \sim \text{Exp}(\theta)$, $\theta > 0$. θ is a parameter of the distribution. As θ is unknown therefore we want to estimate it. The pdf or pmf is sometimes completely unknown or in some situation known down to the parameter only. To draw inferences about the pdf or the parameter(s), our information comes from a sample based on the unknown distribution of X because sample observation have the same distribution as X .

Introduction to Random Sampling

Before starting the properties of the random sampling (r.s), it is important to know about the random sampling.

When we talk about simulating any process with randomness it means that sampling is done by probability distribution. The r.s model describes an experimental situation in which variable of interest has a probability density function $f(x)$. Suppose X is a random variable (r.v) of interest but we do not know its probability density function (pdf) $f(x)$ or probability mass function (pmf) $p(x)$; i.e. one or more parameters associated with pdf or pmf are unknown. The prime objective of statistical inference is to estimate these unknown parameters or to test certain statements about them. For this purpose we take n independent observations on r.v X , where each X_i has a value x_i . This means $X_1, X_2, \dots, X_n$ are independent r.v.s with common pdf and $x_1, \dots, x_n$ are their assumed values. The $X_1, \dots, X_n$ is the sample of size n from a population and $x_1, \dots, x_n$ is called the values or realization of sample. In general, the sample observations $X_1, \dots, X_n$ are mutually independent.

Definition 1: If the r.v.s $X_1, \dots, X_n$ are independent and identically distributed (iid), then these r.v.s constitute a random sample of size n from the common distribution.

Statistic with examples

The information in the sample is summarized by using functions of sample, which are called statistic.

Definition 2: Let $X_1, \dots, X_n$ denote a sample of size n on a r.v. X . Let $T = T(X_1, \dots, X_n)$ be the function of a sample, then T is called a statistic.

The two commonly used statistics are the sample mean and sample variance which are given as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \text{ and } S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Note: The sample summary given by the statistic includes many types of information such as the smallest or largest value in the sample, the average sample value or a measure of the variability in the sample observation.

Since the dependence of statistic on sample is understood therefore we write S rather than $S(X_1, \dots, X_n)$. Generally the observed values of statistic are denoted by lower case so the observed values of $\bar{X}, S^2$ etc are denoted by $\bar{x}, s^2$ etc.

Remarks: The r.s model in definition 1 is sometimes called sampling from infinite population.

Methods of drawing samples

There are many ways of drawing samples but here we discuss only two.

1. Suppose we choose a value from the population in such a way that the probability of selection of each value is same i.e. $1/N$. This type of sampling is called sampling with replacement (w.r) and it meets the condition in definition 1. (This type of sampling is used in bootstrap which will be discussed later).
2. The second method of drawing a r.s from a finite population is called sampling without replacement (w.o.r). In this method the probability of selecting first value is $1/N$. The first value is recorded and second value is chosen from the remaining $N-1$ values. The probability of the remaining $N-1$ values is $1/(N-1)$. The second value is recorded. This process continues for the remaining values till we obtain the samples $X_1, \dots, X_n$. This type of sampling does not satisfy all the conditions of definition 1, i.e. r.v.s $X_1, \dots, X_n$ are not mutually independent.

Example: Suppose a finite population consists of 1000 items, so $N=1000$. A sample of size $n=10$ is drawn without replacement. What is the probability that all ten sample values are greater than 200?

If $X_1, \dots, X_n$ are mutually independent, we will proceed as
 $P(X_1 > 200, \dots, X_n > 200) = P(X_1 > 200) \dots P(X_n > 200)$

$$\frac{\binom{800}{10} \binom{200}{0}}{\binom{1000}{10}} = \frac{800 \times 799 \times \dots \times 790}{1000 \times 999 \times \dots \times 991} \approx 0.1073$$

To calculate this probability exactly, we assume that Y is a random variable representing the number of items in the sample which are greater than 200. Since the sampling is done without replacement therefore we use hypergeometric distribution. So $N = 1000$, $K = 800$, $n = 10$

$$P(X = 200, \dots, X = 200) = P(Y = 10) = \frac{\binom{800}{10} \binom{200}{0}}{\binom{1000}{10}} \approx 0.1062$$

Note: In future we will use Definition 1 as the definition of a random sample from a population.

Sum of random samples from a random sample

When a sample is drawn, some summary of the values is computed. Any well-defined summary of these values is represented by a function $T(x_1, \dots, x_n)$ whose domain is the sample space of the random vector $X_1, \dots, X_n$. The function T may be a real-valued or vector-valued therefore the summary is a random variable or random vector. Since the r.s. $X_1, \dots, X_n$ has a simple probabilistic structure (as of i.i.d) therefore the distribution of Y is observable and called sampling distribution of Y . This distinguishes the probability distribution of Y from the distribution of population, i.e. the marginal distribution of each X_i .

Definition 3: Let $X_1, \dots, X_n$ be a r.s. of size n from a population and let $T(x_1, \dots, x_n)$ be a real-valued or vector-valued function whose domain includes the sample space of $X_1, \dots, X_n$, then the random variable or random vector $Y = T(X_1, \dots, X_n)$ is called a statistic. The probability distribution of a statistic Y is called the sampling distribution of Y .

Some results

Lemma 5.2.5: Let $X_1, \dots, X_n$ be a r.s. from a population and let g be a function such that

Eg X_i and $Var X_i$ exist. Then

$$E(g(X_1, \dots, X_n)) = g(\bar{x}) \quad \text{and} \quad Var(g(X_1, \dots, X_n)) = n \cdot var g(X_1)$$

Proof: To prove the first part we proceed as:

$$E(g(X_1, \dots, X_n)) = \sum_{i=1}^n E(g(X_i)) = n \cdot E(g(X_1))$$

The all X_i s are identically distributed i.e.

assume independence here because it holds for any collection of n identical distributed random variables.

To prove the second part we proceed as:

By the definition of variance

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = E \left(\left(\frac{1}{n} \sum_{i=1}^n X_i - E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \right)^2 \right)$$

(by taking sum as common and applying expectation property)

First we have n terms of $\frac{1}{n} \sum_{i=1}^n X_i - E \left(\frac{1}{n} \sum_{i=1}^n X_i \right)$, $i=1, \dots, n$ and for each term we have, by the definition of variance

$$E \left(\left(\frac{1}{n} \sum_{i=1}^n X_i - E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \right)^2 \right) = \frac{1}{n^2} \sum_{i=1}^n E \left(\left(X_i - E(X_i) \right)^2 \right)$$

Since the r.v.s are identically distributed therefore we have

$$E \left(\left(\frac{1}{n} \sum_{i=1}^n X_i - E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \right)^2 \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}$$

The remaining

, therefore

$$E \left(\left(\frac{1}{n} \sum_{i=1}^n X_i - E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \right)^2 \right) = \frac{\sigma^2}{n}$$

Since we have n identically distributed r.v.s therefore (1) becomes $\frac{\sigma^2}{n}$, which proves the desired result.

Some Results 2 (Theorem 5.2.6)

Theorem: Let $X_1, \dots, X_n$ be a r.s from a population with mean μ and variance σ^2 . Then

(a) $E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \mu$

(b) $\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{\sigma^2}{n}$

(c) $E S_{22} = \frac{(n-1)\sigma^2}{n}$

Proof: (a) $E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \mu$

Assuming that all X_i are identically distributed and using the previous result 1.

(b) Similarly $\text{Var } X = \text{Var } \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n^2} \sum_{i=1}^n \text{Var } X_i = \frac{1}{n} \text{Var } X_1$

$$E X = E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n} \sum_{i=1}^n E X_i = E X_1$$

$$\text{Var } X = \text{Var } \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var } X_i = \frac{1}{n} \text{Var } X_1$$

As $E X_1 = \mu$ and $\text{Var } X_1 = \sigma^2$

And similarly $\text{Var } X_2 = \sigma^2$

Lecture no.2

Topics (1-13)

Moment Generating Function (mgf)

The mgf of a random variable X is a function M_X defined as

$$M_X(t) = E[e^{tX}]$$

The mgf of X exists, if there exists a positive constant a such that M_X is finite for all t in $(-a, a)$.

The mgf of the sum of the independent random variables is obtained as:

Let $Y = \bar{X}$, then

$$M_t = E[e^{tY}] = E[e^{t(\bar{X}_1 + \bar{X}_2 + \dots + \bar{X}_n)/n}] = E[e^{t\bar{X}_1/n} e^{t\bar{X}_2/n} \dots e^{t\bar{X}_n/n}] = E[e^{t\bar{X}_1/n}] \dots E[e^{t\bar{X}_n/n}] = M_{t/n} \dots M_{t/n} = M_{t/n}^n$$

Moment generating function of sample mean

If $Y = \bar{X}$, then using the above result, we have

$$M_{t/n} = E[e^{t\bar{X}_1/n}] = E[e^{(t/n)\bar{X}_1}] = M_{t/n}$$

If $\bar{X}_1, \dots, \bar{X}_n$ are identically distributed, then the following result holds:

Theorem: Let $\bar{X}_1, \dots, \bar{X}_n$ be a r.s from a population with mgf M . Then the mgf of the sample mean is $M_{t/n}^n$.

Example: Let $\bar{X}_1, \dots, \bar{X}_n$ be a r.s from $N(\mu, \sigma^2)$ population. Then the mgf of the sample mean is

$$M_{t/n} = E[e^{(t/n)\bar{X}_1}] = \exp\left\{-\frac{t\mu}{n} + \frac{t^2\sigma^2}{2n}\right\} = \exp\left\{-\frac{t\mu}{n} + \frac{t^2\sigma^2}{2n}\right\}$$

Thus $\bar{X}$ has a $N(\mu, \sigma^2/n)$ distributed.

Exact Sampling distribution: Chi-square distribution

Before going to the Chi-square distribution, we need the following results which will be helpful to proceed:

Definition: The sample variance is defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Result 1: The sum of the squares of the r.v.s $\bar{X}_1, \dots, \bar{X}_n$ is

$$\sum_{i=1}^n (X_i - \bar{X})^2 = (n-1)S^2$$

Proof: By definition

$$(n-1)S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) = \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2 = \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$$

It follows that

$$\sum_{i=1}^n X_i^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n\bar{X}^2$$

Corollary: The sum of the squares of the r.v.s $X_1, \dots, X_n$ centered about mean $\bar{X}$, is

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$$

The proof is simple as above.

Chi-square distribution

The pdf of Chi-square is

$$f(x) = \frac{1}{2^{p/2} \Gamma(p/2)} x^{p/2-1} e^{-x/2}$$

where p is the degree of freedom.

The mgf of Chi-square distribution is

$$M(t) = (1 - 2t)^{-p/2}$$

Lemma 5.3.2 (a): If z is a standard normal r.v., then z^2 is a Chi-square r.v. with one degree of freedom.

Proof: The mgf for z^2 is defined as

$$M_{z^2}(t) = E[e^{tz^2}] = \int_{-\infty}^{\infty} e^{tz^2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2(1/2 - t)} dz$$

This is kernel of a

$$N(0, 1/(1-2t))$$

So

$$M_{z^2}(t) = \frac{1}{\sqrt{1-2t}} = (1-2t)^{-1/2}$$

The above integral is equal to 1. So

This is the same mgf as Chi-square distribution with one d.o.f. so

$$Z_{22}^2 \sim \chi^2_1$$

Some more results(Topic-10)

Lemma 5.3.2 (b): If $X_1, \dots, X_n$ are independent and identically distributed r.v.s, then

$$Y = \sum_{i=1}^n X_i$$

Proof: The mgf of X_i is

It follows that the mgf for Y is

Where Y is sum of random variables

$$M_Y(t) = E[e^{tY}] = E[e^{t(X_1 + \dots + X_n)}] = E[e^{tX_1} \dots e^{tX_n}] = \prod_{i=1}^n E[e^{tX_i}] = \prod_{i=1}^n M_{X_i}(t) = (M_{X_1}(t))^n$$

This is the mgf of Chi-square distribution with nd.o.f. So $Y \sim \chi^2_n$

Sampling from the Normal distribution(Topic-11)

Many distributions came into being in sampling from a normal population i.e. Chi-square, t and F distributions. Sampling from a normal population leads to many useful properties of sample statistics.

Properties of sample mean and variance(topic12 &13)

Theorem 5.3.1: Let $X_1, \dots, X_n$ be a r.s from $N(\mu, \sigma^2)$ and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ and $S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$

Then

- (i) $\bar{X}_n$ and S_{n-1}^2 are independent r.v.s.
- (ii) $\bar{X}_n$ has a $N(\mu, \frac{\sigma^2}{n})$ distribution.
- (iii) $(n-1)S_{n-1}^2$ has a Chi-square distribution with $n-1$ degree of freedom (d.o.f)

Proof:

- (i) Without loss of generality, we assume that $\mu = 0$ and $\sigma^2 = 1$. In (i) we have to show that $\bar{X}_n$ and S_{n-1}^2 are the functions of independent random vectors. We can write S_{n-1}^2 as a function of $X_1, \dots, X_n$ deviations as:

Diagram illustrating the structure of a function f and its associated data structures:

- Top row: f (with subscript 12 and 11), $y, \dots, y$, f (with subscript x and $\dots, x$), J .
- Second row: yy , $\dots$, yy , f (with subscript x and $\dots, x$), n .
- Third row: X_n (with subscript 11 and 2), 1 , 2 .
- Bottom row: Constant.exp , x .

Now

$$X = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$$

$$X = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\frac{1}{\sqrt{n}} \sum_{j=1}^n X_{ij} \right)$$

Since $\sum_{i=1}^n X_i = 0$

So

$$X = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\frac{1}{\sqrt{n}} \sum_{j=1}^n X_{ij} \right)$$

$$Y = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$$

So the pdf of Y , (1) implies that

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = \text{constant} \cdot \exp \left\{ -\frac{1}{2} \sum_{i=1}^n y_i^2 \right\}$$

Or

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = \text{constant} \cdot \exp \left\{ -\frac{1}{2} \sum_{i=1}^n y_i^2 \right\}$$

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = \text{constant} \cdot h(y_1, \dots, y_n) \cdot g(y_1, \dots, y_n)$$

Since the joint pdf of Y can be written as a function of Y_1 and $Y_2, \dots, Y_n$ separately so Y_1 is independent of $Y_2, \dots, Y_n$ and therefore X is independent of S^2 .

(ii) We know that the mgf of Normal distribution is

$$M_{X_i}(t) = E[e^{tX_i}] = \int_{-\infty}^{\infty} e^{tx} f_{X_i}(x) dx$$

$$M_{X_i}(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

Which is the mgf of $N(0, 1)$

(iii) We know that

$$X = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$U = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Now by using lemma 5.3.2 (a), as we assumed that $X_1, \dots, X_n$ are i.i.d. of size n from $N(\mu, \sigma^2)$, therefore X_i follows the standard normal. Similarly by using lemma 5.3.2 (b), S^2 follows the Chi-square distribution with $n-1$ d.o.f. and U follows the t-distribution with $n-1$ d.o.f.

Now assuming that $X_1, \dots, X_n$ are independent, the mgf of U can be written as

$$M_U(t) = \int_{-\infty}^{\infty} e^{tU} f_U(u) du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

$$= \int_{-\infty}^{\infty} e^{t \frac{\bar{X} - \mu}{S/\sqrt{n}}} \frac{1}{\sqrt{n-1}} e^{-\frac{1}{2(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2} du$$

Lecture no.3

(14 to 17)

Exact Sampling distribution: t-distribution

Definition: Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ distribution. The quantity $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ has a t-distribution with $n-1$ degree of freedom. Alternatively, a r.v. T has Student's t distribution with p degree of freedom and we write it as $T \sim t_p$ if it has the following pdf

$$f_T(t) = \frac{\Gamma(\frac{p+1}{2})}{\Gamma(\frac{p}{2})} \left(\frac{p}{2} \right)^{\frac{p+1}{2}} \left(1 + \frac{t^2}{p} \right)^{-\frac{p+1}{2}}$$

$$f_{T_p}(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \quad \text{and} \quad f_{U_p}(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$$

Derivation: Let U and V and they are independent, then the joint pdf of U and V can be written as

$$f_{U,V}(u,v) = \frac{1}{2\pi} e^{-\frac{u^2}{2}} e^{-\frac{v^2}{2}} = \frac{1}{2\pi} e^{-\frac{u^2 + v^2}{2}}$$

Now make the transformation

$$u = \sqrt{t} \cos \theta, \quad v = \sqrt{t} \sin \theta$$

Which implies that $u^2 + v^2 = t$

$$|J| = \begin{vmatrix} \frac{\partial u}{\partial t} & \frac{\partial u}{\partial \theta} \\ \frac{\partial v}{\partial t} & \frac{\partial v}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \frac{1}{2\sqrt{t}} \cos \theta & -\sqrt{t} \sin \theta \\ \frac{1}{2\sqrt{t}} \sin \theta & \sqrt{t} \cos \theta \end{vmatrix} = \frac{1}{2}$$

Now the joint distribution of t and w is

$$f_{T,W}(t,w) = \frac{1}{2\pi} e^{-\frac{t}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} = \frac{1}{2\pi} e^{-\frac{t}{2}} e^{-\frac{w^2}{2}}$$

To find the marginal distribution of t , we need to integrate the above expression over w .

$$\text{Let } w = \sqrt{t} z \quad \Rightarrow \quad dw = \frac{\sqrt{t}}{2} dz$$

$$dw = \frac{\sqrt{t}}{2} dz$$

$$f(t) = \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p}{2}\right)} \frac{1}{\sqrt{p} \pi^{p/2}} \left(1 + \frac{t^2}{p}\right)^{-\frac{p+1}{2}}$$

Note that for $p=1$ the f will become the pdf of Cauchy distribution for sample size 2.

The mgf of Student's t distribution does not exist and the mean and variance are given as

$$E(T) = 0 \text{ for } p > 1 \text{ and } \text{var}(T) = \frac{p}{p-2}, \text{ for } p > 2.$$

Exact Sampling distribution: F-distribution

Definition: Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ population and let $Y_1, \dots, Y_m$ be a

random sample from an independent $N(\mu, \sigma^2)$ population. The r.v. $F = \frac{S_1^2/n}{S_2^2/m}$ has Snedecor's

F distribution with n and m degree of freedom. Alternatively, the r.v. F has the F

distribution with p and q degree of freedom if it has the following pdf

$$f(x) = \frac{\Gamma\left(\frac{p+q}{2}\right)}{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)} \frac{1}{x^{(p+q)/2}} \left(\frac{px}{q}\right)^{p/2} \left(1 + \frac{px}{q}\right)^{-(p+q)/2}$$

A variance ratio may have a F distribution even if the parent populations are not normal. Kelker

(1970) has shown that as long as the parent populations have a certain type of symmetric, then the variance ratio will have a F distribution.

Derivation: Let $U = \frac{S_1^2}{\sigma^2}$ and $V = \frac{S_2^2}{\sigma^2}$ be independent, $U \sim \chi_p^2$ and $V \sim \chi_q^2$, then the joint

distribution of U and V can be written as

$$f(u, v) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{\sqrt{u^2 + v^2}} e^{-\frac{1}{2}(u^2 + v^2)} e^{i(xu + yv)} du dv$$

Now make the transformation

$$x = u/p \text{ and } y = w/v$$

$$v/q$$

which implies that

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{p} & 0 \\ -\frac{u}{p^2} & \frac{1}{v} \end{vmatrix} = \frac{1}{pv}$$

So

$$f(w, x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{\sqrt{w^2 + x^2}} e^{-\frac{1}{2}(w^2 + x^2)} e^{i(xu + yv)} dw dx$$

To find the marginal distribution of x , we need to integrate the above expression over w .

$$f(x) = \int_{-\infty}^{\infty} f(w, x) dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{\sqrt{w^2 + x^2}} e^{-\frac{1}{2}(w^2 + x^2)} e^{i(xu + yv)} dw dx$$

Let

$$w = zx, \quad z = \frac{w}{x}$$

$$dw = x dz$$

(a) If $U \sim N(0, 1)$

$V \sim N(0, 1)$ and U and V are independent. Assuming $V \sim N(0, 1)$ and $U \sim N(0, 1)$ and U and V are independent.

(b) We know the distribution of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Let $Y = X^2$

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-\frac{y}{2}}$$

After simplification, we get

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-\frac{y}{2}}$$

(c) We know the distribution of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$Y = X^2 \Rightarrow X = \sqrt{Y} \text{ and } X = -\sqrt{Y}$$

Thus the pdf of Y is

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-\frac{y}{2}}$$

After simplification, we get

$$b = \frac{1122}{22} - \frac{22 \times 22}{22} = 1122 - 22 = 1100$$

Application of Chi-Square t & F distribution

Q5.16 Let $X_1, X_2, \dots, X_n$ be independent with $N(\mu, \sigma^2)$ distributions. For each of the following situations, use the X_i to construct a statistic with the indicated distribution.

- (a) chi-square with 3 degree of freedom
- (b) t distribution with 2 degree of freedom
- (c) F distribution with 1 and 2 degree of freedom

Solution:

(a) Recall that $Z = \frac{X - \mu}{\sigma/\sqrt{n}}$. As we know that if $Z_1, Z_2, \dots, Z_n$ are independent standard normal variables, then by the addition property of independent chi-squares is $W = Z_1^2 + Z_2^2 + \dots + Z_n^2$, therefore

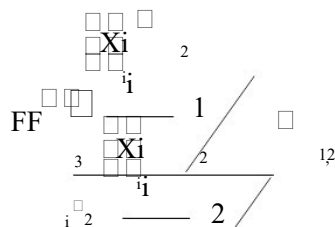
(b) By the definition of t-distribution, $t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. When $X_1, X_2, \dots, X_n$ are independent $N(\mu, \sigma^2)$ variables, $\bar{X}$ and s^2 are independent. (all ready proved).

Now using the given scenario, we have

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{\frac{1}{n} \sum_{i=1}^n X_i - \mu}{\sqrt{\frac{1}{n(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2}}$$

(c) By the definition of the F-distribution,

when u and v are independent chi-square distributions with p and q degree of freedom, $F = \frac{u/p}{v/q}$. (all ready proved). Now using the given scenario, we have



Q5.19(a) Prove that the χ^2 distribution is stochastically increasing in its degree of freedom; i.e. if p_q then for any $a, P_{22} a P_{22} a$ with strict inequality for some a .

Solution: Let χ^2_{22} where χ^2_{22} and χ^2_{22} are independent chi-square random variables with k and $k+2$ degree of freedom. Since $\chi^2_{22} \sim \chi^2_{22}$ for any θ

Lecture no.4 (18 to 23)

Order Statistics

Sample values such as the smallest, largest or middle observation from a r.s provides additional summary information. For example, the highest flood water or lowest winter temperature recorded during the last 50 years might be useful data when planning for future emergencies. Similarly the median price of house sold during the last year might be useful for estimating the cost of living. All these are the examples of order statistics.

Definition: The order statistics of a random sample $X_1, \dots, X_n$ are the sample values placed in a non-decreasing order and they are denoted by $X_{(1)}, \dots, X_{(n)}$.

The order statistics are random variables that satisfy $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$. The following are some statistics that are easily defined in term of order statistics.

$X_{(1)}$ is the distance between the largest and smallest observation. It is the sample range $R = X_{(n)} - X_{(1)}$.

R is a measure of dispersion and reflects the variability in the population.

The sample median, which is denoted by M , is a number which approximate one half observation are less than M and one half observation are greater than M . It is defined as

$$M = \begin{cases} X_{(n/2)} & \text{if } n \text{ is odd} \\ \frac{X_{(n/2)} + X_{(n/2+1)}}{2} & \text{if } n \text{ is even} \end{cases}$$

The median is a measure of location and can be used as an alternative of sample mean. One advantage of the sample median over the sample mean is that it is less affected by extreme observations.

For any number p between 0 and 1, the $(100p)$ th sample percentile is the observation such that approximately np of the observations are less than this observation and $n(1 - p)$ of the observations are greater. The 50th percentile is the sample median, the 25th percentile is called the lower quartile, and the 75th percentile is called the upper quartile. A measure of dispersion that is sometimes used is the inter-quartile range, the distance between the lower and upper quartiles.

Distribution of order statistics

Theorem: 5.4.4: Let $X_{(1)}, \dots, X_{(n)}$ denote the order statistics of a random sample, $X_1, \dots, X_n$ from a continuous population with cdf $F(x)$ and pdf $f(x)$. Then the pdf of $X_{(j)}$ is

Proof: To find the pdf of $X_{(j)}$ we partition the whole set of observation as:

From above we note that:

- (i) $j-1$ observations are less than x .
- (ii) one observation is in the interval $[x, x+dx]$ and 1 observation is zero.
- (iii) $n-j$ observations are greater than x .

Now the probabilities of (i), (iii) and (iv) are $F(x)^{j-1}$, $f(x)dx$ and $[1-F(x)]^{n-j}$ respectively. The number of ways (distinct) in which n observations can be arranged in above three groups are

$$\frac{n!}{(j-1)! 1! (n-j)!} \cdot F(x)^{j-1} \cdot f(x)dx \cdot [1-F(x)]^{n-j}$$

Corollary 1: If put $j = n$ we get the distribution of the largest observation as

$$f(x) [1-F(x)]^{n-1}$$

Corollary 2: If put $j = 1$ we get the distribution of the smallest observation as

$$f(x) F(x)^{n-1}$$

$$f_{X_{(j)}} = \frac{n!}{(j-1)!(n-j)!} x^{j-1} (1-x)^{n-j} \quad 0 \leq x \leq 1$$

Example: (Uniform order statistics pdf) Let $X_1, \dots, X_n$ be iid uniform $[0, 1]$, so $f_X(x) = 1$ for $x \in [0, 1]$ and $F_X(x) = x$ for $x \in [0, 1]$. The pdf of the j -th order statistics is

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} x^{j-1} (1-x)^{n-j} \quad 0 \leq x \leq 1$$

Thus the j -th order statistic from a uniform $[0, 1]$ sample has a beta $(j, n-j+1)$ distribution. From this we can deduce that

$$E X_{(j)} = \frac{j}{n+1} \quad \text{and} \quad \text{Var } X_{(j)} = \frac{j(n-j+1)}{(n+1)^2(n+2)}$$

Joint distribution of i -th and j -th order statistics

Theorem 5.4.6: Let $X_1, \dots, X_n$ denote the order statistics of a random sample, $X_1, \dots, X_n$ from a continuous population with cdf F_X and pdf f_X . Then the joint pdf of $X_{(i)}$ and $X_{(j)}$ is

$$f_{X_{(i)}, X_{(j)}}(u, v) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} F_X^{i-1}(u) f_X(u) [F_X(v) - F_X(u)]^{j-i-1} f_X(v) [1 - F_X(v)]^{n-j} \quad u < v$$

Proof: To find the joint distribution of $X_{(i)}$ and $X_{(j)}$, we partition the observation as:

$$X_1, \dots, X_{i-1} < X_{(i)} = u < X_{(i+1)}, \dots, X_{(j-1)} < X_{(j)} = v < X_{(j+1)}, \dots, X_n$$

Arranging as before, we get

$$f_{X_{(i)}, X_{(j)}}(u, v) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} F_X^{i-1}(u) f_X(u) [F_X(v) - F_X(u)]^{j-i-1} f_X(v) [1 - F_X(v)]^{n-j}$$

Application: Q 5.24: Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f(x) = \begin{cases} 10x^9 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Let $X_{(1)}, \dots, X_{(n)}$ be the order statistics. Show that $X_{(1)}$ and $X_{(n)}$ are independent random variables.

Solution: We have $f(x) = 10x^9$ and $F(x) = x^{10}$. Let $U = X_{(1)}$ and $V = X_{(n)}$. The joint distribution of U and V is

$$f_{U,V}(u,v) = \frac{n!}{(j-1)!(n-j)!} f(u) f(v) F(u)^{j-1} [F(v)-F(u)]^{n-j} f(v)$$

Put $j=1$ and get

$$f_{U,V}(u,v) = \frac{n!}{(n-1)!} f(u) f(v) F(u)^0 [F(v)-F(u)]^{n-1} f(v)$$

$$= n f(u) f(v) [F(v)-F(u)]^{n-1}$$

$$= n \cdot 10u^9 \cdot 10v^9 [v^{10} - u^{10}]^{n-1}$$

$$= 100n u^9 v^9 [v^{10} - u^{10}]^{n-1}$$

Now making transformation as

$$u = z, \quad v = tz$$

$$J = \begin{vmatrix} \frac{\partial u}{\partial z} & \frac{\partial u}{\partial t} \\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial t} \end{vmatrix} = \begin{vmatrix} 1 & z \\ t & tz \end{vmatrix} = 1 \cdot tz - z \cdot t = 0$$

$$f_{T,Z}(t,z) = 100n z^9 (tz)^9 [t^{10}z^{10} - z^{10}]^{n-1} |J|$$

$$= 100n z^{18} t^9 [z^{10}(t^{10} - 1)]^{n-1}$$

$$= 100n z^{18} t^9 z^{10(n-1)} (t^{10} - 1)^{n-1}$$

$$= 100n z^{10n-2} t^9 (t^{10} - 1)^{n-1}$$

Hence the joint pdf of t and z can be written into the functions of t and z , so T and Z are independent.

Chebychev's Inequality and its usage

Chebychev's inequality is a probabilistic inequality which provides a way to know what fraction of data falls within K standard deviation from the mean of any data set. Here K is any positive real number greater than one.

Let $X_1, \dots, X_n$ is a random sample from a population with mean μ and variance σ^2 , the Chebychev's inequality says

$P\{X \geq \mu + k \cdot \frac{1}{\sqrt{2}}\} \leq \frac{1}{2k^2}$
 of the distribution's values can be more than K standard deviations

It says that no more than $\frac{1}{2}$

away from the mean or equivalently, at least $1 - \frac{1}{2k^2}$ of the distribution's values are within K

standard deviation of the mean. It can be applied to any probability distribution whose mean and variance are known.

Example: Let X be the number of screws delivered to a box by an automatic filling device. Assume $\mu = 1000$, $\sigma^2 = 25$. There are problems with too many (giving away free) or too few (potential irritated customers) screws in a box.

- How many units to the right of μ is 1009?
- What X values is 2.6 units to the left of μ ? $P\{X \leq 994\}$ $P\{X \geq 1006\}$
- Use Chebychev's inequality to find a bound on

To convert a specific value of X to units is

$$K = \frac{X - \mu}{\sigma}$$

To convert a units to a specific value of X is

$$X = \mu + K \cdot \sigma$$

- Using the previous formula $X = 1009$ is

$$K = \frac{1009 - 1000}{5} = 1.8 \text{ units to the right of } \mu$$

- Using the previous formula $K = -2.6$ units corresponds to

$$X = 1000 - 2.6 \cdot 5 = 987$$

- Since the interval is symmetric about $\mu = 1000$, the endpoints are

$$K = \frac{1006 - 1000}{5} = 1.2 \text{ units from } \mu \text{ and Chebychev's inequality implies}$$

that

$$P\{1000 - 1.2 \cdot 5 \leq X \leq 1000 + 1.2 \cdot 5\} = 1 - \frac{1}{2 \cdot 1.2^2}$$

$$P\{994 \leq X \leq 1006\} = 0.3056$$

Lecture no.5

(24 to 30)

Convergence concept

In probability theory the concept of convergence is an important topic to study the behavior of the random variables to some limit. In convergence we increase the sample size to approach infinity and investigate the behavior of certain sample quantities as this happens. Although the notion of an infinite sample size is an artificial but it provides us some useful approximations for the large finite sample case as it usually happens that expression becomes simplified in the limit. Generally we are interested to look at the behavior of X_n , the mean of n observations, as n .

Convergence in probability

We call convergence in probability of a sequence of random variables if it converges to a constant, say C , i.e. X_n . This concept is used frequently in statistics, e.g. an estimator is called consistent if it converges in probability to the quantity being estimated.

Definition: A sequence of random variables $X_1, X_2, \dots$ (say X_n) converges in probability to a random variable X if, for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$$

or equivalently,

$$\lim_{n \rightarrow \infty} P(|X_n - X| < \epsilon) = 1$$

The $X_1, X_2, \dots$ in the above definition are generally not independent and identically distributed random variables, as in a random sample. The distribution of X_n changes as the subscript changes and convergence concepts describe different ways in which the distribution of X_n converges to some limiting distribution as the subscript becomes large.

Weak Law of Large Numbers (WLLN)

Theorem 5.5.2: Let $X_1, X_2, \dots$ be iid random variables with $EX_i = \mu$ and $Var X_i = \sigma^2$.

Define $X_n = \frac{1}{n} \sum_{i=1}^n X_i$, then for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - \mu| \geq \epsilon) = 0, \quad \text{that is, } X_n \text{ converges in probability to } \mu.$$

Proof: We have for every $\epsilon > 0$,

$$P(X - \mu)^2 \leq \epsilon^2$$

$$P(X - \mu)^2 \leq \epsilon^2 = P\left(\frac{X - \mu}{\sigma} \leq \frac{\epsilon}{\sigma}\right)$$

$$= P\left(\frac{X - \mu}{\sigma} \leq \frac{\epsilon}{\sigma}\right) = P\left(\frac{X - \mu}{\sigma} \leq \frac{\epsilon}{\sigma}\right)$$

$$= P\left(\frac{X - \mu}{\sigma} \leq \frac{\epsilon}{\sigma}\right) = P\left(\frac{X - \mu}{\sigma} \leq \frac{\epsilon}{\sigma}\right)$$

Hence, $P(X - \mu)^2 \leq \epsilon^2 = P\left(\frac{X - \mu}{\sigma} \leq \frac{\epsilon}{\sigma}\right)$

The WLLN quite elegantly states that under general conditions, the sample mean approached the population mean as $n \rightarrow \infty$.

Example (Consistency of S_n^2)

Suppose we have a sequence $X_1, X_2, \dots$ be iid random variables with $EX_i = \mu$ and $Var X_i = \sigma^2$. If we define

$$S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

Using Chebychev's Inequality, we have

$$P(S_n^2 - \sigma^2 \geq \epsilon) \leq \frac{Var S_n^2}{\epsilon^2}$$

$$= \frac{\sigma^4}{n \epsilon^2}$$

and thus, a sufficient condition that S_n^2 converges in probability to σ^2 is that $\frac{\sigma^4}{n \epsilon^2} \rightarrow 0$ as $n \rightarrow \infty$.

Almost sure convergence

A type of convergence that is stronger than convergence in probability is almost sure convergence. This type of convergence is similar to point wise convergence of a sequence of functions, except that the convergence need not occur on a set with probability 0 (hence the "almost" sure).

To understand the concept of almost sure convergence, we need to recall the definition of random variable; A random variable is a real-valued function defined on a sample space S . If a sample space S has elements denoted by s , then X and Y are all functions defined on S . X converges to Y almost surely if the functions X_n converge to Y for all $s \in S$.

$$P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{X_k \neq Y\}\right) = 0$$

Example 5.5.7: (almost sure convergence) Let the sample space S be the closed interval $[0, 1]$ with the uniform probability distribution. Define random variables X_n and Y as follows. For every $s \in [0, 1]$, $X_n(s) = \frac{s}{n}$ and $Y(s) = s$. However, $X_n(s) \neq Y(s)$ for all $s \in [0, 1]$.

X

But since the convergence occurs
 $\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$ for every $\epsilon > 0$, so X_n converges to X almost surely.
 on the set Ω of probability 1.

Strong Law of Large Number (SLLN)

Theorem 5.5.9: Let $X_1, X_2, \dots$ be iid random variables with $EX_i = \mu$ and $Var X_i = \sigma^2 < \infty$.

Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, then for every $\epsilon > 0$,

$\lim_{n \rightarrow \infty} P\left(\left|\bar{X}_n - \mu\right| > \epsilon\right) = 0$,
 that is, $\bar{X}_n$ converges almost surely to μ .

For both the weak and strong law of large numbers we had the assumption of a finite variance. In fact, both the weak and strong laws hold without this assumption. The only moment condition needed is that $E|X_i| < \infty$.

Convergence in Distribution

Definition: A sequence of random variables $X_1, X_2, \dots$ converges in distribution to a random variable X if $\lim_{n \rightarrow \infty} F_n(x) = F(x)$ at all points x where F is continuous.

Note that although we talk of a sequence of random variables converging in distribution, it is really the cdfs that converge, not the random variables. In this very fundamental way convergence in distribution is quite different from convergence in probability or convergence almost

Theorem 5.5.12: If the sequence of random variables, $X_1, X_2, \dots$, converges in probability to a random variable X , the sequence also converges in distribution to X .

Central limit theorem

Let $X_1, X_2, \dots$ be iid random variables with $EX_i = \mu$ and $0 < Var X_i = \sigma^2 < \infty$. Define

$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$. Then for any x ,
 $\lim_{n \rightarrow \infty} P(Z_n \leq x) = \Phi(x)$ where $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$.

That is, $\frac{nX - r}{\sqrt{rp(1-p)}}$ has a limiting standard normal distribution.

Example (Normal approximation to the negative binomial)

Suppose $X_1, \dots, X_n$ are a random sample from a negative binomial (r, p) distribution. Recall that

$$EX = \frac{r}{p} \quad \text{and} \quad \text{Var}(X) = \frac{r(1-p)}{p^2}$$

and the central limit theorem tells us that $\frac{nX - nr/p}{\sqrt{nr(1-p)/p^2}}$ is approximately standard normal.

approximate probability calculation are much easier than the exact calculations. For example, if $r=10$, $p=0.2$ and $n=30$

The probability mass function of negative binomial is

$$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} \quad \text{for } x = r, r+1, r+2, \dots$$

Now we are interested to find the following exact probability

$$P(X \leq 11) = P(X \leq 30) - P(X \leq 330)$$

Now by using the above probability mass function of negative binomial, we have (as)

$$P(X \leq 11) = \sum_{x=10}^{11} \binom{x-1}{r-1} p^r (1-p)^{x-r} = \binom{10}{9} (0.2)^{10} (0.8)^1 + \binom{11}{9} (0.2)^{10} (0.8)^2$$

which is very difficult calculation. But by using CLT approximation, we get

$$P(X \leq 11) \approx P\left(\frac{30X - 10}{\sqrt{20}} \leq \frac{30 \cdot 11 - 10}{\sqrt{20}}\right) = P(Z \leq 1.225) = 0.8888$$

Lecture no.6 (31 to 37)

Generating a random sample

We have studied different methods of describing the behavior of the random variables- transformation, distributions, moment calculations, limit theorems and Weibull

observation based on random variables, called data, to describe and model the real phenomena. Typically, we observe random variables $X_1, \dots, X_n$ from some probability distribution $f(x)$ and using its properties to describe the behavior of the random variables. Now we focused how to generate random samples $X_1, \dots, X_n$ from a given probability distribution $f(x)$. There are two major concerns; first, we must know how to generate random samples and second, to validate the simulation approximation by using Law of Large Number. Generally the uniform distribution is used to generate the random samples. As our problem is not generating uniform random variables but rather of transforming the uniform random samples to the desired distribution. Two methods are generally used i.e. direct and indirect method.

Direct Method with example

A direct method also known as inverse sampling or inverse probability integral transform is a method of generating samples number at random from any probability distribution provided that if its cdf exist in closed form.

Definition: Let X is a continuous random variable with cdf F_X , then the random variable $Y = F_X(X)$ has a uniform distribution on $[0, 1]$. In other words if U has a uniform distribution on $[0, 1]$, then the random variable $F_X^{-1}(U)$ has the same distribution as X .

Method:

1. Generate random number u from standard uniform distribution $[0, 1]$
2. Compute x such that $F_X(x) = u$
3. Take x as a random number drawn from F_X

Example: If $X \sim \text{Exp}(\lambda)$, then $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.

Suppose $U \sim \text{Uniform}(0, 1)$, then the following is R code to generate random samples.

```
>set.seed(100)
>theta=2
>u=runif(100)
>x=-1*log(1-u)/theta
>mean(x);var(x)
```

Indirect Method with an example

Indirect method or acceptance/rejection sampling is another way to draw random samples from an unknown distribution or where different to implement direct method or F_X is not in proper form. This method generates samples from the target distribution X with pdf $f(x)$ by using

proposal distribution Y with pdf g_x . The proposal distribution is close to the target distribution or pick by its neighbor.

In practice, we assume that the ratio $f(x)/g(x)$ is bounded by a constant C ;

Method:

To generate a random variable Y :

(a) Generate U (independent from Y)

(b) If $u, Cg(x) \geq f(x)$

set $X=Y$ (accept) ; otherwise go back to step (a) (reject).

The choice of C is little tricky, Casell & Berge (2007) suggest that

Example: Suppose we have density with kernel

$$\exp\left(-\frac{1}{2}(1-t)^2\right) \cdot \frac{1}{1+t^2}, x$$

First we need to find the normalizing constant so that it becomes a proper density.

```
>l=function(t) exp(-(1/2)*(1-t)^2)*(1/(1+t^2))
```

```
>c=integrate(l,-Inf,Inf)$value;c
```

```
>td=function(t)l(t)/c
```

Next we plot target density and some proposal density $g(x)$ and choose C so that $Cg(x)$ overlaps the target density

```
>plot(function(x) td(x),from=-4,to=4.5,xlab="x",ylab="density")
```

```
>plot(function(x) dnorm(x,0.425,0.95)*1.33,from=-4, to=4.5,col="red",lty=2,add=T)
```

```
>title(main="Target and Trial densities")
```

```
>legend(-4,0.55, c("Target", "M*Trial"), col = c(1,2), text.col= "black", lty = c(1,2),bg="gray90")
```

```
>set.seed(73)
```

```
>nsim=10000
```

```
>v=rnorm(n=nsim,mean=0.425,sd=0.95)
```

```
>u=runif(n=nsim)
```

```
>ratio=td(v)/(dnorm(v,0.425,0.95)*1.33)
```

```
>ind=I(u<ratio)
```

```
>post=v[ind==1]
```

```
>length(post)
```

```
>plot(density(post),xlab="x",ylab="density",main="")
```

```
>plot(function(x) td(x),from=-4, to =4.5,col="red",lty=2,add=T)
>title(main="Target and Sample densities")
>legend(2,0.55, c("Target", "Sample"), col=c(2,1), text.col="black", lty = c(2,1),bg="gray90")
```

Application Q4.8.5 (direct method)

Determine a method to generate random observations for the logistic pdf. If access is available, write an R function which returns a random sample of observations from a logistic distribution.

Solution: The cdf of logistic distribution is

$$F_x = \frac{1}{1 + e^{-x}}$$
 To determine the inverse function, set $u = F_x$ and then solve it for x . After some algebra, we

get $u = \frac{1}{1 + e^{-x}}$

$$1 - u = \frac{e^{-x}}{1 + e^{-x}}$$

$$\log \frac{u}{1-u} = -x$$
 Thus, if U is uniform $(0,1)$, then $\log \frac{U}{1-U}$ has a logistic distribution with cdf F_x

The following is the R-code to draw n random samples from logistic distribution.

```
>rlogist=function (n){
>u=runif(n)
>rlogist=log(u/(1-u))
>rlogist
}
```

Application Q4.8.7 (direct method)

Determine a method to generate random observations for the Laplace pdf. If access is available, write an R function which returns a random sample of observations from a Laplace distribution.

Solution:

The cdf of the Laplace distribution is given as:

$$F_x = \begin{cases} 1 - \frac{1}{2}e^{-x} & x \geq 0 \\ \frac{1}{2}e^x & x < 0 \end{cases}$$

The inverse of the above cdf is

$$F_u^{-1} = \begin{cases} -\log 2(1-u) & 0 \leq u < \frac{1}{2} \\ \log 2(1-u) & \frac{1}{2} \leq u < 1 \end{cases}$$

Thus, if U is uniform $(0,1)$ then F_U^{-1} has the Laplace pdf. The following R-code is used to generate n observation from Laplace distribution.

```
> uni=runif(n)
> x=rep(0,n)
```

$>x[uni<0.5]=\log(2*uni[uni<0.5])$
 $>x[uni>=0.5]=-\log(2*uni[uni>=0.5])$

Delta Method

The delta method is a general method for deriving the variance of a function of asymptotically normal random variables with known variance.

Theorem 5.5.24: Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ in distribution. For a given function g and a specific value of θ , suppose that $g'(\theta) \neq 0$. Then $\sqrt{n}(g(Y_n) - g(\theta)) \xrightarrow{d} N(0, \sigma^2 (g'(\theta))^2)$ in distribution.

In this situation we take one more term in Taylor expansion. There is a possibility that $g'(\theta) = 0$.

to get

$$g(Y_n) = g(\theta) + g'(\theta)(Y_n - \theta) + \frac{g''(\theta)}{2}(Y_n - \theta)^2 + \text{Remainder}.$$

Setting $g'(\theta) = 0$, we have

$$g(Y_n) = g(\theta) + \frac{g''(\theta)}{2}(Y_n - \theta)^2 + \text{Remainder}.$$

Recall that the square of a standard normal is a χ^2_1 , which implies that $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ in distribution.

Theorem 5.5.26 (Second-order Delta Method).

Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ in distribution. For a given function g and a specific value of θ , suppose that $g'(\theta) = 0$ and $g''(\theta) \neq 0$. Then

$$\sqrt{n}(g(Y_n) - g(\theta)) \xrightarrow{d} N(0, \sigma^2 g''(\theta)) \text{ in distribution.}$$

This approximation technique is very useful whenever more than one parameter to be estimated and more than one variable to be used in the estimator. For example, in growth studies, the ration weight/height is a variable of interest. We have already studied that ratio of two normal random variables has a Cauchy distribution.

Example:

Suppose we observe $X_1, X_2, \dots, X_n$ independent Bernoulli (p) random variables. The typical parameter of interest is p , but another population is $1-p$. As we would estimate p by $\hat{p} = \frac{\sum_{i=1}^n X_i}{n}$, as an estimate of $1-p$

we might consider using $1 - \hat{p}$. How might we estimate the variance of $\frac{1 - \hat{p}}{1 - p}$?

Answer: Using above notation, consider $g(p) = \frac{1-p}{1-p}$.

We know that $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$.
 $\text{Var}\left(\frac{1 - \hat{p}}{1 - p}\right) = \frac{1}{(1-p)^2} \text{Var}(\hat{p}) = \frac{1}{(1-p)^2} \cdot \frac{p(1-p)}{n} = \frac{p}{n(1-p)}$.

Example: Suppose X is a random variable with $E[X] = 0$. If we want to estimate a function

g , a first-order approximation would give us

$$g(X) \approx g(0) + g'(0)X.$$

If we use $g(X)$ as an estimator of g , then

$$E[g(X)] \approx g(0).$$

$$\text{Var}[g(X)] \approx g'(0)^2 \text{Var}[X].$$

Lecture no.7

(38 to 44)

Point Estimation

Let $X_1, \dots, X_n$ be a random sample on a random variable X with pdf or pmf is $f(x)$ or $p(x)$, where $\hat{\theta}$ is a statistic used to estimate θ is called a point estimator. A point estimator is denoted by $\hat{\theta}$ and is a function of random variable X , i.e. $\hat{\theta} = \hat{\theta}(X)$. A point estimate is

one of the possible value of the point estimator when a sample is taken.

Since an estimator is a function of the sample therefore it is a random variable whereas an estimate is the realized value of an estimator so it is a number.

Example1: Suppose that $X_1, \dots, X_n$ are iid from $N(\mu, \sigma^2)$. Then the following are all point estimates of the mean

Sample median
 $\hat{M} = \frac{1}{n} \sum_{i=1}^n X_i$

Example 2: Let θ denote the proportion of the individuals in the population who favor the government policies. To estimate θ , sample $X_1, \dots, X_n$ are iid from $\text{Ber}(\theta)$; i.e. $X_i = 1$ if individual i favors the policies, and $X_i = 0$ otherwise. Then an estimate of θ is the sample mean, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

The basic objective of the point estimation is to seek a method of finding good point estimator. The choice of good estimator depends on many factors such as the data, assumed model etc. It is important to note that under study methods (discuss later) do not guarantee that they provide exact estimator but we are expecting to provide at least some reasonable estimator.

Methods of Point Estimation

Estimation of parameters of a probability function can be done by using sample statistic. The problem arises when we do not have any idea about the form of the estimator that can be used to estimate the population parameter. Various methods are available to obtain a proper estimator for a parameter with a view that the resulting estimator has certain distributional properties. The following are the methods used to estimate the parameters of the certain distribution.

1. Maximum Likelihood (ML) Method
2. Moment Method
3. Bayesian Method

Before studying the Maximum Likelihood Method we must know what is likelihood function?

Likelihood function

Let $X_1, \dots, X_n$ be a random sample on a random variable X with pdf is $f(x; \theta)$, where θ is an unknown parameter and θ is a scalar quantity but it can assume as vector. The likelihood function is

Suppose $X_1, \dots, X_n$ are iid random variables with common pdf $p(x; \theta)$, where θ is a scalar quantity but it can assume as vector. The likelihood function is

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

We treat L as a function of θ and write it as $L(\theta)$. The log of this function is usually more convenient to use and given as

$$\ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta)$$

Note that there is no loss of information in using $\ln L$ because the log is a one-to-one function.

Regularity Conditions

The point estimator of θ is $\hat{\theta}$, where $\hat{\theta}$ maximizes the function L and we call it maximum likelihood estimator (MLE) of θ . To study the asymptotic properties of $\hat{\theta}$, it is assumed that we know its functional form but in practice it is unknown. Fortunately, there exist a set of regularity conditions, if satisfied, indirectly we can make safe statement about the asymptotic properties of the MLE.

Let θ_0 denote the "true value" of θ , let I_0 denote the Fisher Information on θ evaluated at θ_0 and let n denote the sample size. The asymptotic properties of the MLE hold under the following regularity conditions:

1. The pdfs defined by any two values of θ are distinct, i.e. $f(x; \theta_1) \neq f(x; \theta_2)$.
2. The parameter space Θ is an open interval of the real line within which θ_0 lies.
3. For any finite n , at least two derivatives of the log-likelihood function w.r.t θ exist in an open neighborhood of θ_0 .

4. In an open neighbourhood of θ_0 , the Fisher Information identity holds.

I_0 is finite and positive.

5. In an open neighborhood of θ_0 :

- (i) $\frac{1}{n} \log L(\hat{\theta}_n) \xrightarrow{P} \log L(\theta_0)$
- (ii) $\frac{1}{n} \log L(\hat{\theta}_n) - \log L(\theta_0) \xrightarrow{P} 0$
- (iii) For some constant c , $\frac{1}{n} \log L(\hat{\theta}_n) - \log L(\theta_0) \xrightarrow{P} 0$

Maximum Likelihood Method

The maximum likelihood method is the most popular technique for deriving estimators. Recall that if $X_1, \dots, X_n$ be the iid random sample from a population with pdf or pmf $f(x; \theta)$, the likelihood function is defined as

$$L(\theta | X) = L(\theta | x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta)$$

Definition: For each sample point X , $\hat{\theta}_n$ is a maximum likelihood estimator of θ if

$$\hat{\theta}_n = \underset{\theta \in \Theta}{\text{Arg max}} L(\theta | x_1, \dots, x_n)$$

$\hat{\theta}$ achieves its maximum value at $\hat{\theta}$.
 The notation Arg max mean that $\mathbf{X}$. To determine mle,
 we often take the log of the likelihood function and find its value; i.e. letting l
 the mle solves the equation
 $\frac{d}{d\theta} l = 0$, subject to the condition that

In general, MLE is a good point estimator if it has follows some optimal properties, discuss later.
 There are two inherent problems of finding the maximum of a function. The first problem is that
 of actually finding the global maximum and verifying it. The second is that of numerical
 sensitivity, i.e. a small change in data, how sensitive the estimate. It is observed that a slightly
 different sample will provide vastly different MLEs.

Example 1 (Normal likelihood)

Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$ and let $\mathbf{X}$ denote the likelihood function. Then

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

The equation $\frac{d}{d\mu} L = 0$ reduces to

$$\sum_{i=1}^n (x_i - \mu) = 0 \quad \text{which has the solution } \hat{\mu} = \bar{x}.$$

Hence, $\hat{\mu}$ is the MLE of μ . To verify that $\hat{\mu}$ is, in

fact, global maximum of the likelihood, we can use the following argument. First, note that
 $\hat{\mu}$ is the only solution to $\frac{d}{d\mu} L = 0$. $\hat{\mu}$ is only zero of the first derivative. Second,
 verify that $\frac{d^2}{d\mu^2} L < 0$. Thus $\hat{\mu}$ is the only extreme point in the interior and it is
 maximum.

Example 2 (Bernoulli MLE)

Let $X_1, \dots, X_n$ be iid Bernoulli(p). Then the likelihood function is

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

Since the above function is not easy to take its derivative, so we take its log and
 where $y = \sum_{i=1}^n x_i$

the resulting function is much easier to differentiate.
 1

$\log L(p|X)$ and setting it equal to zero, we get y
 $\frac{d}{dp} \log L(p|X) = 0$
 If $y=0$, then $p=1$ is the global maximum in this case. If $y=n$, then $\hat{p} = \frac{y}{n}$.
 If $0 < y < n$, then $\log L(p|X) = \log \left(\prod_{i=1}^n p^{y_i} (1-p)^{1-y_i} \right) = y \log p + (n-y) \log (1-p)$
 $\log L(p|X)$ is a monotone function of p , and simple to verify that $\frac{d}{dp} \log L(p|X) = 0$ in each case.
 In either case $\hat{p}$ is the MLE of p .
 case. Thus $\hat{p} = \frac{\sum_{i=1}^n X_i}{n}$.

Lecture no.8 (45 to 53)

Application Q6.1.2 (a & b) (Hogg, McKean & Craig)

Let $X_1, \dots, X_n$ represent a random sample from each of the distributions having the following pdfs:

- (a) $f(x) = \begin{cases} x^n, & 0 \leq x \leq 1, \\ 0, & \text{zero elsewhere.} \end{cases}$
 (b) $f(x) = \begin{cases} e^{-x}, & x \geq 0, \\ 0, & \text{zero elsewhere.} \end{cases}$ Note this is a non-regular case.

In each case find the mle $\hat{\theta}$ of θ .

Solution:

(a) The likelihood function is

$$L(\theta) = \prod_{i=1}^n x_i^n = n! \theta^n \prod_{i=1}^n x_i^{n-1}$$

The log likelihood function is

$$\log L(\theta) = \log n! + n \log \theta + (n-1) \sum_{i=1}^n \log x_i$$

$$\frac{d}{d\theta} \log L(\theta) = \frac{n}{\theta} = 0$$

$$\ln \log x_n$$

$$\frac{d}{d\theta} \log L(\theta) = \frac{n}{\theta} = 0$$

$$\ln \log x_n$$

$$\frac{d}{d\theta} \log L(\theta) = \frac{n}{\theta} = 0$$

$$L(\mathbf{x}) = \prod_{i=1}^n \frac{1}{x_i} e^{-x_i/x_i}$$

Note that $\ln \frac{1}{x_i} = -\ln x_i \leq 0$.

Thus $L(\mathbf{x})$ is strictly concave. Hence

$$\frac{\partial}{\partial x_i} \ln L(\mathbf{x}) = -\frac{1}{x_i}$$

(b) The likelihood function is

$$L(\mathbf{x}) = e^{-\sum_{i=1}^n x_i / x_i} \text{ provided that } x_i \geq x, \text{ otherwise } L(\mathbf{x}) = 0$$

The log likelihood function is

$$\ln L(\mathbf{x}) = -\sum_{i=1}^n \frac{x_i}{x_i} \text{ i.e. } -\sum_{i=1}^n 1 = -n$$

is an increasing function of x if $x \geq \min_{i=1, \dots, n} x_i$

Application Q6.1.5 (Hogg, McKean & Craig)

Suppose $X_1, \dots, X_n$ are iid with pdf $f(x) = \frac{1}{2} e^{-x/2}$, $x \geq 0$, zero elsewhere. Find the mle of θ .

Solution:

The likelihood function is

$$L(\mathbf{x}) = \left(\frac{1}{2}\right)^n e^{-\sum_{i=1}^n x_i / 2}$$

The log likelihood function is

$$\ln L(\mathbf{x}) = -n \ln 2 - \frac{1}{2} \sum_{i=1}^n x_i$$

$$\frac{\partial}{\partial x_i} \ln L(\mathbf{x}) = -\frac{1}{2}$$

$\frac{\partial}{\partial x_i} \ln L(\mathbf{x}) = -\frac{1}{2}$, which implies that

$$L(\theta) = \prod_{i=1}^n P(X_i = x_i) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} = \frac{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

We know that the pdf of the given function is $P(X=x) = \frac{e^{-\theta} \theta^x}{x!}$ then the mle is

Application Q6.1.8 (Hogg, McKean & Craig)

Let the table

x	0	1	2	3	4	5
Frequency	7	14	12	13	6	3

represent a summary of a random sample of size 55 from a Poisson distribution. Find the maximum likelihood estimate of $P(X=2)$.

Solution:

We know that the maximum likelihood estimate of the parameter θ in Poisson distribution is $\hat{\theta} = \bar{X}$. Now the mean of the given data is

$$\bar{X} = \frac{0 \times 7 + 1 \times 14 + 2 \times 12 + 3 \times 13 + 4 \times 6 + 5 \times 3}{55} = \frac{163}{55} \approx 2.96$$

So the mle of $P(X=2)$ is

$$P(X=2) = \frac{e^{-\hat{\theta}} \hat{\theta}^2}{2!} = \frac{e^{-2.96} (2.96)^2}{2} \approx 0.27$$

Application Q6.1.9 (Hogg, McKean & Craig)

so that we know that 11 Let $X_1, \dots, X_n$ be a random sample from a Bernoulli distribution with parameter p . If p is restricted

to $\frac{1}{2} \leq p \leq 1$, find the mle of this parameter.

Solution: The pdf of Bernoulli distribution is

$$f(x) = p^x (1-p)^{1-x}, \quad x = 0, 1.$$

The likelihood function is

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{\sum x_i} (1-p)^{n - \sum x_i}$$

The log likelihood function is

- (i) The asymptotic behavior of maximum likelihood estimates
- (ii) Variance of the estimators
- (iii) Cramer-Rao inequality
- (iv) Jeffery's prior in Bayes inference

From the previous topic, we know the (1)

$$\log f(x) - \log f(\theta) = 0$$

Differentiate it again, we get

$$2 \log f(x) - \log f(\theta) = 0$$

The 2nd quantity on the left hand side is the Fisher Information and denoted by I . i.e.

$$I = \int \left(\frac{\partial \log f(x)}{\partial \theta} \right)^2 f(x) d\theta$$

So (2) implies that

$$I = \int \left(\frac{\partial \log f(x)}{\partial \theta} \right)^2 f(x) d\theta$$

$$E \left(\frac{\partial \log f(x)}{\partial \theta} \right)^2$$

Using $E \left(\frac{\partial \log f(x)}{\partial \theta} \right) = 0$, the Fisher Information is the variance of the random variable

$$\frac{\partial \log f(x)}{\partial \theta}; \text{ i.e.}$$

$$I = \int \left(\frac{\partial \log f(x)}{\partial \theta} \right)^2 f(x) d\theta$$

Remarks

The information $\int \left(\frac{\partial \log f(x)}{\partial \theta} \right)^2 f(x) d\theta$ or $\int \left(\frac{\partial \log f(x)}{\partial \theta} \right)^2 f(x) d\theta$ is the weighted mean, where the

weights are given by $f(x)$. This means that on the average, the greater value of these derivatives, the more information we have about θ . Clearly if they are equal to zero, it means that we have zero information about θ . The important function $\log f(x)$ is called the score

function. As we know that it actually determines the estimating equations for the mle i.e. the mle

$$\frac{d}{dx} \log f(x) = 0$$

Example 1 (Information for Bernoulli Random Variables)

Let X be a Bernoulli $b, 1$. Thus the pdf of Bernoulli distribution is

$$f(x) = \begin{cases} 1-x & \text{if } x=0 \\ x & \text{if } x=1 \end{cases}$$

The log of the function is

$$\log f(x) = \begin{cases} -\log(1-x) & \text{if } x=0 \\ -\log(x) & \text{if } x=1 \end{cases}$$

So IE

$$E(X) = \frac{1}{2}$$

We know that the $E(X)$ in Bernoulli distribution is $\frac{1}{2}$. So

$$I = \frac{1}{2} \log \frac{1}{1-x} + \frac{1}{2} \log \frac{1}{x}, \text{ which is the larger of } x \text{ values, close to zero or one.}$$

Information for the Location Family

Consider the random samples $X_1, \dots, X_n$, such that

$$X_i \sim f(x; \theta)$$

where $\theta_1, \dots, \theta_n$ are iid having common pdf with support $\mathcal{S}$. Then the common pdf of X is

$f(x; \theta)$. We call model (1) is a location model. Assume that $f(x)$ satisfies the

regularity conditions, then the information is obtained as:

By definition

$$I = \int_{-\infty}^{\infty} \log f(x) f(x) dx$$

$$= \int_{-\infty}^{\infty} \log f(x) f(x) dx$$

, or equivalently

$$= \int_{-\infty}^{\infty} \log f(x) f(x) dx$$

Using the change of variable $z = x$, we get

$$I = \int_{-\infty}^{\infty} \log f(z) f(z) dz$$

So in location model, the information does not depend on μ .

Lecture no.9 (54 to 60)

Example (Laplace Distribution)

Let $X_1, \dots, X_n$ be iid with density

$$f(x) = \frac{1}{2} e^{-|x|}$$

This density is known as Laplace or double exponential distribution.

The log likelihood function is

$$\ln L(\theta) = \sum_{i=1}^n \ln \left(\frac{1}{2} e^{-|x_i|} \right)$$

Where the sgn (signum) function of a real number x is defined as

$$\text{sgn } x = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Alternatively

$$\int_{-\infty}^{\infty} \frac{d}{dx} f(x, \theta) dx = 0.$$

Setting equation (1) to 0, the solution for θ is the median $x_1, x_2, \dots, x_n$, so $\hat{\theta}_{\text{med}}$.

Now using this to find the information for location family, we proceed as:

Let $X_1, \dots, X_n$ be the random sample from this distribution. Then X_i can be expressed as

$$X_i = \theta + e_i \quad \text{where } e_1, \dots, e_n \text{ are iid having common pdf}$$

$$f(z) = \frac{1}{2} e^{-|z|} \quad \text{by using} \quad \int_{-\infty}^{\infty} f(z) dz = 1$$

$$I = \int_{-\infty}^{\infty} \left(\frac{f'(z)}{f(z)} \right)^2 f(z) dz$$

since $f(z)$ is symmetric about 0, $f'(z)$ is an odd function so it is equal to 1.

$$I = \int_{-\infty}^{\infty} f'(z)^2 dz = 1$$

Although Laplace distribution does not satisfy the regularity conditions but this argument can be made rigorous.

Application Q 6.2.7(a)

Let X have a Gamma distribution with $\alpha = 4$ and $\beta = 1$. Find the Fisher Information I .

Solution:

The pdf of Gamma distribution is

$$f(x) = \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-x/\beta}$$

The log of the function is

$$\log f(x) = -\log \Gamma(\alpha) - \alpha \log \beta + (\alpha-1) \log x - \frac{x}{\beta}$$

$$2 \log f(x) = 42x$$

$$\text{So } E\left[\frac{1}{X}\right] = \frac{42}{4} = 10.5$$

We know that the $E(X)$ in Gamma distribution is $\frac{1}{\lambda}$, i.e. $E(X) = \frac{1}{\lambda}$, So

$$\lambda = \frac{1}{10.5} = \frac{2}{21}$$

Application Q 6.2.12(a)

Recall the $\hat{\theta}$ is the mle of θ for a beta $(\alpha, 1)$ distribution. Also, $W = \sum_{i=1}^n \log X_i$ has the gamma distribution

$\frac{1}{2}W$ has a χ^2_{2n} distribution. Show that $2W$ has a χ^2_{2n} distribution.

Solution:

The pdf of W is

$$f_W(w) = \frac{1}{\Gamma(n)} e^{-w} w^{n-1} \quad w > 0$$

Since it is given that $W \sim \text{Gamma}(n, 1)$, so the mgf of it is

given as

$$M_W(t) = E[e^{tW}] = \int_0^\infty e^{tw} \frac{1}{\Gamma(n)} e^{-w} w^{n-1} dw = \frac{1}{\Gamma(n)} \int_0^\infty e^{-w(1-t)} w^{n-1} dw$$

$$= \frac{1}{\Gamma(n)} \int_0^\infty e^{-u} \left(\frac{u}{1-t}\right)^{n-1} \frac{1}{1-t} du \quad (u = w(1-t))$$

$$= \frac{1}{\Gamma(n)} \frac{1}{(1-t)^n} \int_0^\infty e^{-u} u^{n-1} du = \frac{1}{(1-t)^n}$$

Now the mgf of $2W$ is

$$M_{2W}(t) = E[e^{t(2W)}] = E[e^{2tW}] = 1 - \frac{2t^2}{n}$$

which is also the mgf of χ^2_2 with $2n$ degree of freedom.

Write down any three asymptotical properties of MLE.

Solution:

- Asymptotical properties
1. Consistency. For any $\epsilon > 0$, the MLE $\hat{\theta}_n$ converges to θ_0 in probability as $n \rightarrow \infty$. i.e., $\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta_0| < \epsilon) = 1$. In other simple words MLE will give you the true value of the parameter with high probability.
 2. Asymptotical Normal. The MLE asymptotically follows the normal distribution. i.e., for some d ,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0))$$
 This can be stated alternatively as

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0))$$
 3. Efficiency. MLE has asymptotical variance equal to Cramer-Rao lower bound (CRLB). It expresses the lower bound of the variance of estimator. MLE has the optimal lowest asymptotic variance.

Invariance

4. If $\hat{\theta}_n$ is the MLE of θ , then $g(\hat{\theta}_n)$ is the MLE of $g(\theta)$.

Theorem 6.1.2:

Let $X_1, \dots, X_n$ be iid with the pdf $f(x; \theta)$. For a fixed θ_0 , let g be a parameter of interest. Suppose $\hat{\theta}_n$ is the MLE of θ . Then $g(\hat{\theta}_n)$ is the MLE of $g(\theta)$.

Proof: Although the result holds for any function of θ but here we consider g is a one-to-one, then

$g(\hat{\theta}_n)$ and $\hat{\theta}_n$ are both maximized by $\hat{\theta}_n$, so

or

$g(\hat{\theta}_n)$

If g is one-to-one, then $\hat{\theta}$ which maximizes the L still corresponds to $g(\hat{\theta})$, so $g(\hat{\theta})$ still corresponds to the maximum of L .

Explanation of this property: Let the likelihood function $L(x_1, \dots, x_n; \theta)$ is the MLE of θ and g is a one-to-one function, i.e. $g(\theta)$ is the MLE of $g(\theta)$. For example, $\hat{\theta}^2$ is the MLE of θ^2 and if we apply square transformation on it, then $\hat{\theta}^2$ is also the MLE of θ^2 , so $\hat{\theta}^2$ is the MLE of θ^2 . An important point is that, for one-to-one transformation the proof of invariance property of MLE is simple but if extend this property to some arbitrary transformation, this property of MLE still holds.

Example:

Let $X_1, \dots, X_n$ be iid from Bernoulli random variables with probability of success p . We know that $\text{Var}(X) = p(1-p)$, $X \in \{0, 1\}$.

Let consider p as θ . We already know that the MLE of p in Bernoulli distribution is $\bar{X}$, i.e. $\hat{p}$ (sample mean).

Now by apply the invariance property that the $\text{Var}(X)$ is $p(1-p)$.

MLE MLE MLE

Newton-Raphson Method

Generally we face problem when we come across a situation when L is non-linear or the solution cannot be in closed form. In such situations, we need some numerical methods to solve them. Newton Raphson's method is one of the iterative methods that help to solve such equations and able to converge them rapidly. In this method, any differentiable form can be approximated by a linear function which latterly used to develop recursive procedure, based on certain conditions, find the desired solution.

Suppose we want to estimate the unknown parameter θ (here is scalar, extended for vector on the same line). Our basic objective is to find the solution of $L'(\theta) = 0$. This method follows as: first we have an initial guess θ_0 then approximate L by a linear function at θ_0 using Taylor series expansion as

$L(\theta) \approx L(\theta_0) + L'(\theta_0)(\theta - \theta_0) + \frac{L''(\theta_0)}{2}(\theta - \theta_0)^2 + \dots$ higher term

Now solve for θ_1 and calling it θ_1 , we get

$$\theta_1 = \theta_0 - \frac{L'(\theta_0)}{L''(\theta_0)}$$

We then substitute for θ and repeat the process until it converge. So the method is to take a reasonable starting value of θ and iterate it at x by using and stop the algorithm when x is large or

$$f(x) = \frac{e^{-x}}{1 + e^{-x}}$$

$$f(x) = \frac{e^x}{1 + e^x}$$

Lecture no.10 (61 to 67)

Example using R

Let $X_1, \dots, X_n$ be the iid random sample from logistic distribution with pdf

$$f(x) = \frac{e^{-x}}{1 + e^{-x}} \quad , \quad x \in \mathbb{R}$$

The log likelihood can be written as

$$l(\theta) = \sum_{i=1}^n \log f(x_i; \theta) = -n \sum_{i=1}^n \log(1 + e^{x_i})$$

Differentiating w.r.to θ and putting it equal to zero, we get

$$\frac{\partial l(\theta)}{\partial \theta} = -n \sum_{i=1}^n \frac{e^{x_i}}{1 + e^{x_i}}$$

The above equation does not have any unique solution so we apply Newton's method. The 2nd derivative of log likelihood is

$$l''(\theta) = -n \sum_{i=1}^n \frac{e^{x_i}}{(1 + e^{x_i})^2}$$

R-code to estimate the θ .

```
x <- rlogis(50,1)
mlelogistic=function(x,theta0=mean(x),numstp=100,eps=0.0001){
  n=length(x)
  numfin=numstp
  small=1.0*10^(-8)
  ic=0
```

```

istop=0
while(istop==0){
  ic=ic+1
  expx=exp(-(x-theta0))
  lprime=n-2*sum(expx/(1+expx))
  l2prime=-2*sum(expx/(1+expx)^2)
  theta1=theta0-(lprime/l2prime)
  check=abs(theta0-theta1)/abs(theta0+small)
  if(check<eps){istop=1}
  theta0=theta1
}
list(theta1=theta1,check=check,realnumstps=ic)
}

```

Method of Moments

The method of moment (MM) is one of the oldest and easiest method of finding point estimators. This method was suggested by Karl pearson in the 18th century. The beauty of this method that it will almost always provide some estimates when other methods are intractable (hard to use).

Suppose $X_1, X_2, \dots, X_n$ be the random sample from a population having pdf or pmf $f(x)$.

In method of moments we equate first k sample moments to the corresponding k population moments and solving the resulting equations simultaneously. Thus the k -th sample moments of a set of n random sample $X_1, X_2, \dots, X_n$ is given as:

$$m_k' = \frac{1}{n} \sum_{i=1}^n X_i^k$$

The k -th moment of a random variable X is

$$\mu_k' = E(X^k)$$

Generally the population moment μ_k' is the function of $\theta_1, \theta_2, \dots, \theta_r$, i.e. $\mu_k' = \mu_k'(\theta_1, \theta_2, \dots, \theta_r)$.

estimators m_k' of μ_k' is obtained by solving the following system of equations

$$m_k' = \mu_k'(\theta_1, \theta_2, \dots, \theta_r) \quad k=1, 2, \dots, r$$

$$m_k' = \mu_k'(\theta_1, \theta_2, \dots, \theta_r) \quad k=1, 2, \dots, r$$

Thus the estimators are obtained by replacing the population moments by the sample moments.

Example 1 page 313 (Casella & Berger) Normal distribution

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and variance σ^2 .
Find the moment estimators for μ and σ^2 .

The pdf of normal distribution is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}, \quad -\infty < x < \infty.$$

The first and second moments about origin are

$$\mu = E(X) \quad \text{and} \quad \sigma^2 = E(X^2) - \mu^2.$$

Since there are two parameters μ and σ^2 , therefore we need two equations to equate population moments by the sample moments

$$\begin{aligned} \mu &= \bar{X} \\ \sigma^2 &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \end{aligned}$$

For this example the MM estimators are the same as the ML estimators.

Example 2 page 313 (Casella & Berger) Binomial distribution

Let $X_1, X_2, \dots, X_n$ be iid binomial k, p , that is,

$$P(X = x) = \binom{k}{x} p^x (1-p)^{k-x}, \quad x = 0, 1, \dots, k.$$

By assuming that both k and p are unknown.

The first and second moments about origin are

$$E(X) = kp \quad \text{and} \quad E(X^2) = kp + k(k-1)p^2.$$

$$E(X) = kp$$

$$\square \square \quad (1)$$

$$p \vdash p \quad \square \square \quad (2)$$

From (1) $\square$ putting this in (2), we get

$$\begin{array}{ccccccc} \square & \square & \square & \square & \square & \square & \square \\ mk & mm & 1 & m & m & m & m \\ 21 & kk & k & k & m & m & m \\ & & & & 1 & k & 22 \\ & & & & & & \square \end{array}$$

Solving the above equation for k,we get

$$\hat{k} \cdot \frac{mX}{mm} \frac{m_1^2}{m_2^2} \frac{X}{n} \frac{(X-X)}{i} \frac{X}{-2}$$

In the above example, the following are the important points:

1. It is possible to get negative estimates of k and p , which must be positive. Actually this is the case where the range of the estimator does not coincide with the range of the parameter itself estimating.
2. Talking to the fairness of this method, negative estimates will occur only when the sample mean is less than the sample variance showing great variability in the data.

Application Q 7.9 (Casella & Berger)

Let $X_1, X_2, \dots, X_n$ be iid with pdf

$$f(x) = \frac{1}{x^2} = x^{-2}, \quad f'(x) = -2x^{-3} = -\frac{2}{x^3}.$$

This is a uniform distribution 0, . The method of moment equation is



$$x \text{ d}; \quad x \text{ Ef } Xx \quad x \quad 2$$

Thus
Since

$\mu_{\hat{\theta}}$ is the function of mean so we can find its mean and variance as:

$$X \otimes_{\mathbb{K}} E \cong \bigoplus_{i=1}^n \mathbb{K} \otimes_{\mathbb{K}} E \cong \bigoplus_{i=1}^n E \cong E^{\oplus n}$$

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2} \text{Var} \left(\sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n} \text{Var}(X)$$

(1) Implies that $\text{var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n} \text{var}(X)$.

Application Q 7.12 (Casella & Berger)

Let $X_1, X_2, \dots, X_n$ be a random sample from a population with pmf

$$P(X = x) = \begin{cases} 1-x, & x = 0 \text{ or } 1, \\ x^2, & x = 2 \end{cases}$$

The method of moment equation is

so

$$m_1 = E(X) = \sum_{x=0}^2 x P(X=x) = 0(1-x) + 1(1-x) + 2x^2$$

$$= 1 - x + 2x^2$$

$$= 1 - \frac{1}{n} \sum_{i=1}^n X_i + 2 \left(\frac{1}{n} \sum_{i=1}^n X_i^2 \right)$$

$$\text{MSE} = \text{var}(\hat{\theta}) + \text{bias}^2$$

Now

$$E(X) = 1 - \frac{1}{n} \sum_{i=1}^n X_i + 2 \left(\frac{1}{n} \sum_{i=1}^n X_i^2 \right)$$

$$\text{var}(X) = E(X^2) - (E(X))^2 = \sum_{x=0}^2 x^2 P(X=x) - (1 - x + 2x^2)^2$$

$$= 0^2(1-x) + 1^2(1-x) + 2^2x^2 - (1 - x + 2x^2)^2$$

$$= 1 - x + 4x^2 - (1 - x + 2x^2)^2$$

$$= 1 - x + 4x^2 - (1 - 2x + x^2 + 2x^2 - 4x^3 + 4x^4)$$

$$= 1 - x + 4x^2 - 1 + 2x - x^2 - 2x^2 + 4x^3 - 4x^4$$

$$= 2x - 3x^2 + 4x^3 - 4x^4$$

$$\text{bias} = E(\hat{\theta}) - \theta = 0$$

$$\text{MSE} = \frac{1}{n} \text{var}(X) = \frac{1}{n} (2x - 3x^2 + 4x^3 - 4x^4)$$

Properties of Moment Method

The following are some important properties of moment method:

1. The MM is easy to use and provide consistent estimator under the weak law of large number, i.e. $n \sum_{i=1}^n \frac{X_i}{n} \rightarrow \mu$
2. In some cases, the other method like MLE may be intractable without computer whereas MM estimators can be easily and quickly calculated.
3. This method is useful to access the initial values of the parameters which may be useful for the solution of ML equations and further improvement can be achieved by using different simulation methods, e.g. Newton-Raphson method etc.
4. This is also useful in obtaining approximation to the distribution of the statistics. This technique is called "moment matching". The moment of the distribution of any statistic can be matched with those of any distribution which are similar, e.g. Satterthwaite approximation.

Lecture no.11 (68 to 71)

Bayesian Method

Bayesian method is an attractive technique in statistics to handle a variety of problems in these days. This approach to statistics is quite different from the classical approach. In classical approach, a random sample $X_1, X_2, \dots, X_n$ is drawn from a population having unknown parameter

which is a fixed quantity. On the basis of sample information we estimate the unknown parameter.

In Bayesian approach θ is considered a random quantity whose variation is observed by the probability distribution, called prior distribution. This is a subjective distribution based on the experimenter's belief and is formulated before the data. A sample is then taken by the population having a parameter θ and the prior distribution is updated with the sample information. This updated prior is called the posterior distribution.

Thus there are three basic ingredients of Bayesian learning:

- 1- Prior distribution. If Ω represents the parametric space, then for each value of $\theta \in \Omega$, the prior distribution describes our belief that θ represents the true population characteristic.
- 2- Sampling model. If Ω represents the sample space which consists of all possible dataset of X , then for all $\theta \in \Omega$ and $X \in \Omega$, the sampling model describes our belief that X be the outcome if we know θ .
- 3- Posterior distribution. For each numerical value of $\theta \in \Omega$, the posterior distribution describes our belief that θ is the true value having some observed dataset. Ω

If we denote the prior distribution by $\pi(\theta)$ and the sampling distribution by $f(X|\theta)$, then the posterior distribution which is the conditional distribution of θ given the sample X , is

$$f(x) = \frac{f(x, \theta)}{m(x)}, \quad \text{where } m(x) = \int f(x, \theta) d\theta$$

where $m(x)$ is the marginal of x , and is given as

$m(x) = \int f(x, \theta) d\theta$. Since $m(x)$ is not a function of θ , that is why we use the sign of proportionality in the Bayes formula and is used in most common as

$$f(\theta|x) \propto f(x|\theta) f(\theta)$$

Posterior distribution = Prior distribution * likelihood of the data.

In Bayes, posterior distribution is used to make inferences about the parameter θ and we know that it is a random quantity so that mean of the posterior is generally used as a point estimate of θ .

Bayes provides a recursive mechanism for updating the posterior distribution and it is very useful where the observations are obtained sequentially. As new observation is obtained, the posterior is updated treating previous posterior as the prior and obtaining a chain as:

$$P(\theta) \rightarrow P(\theta|y_1) \rightarrow P(\theta|y_1, y_2) \rightarrow \dots$$

This sequential application of Bayes is very useful in some engineering application.

Conjugate Prior

A prior is said to be a conjugate prior for a family of distributions if the prior and the posterior distributions are from the same family. This means that the form of the posterior has the same distributional form as the prior distribution. So conjugacy is a closure property.

Closure property: A set is closure under an operation if the result of the operation on numbers always produces a number of same set e.g. real numbers are closure under subtraction but the natural number are not i.e. 3 and 8 are both natural number but 3-8 is not a natural number.

If $\mathcal{C}$ is a class of sampling distribution $f(y|\theta)$ and $\mathcal{P}$ is a class of prior distribution for θ , $f(\theta)$, then $\mathcal{C}$ is said to be conjugate for $\mathcal{P}$ if

$$f(\theta|y) \in \mathcal{P} \quad \text{for all } y$$

For example, if the likelihood is binomial, $f(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$, a conjugate prior on θ is the beta distribution: so it follows that the posterior distribution on θ is also a beta distribution.

The pmf of binomial is

$$P_y = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

The likelihood function of binomial is

$$L(\theta) = \prod_{i=1}^n \theta^{y_i} (1-\theta)^{1-y_i} = \theta^{\sum y_i} (1-\theta)^{n - \sum y_i}$$

The beta density with parameters a and b is

$\pi(a, b)$

$\pi(b, a)$; $\pi(a, b)$ implies that $\pi(b, a)$ which is a beta density with parameters a, n and b, n .

Other examples of conjugate prior/likelihood combinations include the normal/normal, gamma/Poisson, gamma/gamma, gamma/beta etc. The main advantage of conjugate prior is that it simplifies the computations.

Other types of Priors

There are many types of prior distribution but here we discuss three of them:

1- Non-informative Priors

It is a class of non-conjugate prior and tries to provide a situation where it does not provide any information about the parameter. In simple words, a prior distribution is non-informative if the prior is “flat” relative to the likelihood function. Thus a prior P is non-informative if it has minimal impact on the posterior distribution. Non-informative prior is also called vague, diffuse, objective and flat prior.

Several statisticians favor the non-informative prior because it appears to be more objective. However it is not appropriate that non-informative prior provide total ignorance about the parameter of interest. The main issue is when we use non-informative prior that it is not easy to prove that the posterior is proper and it leads to improper posterior. Generally non-informative priors are not invariant under transformation. A common choice of non-informative prior is the flat prior. In flat prior assign equal likelihood on all possible values of the parameters.

2- Improper Prior

A prior P is said to be improper if

$$\int P d\mu = \infty$$

For example, a uniform prior distribution on real line, $\pi(\theta) = 1$, is an improper

prior. Improper priors are often used in Bayesian inference and they yield non-informative prior and proper posterior distributions. To determine whether a posterior distribution is proper, we need to make sure that the normalizing constant $\int P d\mu$ is finite for all values of y .

The inference based on improper posterior do not consider valid.

3- Jeffreys' prior

A very commonly used prior is the Jeffreys prior. It is defined as the square root of the determinant of the Fisher Information matrix and is given as

$$PI, \quad \frac{1}{\sqrt{I(\theta)}}$$

where I is the Fisher Information based on the likelihood function $P(y)$:

$$I = E \left[\frac{\partial \log P(y)}{\partial \theta} \right]^2$$

Jeffreys prior fulfill the local uniformity property i.e. it does not change much over the region where the likelihood is significant and does not assume large values outside that range. Another important property of Jeffreys prior is that it is invariant under one-to-one transformation. It has some short coming as well i.e. the prior is improper for many models, it can be difficult to use specially in high dimensions situation.

Example 1 (Binomial Bayes estimation)

Let $X_1, \dots, X_n$ iid Bernoulli (p). Then $Y = \sum_{i=1}^n X_i$ is binomial n, p . We want

distribution on p is beta, Find the posterior distribution along with its mean.

Solution:

The likelihood function of binomial is

$$f(y) = \binom{n}{y} p^y (1-p)^{n-y}$$

It is also given that the prior density is beta, which is given as

$$p(p) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}$$

So the joint distribution of y and p is

$$p(y, p) = \binom{n}{y} p^y (1-p)^{n-y} \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}$$

$$= \frac{\binom{n}{y}}{B(\alpha, \beta)} p^{\alpha+y-1} (1-p)^{\beta+n-y-1}$$

The marginal of y is

$$f(y) = \frac{\binom{n}{y}}{B(\alpha, \beta)} \int_0^1 p^{\alpha+y-1} (1-p)^{\beta+n-y-1} dp$$

$$= \frac{\binom{n}{y}}{B(\alpha, \beta)} \frac{B(\alpha+y, \beta+n-y)}{B(\alpha, \beta)}$$

The posterior distribution, distribution of p given y is

$$f(p|y) = \frac{f(y, p)}{f(y)}$$

Using (1) and (2), we get which is a beta distribution.

$$f(p|y) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}$$

The natural estimate of p is the mean, so the mean of the posterior distribution is

$$\hat{p}_B = \frac{\alpha + \sum y_i}{\alpha + \beta + n}$$

This also called the Bayes estimator of p . The Bayes estimator $\hat{p}_B$ can be written as

$$\hat{p}_B = \frac{\alpha}{\alpha + \beta + n} + \frac{\sum y_i}{\alpha + \beta + n}$$

So $\hat{p}_B$ can be written as a linear combination of prior mean and the sample mean with weights being determined by α and β and n . Since the prior, beta distribution, and the posterior distribution is a beta distribution so it is a conjugate prior.

Lecture no.12 (72 to 75)

Example 2 (Normal Bayes estimators)

Let $X_1, \dots, X_n$ suppose our prior distribution on μ is $N(\mu_0, \sigma_0^2)$. Here we assume that σ^2 and σ_0^2 are all known. Show that the posterior distribution of μ is also normal with mean and variance are given by

$$E[\mu|x] = \frac{\sigma_0^2 \sum x_i + n \sigma^2 \mu_0}{\sigma_0^2 + n \sigma^2}$$

$$\text{Var}[\mu|x] = \frac{\sigma_0^2 \sigma^2}{\sigma_0^2 + n \sigma^2}$$

Solution:

Since it is given that $X_1, \dots, X_n$ then the pdf of X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{x^2}{2\sigma^2}\right\}$$

It is given that the prior distribution of μ is $N(\mu_0, \sigma_0^2)$ the pdf is

$$f(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left\{-\frac{\mu^2}{2\sigma_0^2}\right\}$$

The posterior density is

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\text{constant} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Now first we evaluate the exponents as:

$$-\frac{(x-\mu)^2}{2\sigma^2} = -\frac{x^2 - 2x\mu + \mu^2}{2\sigma^2} = -\frac{x^2}{2\sigma^2} + \frac{x\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2}$$

Writing above equation in quadratic form of x , we get

where

$$a = -\frac{1}{2\sigma^2}, \quad b = \frac{\mu}{\sigma^2}, \quad \text{and } c = -\frac{\mu^2}{2\sigma^2}$$

Ignoring c because it does not contain x . So by completing square, we get

$$-\frac{x^2}{2\sigma^2} + \frac{x\mu}{\sigma^2} = -\frac{1}{2\sigma^2} \left(x^2 - 2x\mu \right) = -\frac{1}{2\sigma^2} \left(x^2 - 2x\mu + \mu^2 - \mu^2 \right) = -\frac{1}{2\sigma^2} \left((x-\mu)^2 - \mu^2 \right) = -\frac{(x-\mu)^2}{2\sigma^2} + \frac{\mu^2}{2\sigma^2}$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \exp\left(\frac{\mu^2}{2\sigma^2}\right)$$

$$p(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

and standard deviation σ .

The above is exactly the same shape as the normal with mean μ and standard deviation σ .

Since the probability distribution is determined by their shape therefore $f(x)$ is a normal distribution with mean and variance are:

$$E(x) = \mu \quad \text{and} \quad \text{variance} = \sigma^2$$

$$\text{Var} = \frac{1}{a} \left(\frac{1}{22} + \frac{1}{22} \right)$$

Application Q 11.3.1 (H M & C)

Let $X_1, X_2, \dots, X_n$ be a random sample from a Cauchy distribution with pdf

$$f(x) = \frac{1}{\pi} \frac{1}{1 + x^2}, \quad x \in \mathbb{R},$$

where $\mu = 0$. Use the non-informative prior $\pi(\mu) = 1$.

- Find the posterior pdf of μ , other than the constant of proportionality.
- Evaluate this posterior pdf if $x_1 = 1, x_2 = 4, x_3 = 12, x_4 = 34, x_5 = 0.5, x_6 = 1.5, x_7 = 2.0, x_8 = 1.2, x_9 = 0.1, x_{10} = 0.5, x_{11} = 1.0, x_{12} = 1.5, x_{13} = 2.0, x_{14} = 2.5, x_{15} = 3.0, x_{16} = 3.5$.
- From the 16 values in part (b), where does the maximum of the posterior pdf seem to be?

Solution: (a) The likelihood of the Cauchy distribution other than the constant of proportionality is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\pi} \frac{1}{1 + (x_i - \mu)^2}$$

Since the prior is the non-informative prior so

Thus the posterior pdf is

$$g(\mu) = \frac{1}{\pi^{n+1}} \frac{1}{\prod_{i=1}^n (1 + (x_i - \mu)^2)}$$

(b) The posterior pdf is if $x_1 = 1, x_2 = 4, x_3 = 12, x_4 = 34, x_5 = 0.5, x_6 = 1.5, x_7 = 2.0, x_8 = 1.2, x_9 = 0.1, x_{10} = 0.5, x_{11} = 1.0, x_{12} = 1.5, x_{13} = 2.0, x_{14} = 2.5, x_{15} = 3.0, x_{16} = 3.5$

$g(1,0.5)=0.108$	$g(2,0.5)=0.047$	$g(3,0.5)=0.047$	$g(4,0.5)=0.108$
$g(1,1.0)=0.10$	$g(2,1.0)=0.10$	$g(3,1.0)=0.10$	$g(4,1.0)=0.10$
$g(1,1.5)=0.088$	$g(2,1.5)=0.111$	$g(3,1.5)=0.111$	$g(4,1.5)=0.088$
$g(1,2.0)=0.0769$	$g(2,2.0)=0.10$	$g(3,2.0)=0.10$	$g(4,2.0)=0.0769$

(c) From part (b), the maximum of the posterior pdf looks at 0.111.

Application Q 11.3.7(a) (H M & C)

Consider the Bayes model

$X_1, X_2, \dots, X_n$ iid with distribution $b, 1$. Obtained the Jeffreys' prior for this model. Obtained the Jeffreys' prior for this model.

Let X be a Bernoulli $b, 1$. Thus the pdf of Bernoulli distribution is

$$f(x) = b^x (1-b)^{1-x}.$$

The log of the function is

$$\begin{aligned} \log f(x) &= x \log b + (1-x) \log (1-b) \\ \log f(x) &= x \log b + (1-x) \log (1-b) \\ \frac{d}{dx} \log f(x) &= \log b - \log (1-b) \\ \frac{d^2}{dx^2} \log f(x) &= -\frac{1}{b} - \frac{1}{1-b} \end{aligned}$$

So IE

$$\begin{aligned} E(X) &= \frac{1}{2} \\ E(X) &= \frac{1}{2} \end{aligned}$$

We know that the $E(X)$ in Bernoulli distribution is $\frac{1}{2}$. So

$$\begin{aligned} I &= \frac{1}{b(1-b)} \\ I &= \frac{1}{b(1-b)} \end{aligned}$$

Since the Jeffreys' prior is the square root of the of the Fisher information, so

which is a beta distribution with parameters $\frac{1}{2}$ and $\frac{1}{2}$.

P,

$$\sqrt{\frac{1}{b(1-b)}}$$

Application Q 11.3.7(b) (H M & C)

Assuming squared-error loss, and obtain the Bayes estimate of θ .

Under the squared-error loss function, the Bayes estimate is the posterior mean. Now the log likelihood function of the sampling model is

The prior distribution is

So the posterior distribution is

So the Bayes estimate is

$$\frac{1}{B} \cdot \frac{nX}{n} = \frac{1}{2}$$

Bayesian Predictive distribution

In Bayesian analysis posterior predictive distribution is used to predict future values on the conditional observed values. It is given as

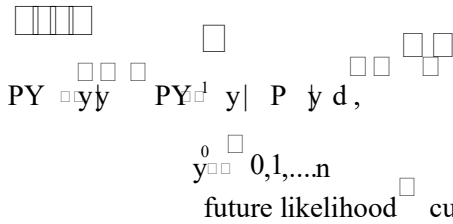
$$P(y|y) \quad P(y|y) \quad P(y|y) \quad P(y|y)$$

Let $y_1, \dots, y_n$ be the outcomes from a sample of size n from binary random variables and let y_{n+1} be an additional outcome from the same population that has yet to be observed. The predictive distribution of y_{n+1} is the conditional distribution of y_{n+1} given $y_1, \dots, y_n$.

Probability of some particular number y of success in this sample given θ is

PY y | C ny . 1ny

Recall that our knowledge contained in the posterior distribution i.e. $\mathcal{P}[\mathbf{y}|\mathbf{X}]$, the posterior predictive distribution is given as



If the posterior predictive distribution is some sort of beta distribution, then it is implemented in R by using “pbetap” in library “LearnBayes” which calculate predictive probabilities. For

example, if $n=25$ and $\alpha=1$, $\beta=21$

`pbetap(c(1,21), 25, 0.5)` : gives output

prob	0.4565	0.2536	0.1383	0.0740	0.0388	0.0199

Properties of point estimate

Let $X_1, \dots, X_n$ are the iid random samples having pdf or pmf $f(x)$. In point estimation we are going to estimate a quantity θ , called an estimator. The estimation of θ depends on the data, the assumed model and the definition of “good”. Generally “good” means that $\hat{\theta}$ is close to θ , but θ itself is unknown. Now the question is how we can judge that our estimator $\hat{\theta}$ is good or not. Indeed, $\hat{\theta}$ is not exactly equal to θ , and it is important to know about this difference which can be obtained by the sampling distribution of $\hat{\theta}$ in term of standard error or confidence interval. The properties of estimator $\hat{\theta}$ depends on its sampling distribution and one of the important objective of statistical inference is to identify that the given estimator is an “optimal” and fulfill the definition of “good”. Therefore, it is important that it should have possessed the following properties. As it is already mentioned that that all the outcomes of $\hat{\theta}$ depends on its sampling distribution and most of the times the full sampling distribution of $\hat{\theta}$ is

not available therefore we focus on the properties which do not required the complete sampling distribution of it. These properties are summarized as:

- 1- Unbiasedness
- 2- Consistency
- 3- Efficiency
- 4- Sufficiency

Unbiasedness

An estimator is said to be unbiased if the expected value is equal to the corresponding value of population parameter.

Definition: Let $X_1, \dots, X_n$ denote a random sample drawn from X with pdf $f(x)$. Let T be a statistic. We say that T is an unbiased estimator of θ if $E(T) = \theta$.

No matter what is the actual value of μ , if we apply $\bar{X}$ to many data sets obtained from $X_1, \dots, X_n$, then the average of all these values will equal to μ . In other words, $E(\bar{X}) = \mu$, for all μ , otherwise it is a biased estimator and it is given as

$$\text{Bias} = E(\bar{X}) - \mu.$$

To understand this concept, let's take a numerical example by computing the mean and standard deviation of the normal distribution. Suppose we have an example of IQ scores, we draw 100000 scores from a normal distribution with mean score 50 and standard deviation is 5 score.

```
Population<-rnorm(100000, mean=50, sd=5)
```

```
Mean(population)=49.99
```

```
Sd(population)=4.99
```

So one could see that the expected value is almost equal to the population mean and same is for standard deviation.

Example 1

Let $X_1, \dots, X_n$ be iid from normal distribution having mean μ and variance σ^2 . Let $\bar{X}$ be the sample mean and s^2 be the sample variance. Then $\bar{X}$ and s^2 are unbiased estimators of μ and σ^2 respectively.

Let $X_1, \dots, X_n$ be a random sample of size n from a population with mean μ , then

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$E(\bar{X}) = \frac{E(X_1) + E(X_2) + \dots + E(X_n)}{n}$$

$$= \frac{\mu + \mu + \dots + \mu}{n}$$

$$= \frac{n\mu}{n} = \mu$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Recall the following facts

$$E(X) = \mu, \text{ var}(X) = \sigma^2 \text{ and } E(X^2) = \sigma^2 + \mu^2$$

$$E(\bar{X}^2) = \frac{1}{n^2} E\left(\sum_{i=1}^n X_i^2 - 2\sum_{i < j} X_i X_j + \sum_{i=1}^n X_i^2\right)$$

$$E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \mu$$

$$E\left[\frac{1}{n} \sum_{i=1}^n X_i^2\right] = \mu^2 + \sigma^2$$

$$E\left[\frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i\right)^2\right] = \sigma^2$$

Thus sample mean and sample variance are unbiased estimators of population mean and population variance.

Example 2

Let $X_1, \dots, X_n$ be iid $\text{Ber}(p)$ and suppose we want to estimate p .

Suppose $\hat{p}$ is an unbiased estimator of p , so $E[\hat{p}] = p$ for all p or, equivalently

$$E[\hat{p} - p] = 0 \text{ for all } p.$$

The pmf of Bernoulli is

$$f(x) = p^x (1-p)^{1-x}$$

For $X_1, \dots, X_n$, the joint pmf is

$$f(x_1, \dots, x_n) = p^{\sum x_i} (1-p)^{n - \sum x_i}$$

$$E[\hat{p} - p] = \sum_{\mathbf{x}} (\hat{p} - p) f(\mathbf{x}) = 0 \text{ for all } p.$$

The quantity on the left side is a polynomial in p of degree $n-1$. By the fundamental rule of algebra that there are at most $n-1$ real roots of the above equation. However, unbiasedness requires that there be infinitely many roots which contradict the fundamental theorem. Thus we conclude that there are no unbiased estimators of p .

Lecture no.14 (81 to 85)

Remarks (Hogg, McKean & Criag)

Let X be a random sample from Poisson distribution. Suppose the goal is to estimate $e^{-\lambda}$ itself. We know that $\hat{X}$ is an unbiased estimator of λ . However the natural estimator $e^{-\hat{X}}$ is not an unbiased estimator of $e^{-\lambda}$.

$$E(e^{-\hat{X}}) = \sum_{x=0}^{\infty} e^{-x} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1$$

We know that

$$E(e^{-\lambda \hat{X}}) = \sum_{x=0}^{\infty} e^{-\lambda x} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1$$

But if we consider $e^{-\lambda \hat{X}}$, then it is unbiased estimator of $e^{-\lambda}$ as

$$E(e^{-\lambda \hat{X}}) = \sum_{x=0}^{\infty} e^{-\lambda x} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1$$

Application Q 6.2.9

Let $X_1, \dots, X_n$ is a random sample from a distribution with pdf

$$f(x) = \frac{1}{\Gamma(x)} e^{-x} x^{x-1}, \quad x > 0$$

Show that $\bar{X}$ is unbiased estimator of λ .

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \lambda = \lambda$$

Let t

$$\frac{dx}{dt} = x^2, \quad \text{if } x > 0, \text{ then } t = \frac{1}{x}$$

So $2\bar{X}$ is an unbiased estimator of μ .

Consistency

An estimator T_n , computed from a sample of n observation is said to be consistent estimator if and only if it converges in probability to θ as $n \rightarrow \infty$.

Definition: Let X be a random variable with cdf F_X , $\theta = \int_{-\infty}^{\infty} x f_X(x) dx$. Let $X_1, \dots, X_n$ be a sample from the distribution of X and let T_n denote a statistic. We say T_n is a consistent estimator of θ if, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|T_n - \theta| < \epsilon) = 1, \text{ or equivalently } \lim_{n \rightarrow \infty} P(|T_n - \theta| \geq \epsilon) = 0.$$

Consistency is an asymptotic property and depends on the sample size n ; one should get close to the true value of θ as n gets larger. There are many examples of consistent estimator i.e, the sample mean $\bar{X}$ is a consistent estimator of μ , sample variance s^2 is consistent estimator of σ^2 , sample proportion is a consistent estimator of p in binomial distribution etc. On the other hand, the smallest observation, the largest observation and the mean of the smallest and largest observations are not consistent estimators because to do not approach the population values as sample sizes are increased.

Generally a consistent estimator is an unbiased in limit but an unbiased estimator may or may not be a consistent estimator. But this not necessarily true, e.g.

$$T_n = \frac{1}{n} \sum_{i=1}^n X_i$$

So it is a biased estimator, but if we apply,

$$\lim_{n \rightarrow \infty} E(T_n) = \mu$$

estimator. On the other hand, we know that $E(X) = \mu$, here X is a sequence of random variable.

But if we consider only the first value of the X i.e. X_1 , then $E(X_1) = \mu$ i.e. for every X is expected value is μ , so it is unbiased estimator of μ . But $\lim_{n \rightarrow \infty} X_1 = X_1$, as X_1 is a number

without regarding a small or large number and when we apply limit it remains a number. So it is not a consistent estimator.

Example 1(Hogg, McKean & Criag)

Let X be any random variable such that EX and $E X^2$ are finite, then show that s_2 is a consistent estimator for the variance of X .

To prove this, first it is important to recall the concept of convergence in probability. A sequence of random variables $X_1, X_2, \dots$ (say X_n) converges in probability to a random variable X if, for every $\epsilon > 0$,

$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$, where ϵ is a small quantity greater than 0. And on the same line the

SLLN states that if $X_1, X_2, \dots$ be iid random variables with $EX_i = \mu$ and $\text{Var } X_i = \sigma^2$. Then

$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mu$ almost surely, that is, X_n converges almost surely to μ . Now recall that

X

$$\begin{aligned} \hat{s}_2^2 &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \\ &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2 \end{aligned}$$

is an unbiased estimator of σ^2 ,

Note that $\frac{1}{n} \sum_{i=1}^n X_i^2$ is the second moment of X . Also observe that

$$\begin{aligned} \text{var} \left(\frac{1}{n} \sum_{i=1}^n X_i^2 \right) &= \frac{1}{n} \text{var}(X^2) \\ &= \frac{1}{n} (E X^4 - (E X^2)^2) \end{aligned}$$

Now clearly $\frac{1}{n} \sum_{i=1}^n X_i^2 \rightarrow E X^2$ as $n \rightarrow \infty$, therefore as variance approaches to zero,

in probability. Thus $\hat{s}_2^2 \rightarrow \sigma^2$ in probability. Therefore $\hat{s}_2$ is a consistent estimator of σ .

$$\hat{s}_2^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \rightarrow E X^2 - \mu^2 = \sigma^2$$

Example 2 (Hogg, McKean & Criag)

Let $X_1, \dots, X_n$ is a random sample from uniform distribution i.e. $Uni(0, 1)$. Also suppose that $Y = \max\{X_1, \dots, X_n\}$ is unknown. An intuitive estimate of θ is the maximum of the sample. Let

Let the pdf of Y_n is

Now

$$f_{Y_n}(t) = n t^{n-1} \quad 0 \leq t \leq 1$$

$$EY_n = \int_0^1 t \cdot n t^{n-1} dt = n \int_0^1 t^n dt = n \left[\frac{t^{n+1}}{n+1} \right]_0^1 = \frac{n}{n+1}$$

But one can easily see that

Thus Y_n is biased estimator of θ .

$\frac{n+1}{n} Y_n$ is an unbiased estimator of θ .

Further if we use limit, $\lim_{n \rightarrow \infty} \frac{n+1}{n} Y_n = Y$, hence the maximum of the sample is a

consistent estimator of θ . $\frac{n+1}{n} Y_n$ is also consistent estimator of θ .

Lecture no.15 (86 to 89)

Application from the exercise

Exercise 5.1.2. Let the random variable Y_n have a distribution that is $bp(1-p)^n$. This result is one form of the weak law of large

a) Prove that $\frac{Y_n}{n}$ converges in probability to p .

b) Prove that $1 - \frac{Y_n}{n}$ converges in probability to $1-p$.

c) Prove that $\frac{Y_n}{n} \pm \frac{1}{\sqrt{n}}$ converges in probability to $p \pm \frac{1}{\sqrt{n}}$.

Solution:

(a) Let $X_1, \dots, X_n$ are the iid from Binomial distribution with parameters n and p . We know that the expected value and the variance of binomial distribution are np and $np(1-p)$.

respectively, which are finite. Since $\sum_{i=1}^n X_i$ has a b.p., therefore $\sum_{i=1}^n Y_i$ also has the same distribution. By using the Chebyshev's Inequality

$$P\left\{\left|\frac{\sum_{i=1}^n Y_i}{n} - \mu\right| \geq \frac{1}{10}\right\} \leq \frac{\text{var}\left(\frac{\sum_{i=1}^n Y_i}{n}\right)}{\left(\frac{1}{10}\right)^2} = \frac{\frac{\text{var}(Y)}{n}}{\frac{1}{100}} = \frac{\text{var}(Y)}{10n}$$

Hence Y_p .

d n by using Theorem 5.1.2, and 5.1.3 in Hogg, McKean & Criag, p-290-291, it follows that 11

(a) Since $\frac{\sum_{i=1}^n Y_i}{n}$ is unbiased for μ , it follows that 11 Y_p .
(c) Similarly, it follows from Theorem 5.1.5 in Hogg, McKean & Criag, p-291 that

Efficiency

Let $\hat{\mu}_1$ and $\hat{\mu}_2$ be two unbiased estimators of μ with $E\hat{\mu}_1 = \mu$ and $E\hat{\mu}_2 = \mu$. Then $\hat{\mu}_1$ is more efficient estimator of μ than $\hat{\mu}_2$ if $\text{var} \hat{\mu}_1 < \text{var} \hat{\mu}_2$. The relative efficiency of $\hat{\mu}_1$ to $\hat{\mu}_2$ is given as:

$$\text{Eff} \hat{\mu}_1 = \frac{\text{var} \hat{\mu}_2}{\text{var} \hat{\mu}_1}$$

In general, it is not necessary that

Thus $\hat{\mu}_1$ is more efficient than $\hat{\mu}_2$ if and only if $\text{Eff} \hat{\mu}_1 > 1$.

we compare the efficiency of two unbiased estimators but we can also find the efficiency of biased estimators as well in term of their mean square errors (study in the coming topics) as:

$$\text{Eff} \hat{\mu}_1 = \frac{\text{MSE} \hat{\mu}_2}{\text{MSE} \hat{\mu}_1}, \quad \text{MSE} \hat{\mu}_2 = E(\hat{\mu}_2 - \mu)^2 \quad \text{with } E\hat{\mu}_2 = \mu$$

For illustrative purpose, suppose $X_1, \dots, X_n$ are random sample from $N(\mu, \sigma^2)$, then we know that $\bar{X}$ and S^2 are the unbiased estimators of μ and σ^2 respectively.

Clearly, $\text{var } X$ is more efficient than $\bar{X}$. Also the efficiency of $\bar{X}$ relative to X is 2.

In addition, we can also define a most efficient estimator who has the smallest variance in the class of all unbiased estimators e.g. suppose t is called most efficient estimator if and only if there is no other member of the class of unbiased estimators other than t .

t.

In case of point estimate, the efficiency of an unbiased estimator can be defined as:

Let Y be an unbiased estimator of θ . The statistic is called an efficient estimator of θ if and only if the variance of Y attains the Rao-Cramers lower bound. The Rao-Cramers lower bound is defined as:

$\text{var } Y \geq \frac{1}{I(\theta)}$ is the Fisher Information in the sample.

Example 1 (Hogg, McKean & Criag) p-332

Let $X_1, \dots, X_n$ denote a random sample from a Poisson distribution that has mean θ . It is known that $\bar{X}$ is an MLE of θ ; we shall prove that it is also an efficient estimator of θ .

We know that the pmf of Poisson distribution is

$$p(x) = \frac{e^{-\theta} \theta^x}{x!}, \quad x = 0, 1, \dots$$

Suppose $X_1, \dots, X_n$ are the n independent observations, then the likelihood function is

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!}$$

The log likelihood function is

$$\ln L(\theta) = \sum_{i=1}^n \ln \left(\frac{e^{-\theta} \theta^{x_i}}{x_i!} \right)$$

$$= \sum_{i=1}^n \left(-\theta + x_i \ln \theta - \ln x_i! \right)$$

$$= -n\theta + \left(\sum_{i=1}^n x_i \right) \ln \theta - \sum_{i=1}^n \ln x_i!$$

$$I = \frac{E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right]}{n}$$

$$\text{var}(\bar{X}) = \frac{1}{n}$$

But $\frac{1}{n}$ is the variance of $\bar{X}$. Hence $\bar{X}$ is an efficient estimator of μ .

Example 2 (Hogg, McKean & Criag) p-332.

Let $X_1, \dots, X_n$ denote a random sample of size n from a distribution with pdf

$$f(x) = \frac{1}{x^2}, \quad 0 < x < 1.$$

Also $I = \frac{1}{n}$

$$\text{And } E\left[\frac{1}{\bar{X}^2}\right] = \frac{1}{n}$$

- When $\frac{1}{\bar{X}^2}$ will become unbiased estimator?
- Find the variance of unbiased estimator.
- Is unbiased estimator is an efficient estimator?

Where parameter space is $(0, 1]$. This is a beta distribution with parameters α and 1 and we denote it by beta $(\alpha, 1)$.

Suppose $X_1, \dots, X_n$ are the n independent observations, then the likelihood function is

$$L(x) = \prod_{i=1}^n \frac{1}{x_i^2}$$

The log likelihood function is

$$\ln L(x) = \sum_{i=1}^n \ln \left(\frac{1}{x_i^2} \right) = -2 \sum_{i=1}^n \ln x_i$$

$$\frac{d}{dx} \ln L(x) = -2 \sum_{i=1}^n \frac{1}{x_i} = 0$$

$$\sum_{i=1}^n \frac{1}{x_i} = 0$$

□ □ □ □

$$I = E \left[\frac{1}{n} \sum_{i=1}^n \left(\frac{d}{dx} \ln f(x) \right)^2 \right]$$

Hence the information is $\frac{n}{2}$.

Now we investigate its efficiency.

Solving $\frac{d}{dx} \ln f(x) = 0$ is $\frac{1}{2}$. To obtain the distribution of $\hat{\theta}$, let

$$Y = \frac{1}{n} \sum_{i=1}^n \ln X_i$$

$$f(y) = \frac{1}{\Gamma(n)} e^{-ny} \quad \text{for } y \geq 0$$

Since X_i 's are independent, so it is state forward that $W = \sum_{i=1}^n Y_i \sim \Gamma(n, 1)$.

$$E[W] = n, \quad \text{For } W = \sum_{i=1}^n Y_i$$

$\hat{\theta} = \frac{W}{n}$ is a biased estimator but if $n \rightarrow \infty$, this biased vanishes.

$\hat{\theta}$ is an unbiased estimator. Now

From here we can visualize that

$$E[\hat{\theta}] = \frac{1}{n} E[W] = \frac{1}{n} \cdot n = 1, \quad \text{and}$$

$$\text{var}(\hat{\theta}) = \frac{1}{n^2} \text{var}(W) = \frac{1}{n^2} \cdot n = \frac{1}{n}$$

From this we can also find the variance of unbiased estimator as

$$\text{var}(\hat{\theta}) = \frac{1}{n}$$

Since $\text{var}(\hat{\theta}) = \frac{1}{n}$, i.e. $\frac{1}{n}$, so unbiased estimator is not an efficient estimator. But if $n \rightarrow \infty$, it becomes efficient.

Lecture no.16 (90 to 95)

Application 6.2.7 (Hogg, McKean & Criag)

Let X have a gamma distribution with $k=4$ and $\theta > 0$.

(a) Find the Fisher information I .

(b) If $X_1, \dots, X_n$ is a random sample from this distribution, show that the MLE of θ is an efficient estimator of θ .

We know that the pdf of Gamma distribution with parameters k and θ is

$$f(x; k, \theta) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}, \quad x > 0, \quad k > 0, \quad \theta > 0,$$

with mean is $EX = k\theta$ and variance is $\text{var } X = k\theta^2$.

For $k=4$ and θ , the new Gamma distribution is

$$f(x) = \frac{1}{4^4 \Gamma(4)} x^3 e^{-x/4}$$

The likelihood function is

$$\begin{aligned} \ln f &= \text{constant} - 4 \ln \theta - \frac{x}{\theta} \\ \ln f &= \text{constant} - 4 \ln \theta - \frac{x}{\theta} \\ \ln f &= \text{constant} - 4 \ln \theta - \frac{x}{\theta} \end{aligned}$$

We know that the Fisher information is

$$I = -E \left[\frac{\partial^2 \ln f}{\partial \theta^2} \right] = -E \left[\frac{4}{\theta^3} \right] = \frac{4}{\theta^3}$$

The Rao-Cramer Lower Bound for variance of any estimator $\hat{\theta}$ is

$$\frac{k^2}{nI}, \text{ where } k = \theta$$

The log-likelihood function is

$$\ln L(x; \theta) = \text{constant} - 4n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n X_i$$

$$\ln L(\mathbf{x}; \theta) = -\frac{n}{2} \ln |C| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \mu)^T C^{-1} (\mathbf{x}_i - \mu)$$

Thus

$$E \left[\frac{\partial \ln L}{\partial \theta} \right] = 0$$

So $\hat{\theta}$ is an unbiased estimator of θ . Now

$$\text{var} \left(\frac{\partial \ln L}{\partial \theta} \right) = \frac{1}{n} \text{var} \left(\frac{\partial \ln L}{\partial \theta} \right) = \frac{1}{n} \text{var} \left(\frac{\partial \ln L}{\partial \theta} \right)$$

The Rao-Cramer Lower Bound for $\hat{\theta}$ is $\frac{1}{nI}$ which is the same as $\text{var} \left(\frac{\partial \ln L}{\partial \theta} \right)$, so $\hat{\theta}$ is an efficient estimator.

Application 6.2.7 (c) (Hogg, McKean & Criag)

What is the asymptotic distribution of $\hat{\theta}_n$.

We know that from the Theorem 6.2.2 (Hogg, McKean & Criag) about the asymptotics of MLE that if $X_1, \dots, X_n$ are iid with pdf $f(x; \theta)$ for $\theta \in \Theta$ under the regularity conditions of MLE are hold, with the Fisher information $I(\theta) > 0$, then any consistent sequences of solution of the MLE equations satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, I^{-1}(\theta))$$

Thus using this, we have an $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, I^{-1}(\theta))$

We know that from the previous part that $I(\theta) = \frac{4}{\theta^2}$.

$$\text{So } \sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, \frac{\theta^2}{4})$$

Application 6.2.8 (a) (Hogg, McKean & Criag)

Let X be $N(0, \sigma^2)$. Find the Fisher information $I(\sigma^2)$.

We know that the pdf of Normal distribution is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

with mean is EX and variance is $\text{var } x$.

The likelihood function is

$$\ln f = \text{constant} - \frac{1}{22} \ln x_2$$

$$\ln f = \frac{1}{22} x_2$$

$$\ln f = \frac{1}{22} x^2$$

We know that the Fisher information is

$$I = E \left[\frac{\partial}{\partial \theta} \ln f \right]^2 = \frac{1}{22} EX_2^2$$

Application 6.2.8 (b) (Hogg, McKean & Criag)

Let X be $0, 0$. Find the Fisher information I .
 If $X_1, \dots, X_n$ is a random sample from $0, 0$,
 MLE of θ is an efficient estimator of θ .

Solution:

The log-likelihood function is

$$\ln L(x; \theta) = \text{constant} - \frac{n}{22} \ln X_2$$

$$\ln L(x; \theta) = \frac{n}{22} X_2$$

So $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_{2i}$ is an unbiased estimator of θ . Now

$\text{var } \hat{\theta} = \frac{1}{n} \text{var } X_2$. Now

$$\text{var } X = EX^2 - (EX)^2$$

□ □

d

e

$$E X_{44} = \frac{1}{2} \frac{x_1}{2}$$

Substituting $\frac{x_1}{2}$ and solving it we get

$$EX = \frac{2}{4} \frac{3}{2}$$

EX

EX

$$\frac{2}{2} \frac{2}{24} \frac{2}{2} \cdot \frac{2}{2} \frac{32}{2}$$

Thus

$$\text{var} \frac{2}{n}$$

The Rao- Cramer Lower Bound for $\hat{\theta}$ is $\frac{1}{nI}$ which is the same as $\text{var} \hat{\theta}$, so $\hat{\theta}$ is an

efficient estimator.

Application 6.2.8 (c) (Hogg, McKean & Criag)

What is the asymptotic distribution of $\hat{\theta}_n$.

We know that from the Theorem 6.2.2 (Hogg, McKean & Criag) about the asymptotics of MLE that if $X_1, \dots, X_n$ are iid with pdf f_X for θ under the regularity conditions of MLE are hold, with the Fisher information satisfies $0 < I(\theta) < \infty$, then any consistent sequences of solution of the MLE equations satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, \frac{1}{I(\theta)})$$

Thus using this, we have as $n \rightarrow \infty$, $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, \frac{1}{I(\theta)})$

We know that from the previous part that $I(\theta) = \frac{1}{2}$. So according to the above theorem

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, 2)$$

Asymptotically efficiency

Let $X_1, \dots, X_n$ be identically and independently distributed with pdf f_X ; θ . Suppose that $\hat{\theta}_n$ is an estimator of θ , such that $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} N(0, \frac{1}{I(\theta)})$

1. The asymptotic efficiency of $\hat{\theta}_n$ is defined as $e = \frac{1/I(\hat{\theta}_n)}{1/I(\theta)}$

2. The estimator $\hat{\theta}_1$ is said to be asymptotically efficient if one or greater than one.

3. Let $\hat{\theta}_2$ be another estimator of θ , then asymptotic relative efficiency (ARE) of $\hat{\theta}_1$ to $\hat{\theta}_2$ is the reciprocal of the ratio of their respective variances, i.e. $\frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$. Thus if ratio is greater than, $\hat{\theta}_1$ is better than $\hat{\theta}_2$.

For example consider the normal distribution with mean μ and variance σ^2 . We also know that the $\bar{X}$ is an efficient estimator by the Rao-Cramer inequality. Consider sample median $\tilde{X}$ and variance σ^2 . another estimator which an unbiased and consistent estimator of μ and we know that for large n

sample median is approximately normally distributed with mean μ and variance $\frac{\pi}{2} \frac{\sigma^2}{n}$. asymptotic relative efficiency (ARE) of $\tilde{X}$ to $\bar{X}$ is

$\frac{\text{Var}(\bar{X})}{\text{Var}(\tilde{X})} = \frac{\sigma^2/n}{\frac{\pi}{2} \frac{\sigma^2}{n}} = \frac{2}{\pi} \approx 0.637$. This means that sample mean is more efficient estimator than sample median.

Lecture no.17 (96 to 99)

Application 6.2.11 (Hogg, McKean & Criag)

Let $X_1, X_2, \dots, X_n$ be the mean of a random sample of size n from a $N(\mu, \sigma^2)$ distribution, where μ is unknown. Show that $\bar{X}_n$ is an unbiased estimator of μ and find its efficiency.

$$E(\bar{X}_n) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

$$E(\bar{X}_n^2) = E\left(\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n X_i X_j\right) = \frac{1}{n^2} \left(\sum_{i=1}^n E(X_i^2) + \sum_{i \neq j} E(X_i X_j) \right)$$

$$= \frac{1}{n^2} \left(n(\sigma^2 + \mu^2) + n(n-1)\mu^2 \right) = \frac{\sigma^2 + \mu^2}{n} + \mu^2 = \frac{\sigma^2}{n} + \mu^2$$

But we know that

$$E\left[\sum_{i=1}^n X_i^2\right] = \sum_{i=1}^n E[X_i^2]$$

$$= \sum_{i=1}^n (E[X_i]^2 + \text{var}(X_i))$$

$$= n(E[X]^2 + \sigma^2)$$

as X's are independent. After simplifying we

$$E\left[\frac{1}{n} \sum_{i=1}^n X_i^2\right] = E[X^2]$$

arrived at the following result:

$$E\left[\frac{1}{n} \sum_{i=1}^n X_i^2\right] = E[X^2]$$

So $\frac{1}{n} \sum_{i=1}^n X_i^2$ is an unbiased estimator of $E[X^2]$.

Now we find the variance of $\frac{1}{n} \sum_{i=1}^n X_i^2$.

$$\text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i^2)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{var}(X^2)$$

$$\text{var}(X^2) = E[X^4] - (E[X^2])^2$$

$$\text{var}(X^2) = E[X^4] - \sigma^4$$

By using Goodman's expression and assuming iid and independent random variables.

$$E[X^4] = E[X^2]^2 + 3\sigma^4$$

$$\text{var}(X^2) = 2\sigma^4$$

$$\text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{2\sigma^4}{n}$$

Example 1 (Casella & Berger) p-331

Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$. Then we know that the statistic $\bar{X}$ and S^2 are unbiased estimators.

Since $E\bar{X} = \mu$ and $ES^2 = \sigma^2$ for all n and σ^2 . This is true without the normality assumption. The MSE of these estimators are

$$a) E(\bar{X} - \mu)^2 = \frac{\sigma^2}{n}$$

$$a) E(S^2 - \sigma^2)^2 = \frac{2\sigma^4}{n-1}$$

Suppose the samples are taken from the normal distribution. We know that $\frac{(n-1)S^2}{\sigma^2}$

The first proof is straightforward, for second we can proceed as:

$\frac{(n-1)S^2}{\sigma^2}$ is distributed

as a Chi-square random variable with $n-1$ degree of freedom. So

$$\begin{aligned} \text{var}(\bar{X}) &= \frac{\sigma^2}{n} \\ \text{var}(S^2) &= \frac{2\sigma^4}{n-1} \end{aligned}$$

The MSE of $\bar{X}$ remains $\frac{\sigma^2}{n}$ if the normality assumption is dropped. MCQS However, the MSE of S^2 does not same if the normality assumption is relaxed.

Further consider an alternative estimator for σ^2 is the ML estimator

$$\hat{\sigma}^2_{ML} = \frac{1}{n} \sum_{i=1}^n X_i^2 = \frac{1}{n} (n\bar{X}^2 + (n-1)S^2)$$

is a biased estimator of σ^2 . The variance of $\hat{\sigma}^2_{ML}$ can be calculated as

$$\text{var}(\hat{\sigma}^2_{ML}) = \frac{1}{n^2} \text{var}(n\bar{X}^2 + (n-1)S^2) = \frac{1}{n^2} (4\sigma^4 + (n-1)^2 \frac{2\sigma^4}{n-1})$$

Hence its MSE is

$$E(\hat{\sigma}^2_{ML} - \sigma^2)^2 = \frac{2\sigma^4}{n} + \frac{(n-1)^2 \sigma^4}{n^2} = \frac{2\sigma^4}{n} + \frac{(n-1)^2 \sigma^4}{n^2}$$

Thus $E[\hat{\theta}] = \theta$, showing that $\hat{\theta}$ has the smaller MSE than $\hat{\theta}_2$.
 S. Thus, by trade-off variance for biased, the MSE can be improved.

Application 7.12 (Casella & Berger)

Let $X_1, \dots, X_n$ be a random sample from a population with pmf

$$P(X = x) = \frac{1}{n} \left(\frac{x}{n} \right)^{n-1}, \quad x = 1, 2, \dots, n.$$

- Find the moments and MLE estimator of θ .
- Find the mean squared errors of each of the estimators.
- Which estimator is preferred?

Solution (a)

Moment Method

$$E[X] = \sum_{x=1}^n x \cdot \frac{1}{n} \left(\frac{x}{n} \right)^{n-1} = \frac{1}{n} \sum_{x=1}^n x^n = \frac{1}{n} \cdot \frac{n^{n+1}}{n+1} = \frac{n^n}{n+1}.$$

$$\text{So } \hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n X_i.$$

MLE

$$L(\theta) = \prod_{i=1}^n \frac{1}{n} \left(\frac{X_i}{n} \right)^{n-1} = \frac{1}{n^n} \left(\frac{\prod_{i=1}^n X_i}{n^n} \right)^{n-1} = \frac{1}{n^{n^2}} \left(\prod_{i=1}^n X_i \right)^{n-1}.$$

So $\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n X_i$. The function $L(\theta)$ is an increasing function for $\theta < \frac{1}{2}$ and is a decreasing for $\theta > \frac{1}{2}$. Since it is given that $\theta \in (0, 1)$, $\hat{\theta}_2$ is the MLE of θ because it is the overall maximum of $L(\theta)$. When $\theta = \frac{1}{2}$, $L(\theta)$ is an increasing for $\theta < \frac{1}{2}$ and is a decreasing for $\theta > \frac{1}{2}$. So MEL of θ with is $\frac{1}{2}$. So MEL of θ is $\frac{1}{2}$.

(b)

$$\text{MSE} = \text{Var}[\hat{\theta}] + \text{bias}^2[\hat{\theta}] = \frac{1}{n} \left(\frac{n^n}{n+1} \right)^2 + 0 = \frac{n^{2n}}{n(n+1)^2}.$$

$$\text{MSE} = E \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]$$

$$= \frac{1}{n} \sum_{i=1}^n E \left[(y_i - \hat{y}_i)^2 \right]$$

where y_i is binomial n, p .

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n E \left[(y_i - \hat{y}_i)^2 \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{n} \sum_{j=1}^n C_{ny} \right]$$

If n is even then $\frac{n}{2}$ and if n is odd then $\frac{n-1}{2}$.

(c)

Now

$$\text{MSE} = E \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]$$

$$= \frac{1}{n} \sum_{i=1}^n E \left[(y_i - \hat{y}_i)^2 \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{n} \sum_{j=1}^n C_{ny} \right]$$

If

as the first part cancelled out.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{n} \sum_{j=1}^n C_{ny} \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{n} \sum_{j=1}^n C_{ny} \right]$$

So if

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{n} \sum_{j=1}^n C_{ny} \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{n} \sum_{j=1}^n C_{ny} \right]$$

Which estimator is preferred?

Lecture no.18 (100 to 103)

Best unbiased estimator

In the Mean Square Error (MSE) module, the comparison based on MSE may not provide clear result because biased varies according to the estimator and there may chance that some estimator

has smaller variance but the amount of bias may make it having larger variance. So, indeed there is no one "best MSE" estimator as this type of estimators comprises on too large class, e.g. an estimator cannot be beaten in MSE.

On way to fix this problem is to limit the class of estimators. A promising way is to restrict the class of estimators to only unbiased estimators.

If W and $\bar{W}$ are both unbiased estimators of a parameter θ , i.e. $E W = E \bar{W} = \theta$, then the MSEs are equal to their variances. So we should choose the estimator with the smaller variance. And if we find an estimator which has uniformly smallest variance, then we obtain our objective.

Although we are dealing with unbiased estimators but this result can be expressed as a general.

Suppose there is an estimator W of θ with $E W = \theta$, and we are interested in

investigating the worth of W . Consider the class of estimators

$W + C W$.

For any $W + C W$, $Bias W + C Bias W = Bias W$. So

$$E W = E W + C E W = E W + C W = W + C W,$$

and MSE comparison can be based on variance alone within the class C . Hence although we speak in terms of unbiased estimators, we actually are comparing the estimators that have the same expected value i.e. θ .

Definition: An estimator W is the best unbiased estimator of θ if it satisfies $E W = \theta$ and for any other estimator $\bar{W}$ with $E \bar{W} = \theta$, we have

$var W \leq var \bar{W}$. The W is called a uniform minimum variance unbiased estimator (UMVUE) of θ .

Finding the best unbiased estimator (if exists) is not an easy task, we will discuss some issues faced while finding best unbiased estimator in the next example.

Example

Let $X_1, \dots, X_n$ be iid Poisson and let $\bar{X}$ and S^2 be the sample mean and variance respectively, then find out which one is the best unbiased estimator of θ .

We know that for Poisson distribution the mean and the variance are equal i.e. θ . So

$$E \bar{X} = \theta, \text{ for all } n \text{ and } E S^2 = \theta, \text{ for all } n. \text{ So both } \bar{X} \text{ and } S^2 \text{ are unbiased estimators of } \theta.$$

To determine which one is the best estimator we need to find their variances. As we know that

$Var \bar{X} = \frac{\theta}{n}$ but the $Var S^2$ is quite lengthy and in some situations it is difficult to find out.

Thus even if we can establish that $\bar{X}$ is better than S^2 by considering a class of estimators, say

where a is a constant. So again there are infinite cases and there may be chances that they are unbiased but we are not sure for which one is better. To overcome this type of situation we have Cramer-Rao Inequality which gives guaranty that it provides lower bound of the variance of an unbiased estimator.

This example shows some of the problems that might be faced when we are trying to find the best unbiased estimator and therefore we need a comprehensive approach. Suppose for estimating some parameter say θ for a pdf $f(x)$, we can specify a lower bound, say B , on the variance of any unbiased estimator of θ . Thus if we have an unbiased estimator W satisfying $\text{Var } W = B$, then we found a best unbiased estimator. This approach is given in Rao-Cramer Lower Bound.

Rao-Cramer Inequality

Theorem: Let $X_1, \dots, X_n$ be iid random sample from with common pdf $f(x)$ for $x \in \mathcal{X}$. Let Y be any other estimator with mean, $E Y = E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = k$ and $\text{Var } Y = \frac{1}{n}$, then

$$\text{Var } Y \geq \frac{k^2}{E \left(\frac{1}{n} \sum_{i=1}^n \log L(x_i) \right)^2}.$$

Proof: The mean of Y is

$$E Y = k = \int_{\mathcal{X}} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) f(x_1, \dots, x_n) dx_1, \dots, dx_n.$$

Let $L_x = f(x)$, so

$$k = \int_{\mathcal{X}} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) L_x dx_1, \dots, dx_n,$$

Differentiating both sides w.r to θ .

$$k = \int_{\mathcal{X}} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \frac{dL_x}{d\theta} dx_1, \dots, dx_n,$$

$$= \int_{\mathcal{X}} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \frac{1}{L_x} \frac{dL_x}{d\theta} L_x dx_1, \dots, dx_n,$$

□ □ □ □ □ □

(1)

$$\int_{\mathbb{R}^n} \log L(x) \, d\mu(x) = \int_{\mathbb{R}^n} L(x) \, d\mu(x) \quad (1)$$

We know that

$$\int_{\mathbb{R}^n} f(x) \, d\mu(x) = \int_{\mathbb{R}^n} f(x) \, d\mu(x) \quad (2)$$

Differentiating both sides w.r. to

$$\frac{d}{dt} \int_{\mathbb{R}^n} L(x) \, d\mu(x) = 0.$$

$$\frac{d}{dt} \int_{\mathbb{R}^n} L(x) \, d\mu(x) = 0.$$

$$\frac{d}{dt} \int_{\mathbb{R}^n} \log L(x) \, d\mu(x) = 0.$$

Multiplying both sides by k

$$\int_{\mathbb{R}^n} k \log L(x) \, d\mu(x) = \int_{\mathbb{R}^n} L(x) \, d\mu(x) = 0.$$

$$E_k \left(\log L(x) \right) = 0.$$

Using (2) in (1), we get

$$k \int_{\mathbb{R}^n} \log L(x) \, d\mu(x) = \int_{\mathbb{R}^n} L(x) \, d\mu(x).$$

$$= E_k \left(\log L(x) \right) = \int_{\mathbb{R}^n} L(x) \, d\mu(x).$$

Using Cauchy Shewarz's inequality that for two random variables X and Y

$$E(XY) \leq \sqrt{E(X^2) E(Y^2)}.$$

$$E_k \left(\log L(x) \right) \leq \sqrt{E_k \left(\log^2 L(x) \right) E_k \left(L^2(x) \right)}.$$

$$E_k \quad t \quad kE \quad 22 \quad n \quad \log L_{\theta}; \quad 2$$

$$E t \quad k \quad n \quad 2 \quad k \quad 2 \quad \log L_{\theta}; \quad 2 \quad \text{or}$$

$$\text{var} \quad t \quad n \quad k \quad 3 \quad \left(\frac{1}{E \log L_{\theta}; \quad 2} \right)$$

Corollary(Topic 110)

If t is an unbiased estimator of θ , i.e. $E t = \theta$, implies that $k, 1$ then

$$\text{var} \quad t \quad n \quad \left(\frac{1}{E \log L_{\theta}; \quad 2} \right)$$

To illustrate this we consider the Bernoulli model with probability of success θ which was treated in our previous example where we find the Fisher Information for this model. We know that $\frac{1}{n} \sum_{i=1}^n X_i$ We have already proved that the MLE of θ is $\bar{X}$. The mean and variance of Bernoulli distribution are θ and $\theta(1-\theta)$ respectively. Hence the mean and variance of $\bar{X}$ are θ and $\frac{\theta(1-\theta)}{n}$, which is the same. Thus in this case the variance of the MLE has attained the

Rao–Cramer lower bound. The following definitions are stated:

Definition: (Efficient Estimator). Let Y be an unbiased estimator of a parameter θ in the case of point estimation. The statistic Y is called an efficient estimator of θ if and only if the variance of Y attains the Rao–Cramer lower bound.

Definition: (Efficiency). In cases in which we can differentiate with respect to a parameter under an integral or summation symbol, the ratio of the Rao–Cramer lower bound to the actual variance of any unbiased estimator of a parameter is called the efficiency of that estimator.

Lecture no.19 (104 to 107)

Example Topic 111)

Before discussing the example, we have an alternative result, which we use in the example, is:

If $f(x)$ satisfies

then

$$E \left[\frac{d}{dx} \log f(x) \right] = 0, \quad E \left[\left(\frac{d}{dx} \log f(x) \right)^2 \right] = \frac{1}{f(x)}$$

and it is true for an exponential family.

Suppose we have a random sample from the distribution

$$f(x) = \frac{1}{2k} \exp\left(-\frac{x^2}{2k}\right), \quad x > 0.$$

This is the density of Rayleigh distribution. We now find the lower bound to the variance of an unbiased estimator of k . For this first we note that $k = \frac{1}{2} E[X^2]$. Now the lower bound is:

$$\text{var } t \geq \frac{k^2}{n E \left[\left(\frac{d}{dx} \log f(x) \right)^2 \right]}.$$

Now $\log f(x) = -\frac{1}{2} \log 2k - \frac{x^2}{2k}$

$$\frac{d}{dx} \log f(x) = -\frac{x}{k}$$

$$\left(\frac{d}{dx} \log f(x) \right)^2 = \frac{x^2}{k^2}$$

$$E \left[\left(\frac{d}{dx} \log f(x) \right)^2 \right] = \frac{1}{k^2} E[X^2]$$

$$= \frac{1}{k^2} \cdot 2k = \frac{2}{k}$$

$$\text{So } E \left[\left(\frac{d}{dx} \log f(x) \right)^2 \right] = \frac{2}{k}$$

Hence the lower bound to the variance is:

$$\text{var } t \geq \frac{k^2}{n \cdot \frac{2}{k}} = \frac{k^3}{2n}.$$

Corollary

In the above theorem the quantity $E \left[\left(\frac{1}{n} \sum_{i=1}^n \log f(x_i) \right)^2 \right]$ is called Fisher information or information number.

$$\begin{aligned} \text{Proof: } E \left[\left(\frac{1}{n} \sum_{i=1}^n \log f(x_i) \right)^2 \right] &= \frac{1}{n^2} E \left[\sum_{i=1}^n \log f(x_i) \sum_{j=1}^n \log f(x_j) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n E \left[\log f(x_i) \log f(x_j) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n E \left[\log f(x_i) \right] E \left[\log f(x_j) \right] \quad \text{(by independence)} \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n 0 = 0 \end{aligned}$$

The Rao-Cramer Lower Bound (RCLB) is stated here for continuous random variables but it can also apply to the discrete random variables. The key condition is that the interchangeability of integration and differentiation. If $f(x)$ is a pmf, then we must be able to interchange the differentiation and summation.

The quantity $E \left[\left(\frac{1}{n} \sum_{i=1}^n \log f(x_i) \right)^2 \right]$ is called Fisher information or information number. Indeed, it reflects that information number gives a bound on the variance of the best unbiased estimator of θ . As the information number gets larger we have more information about θ and hence a smaller bound on the variance of the best unbiased estimator.

Example 1

Now considering the Poisson distribution and using the results which we have already done in one of the previous example regarding Poisson distribution, we have k , so k . Since Poisson distribution belongs to the exponential family so by using the above result, we get

$$E\left[\frac{1}{x^2}\right] = \int_0^1 \frac{1}{x^2} \cdot 2x \, dx = 2 \int_0^1 \frac{1}{x} \, dx = 2 \left[\ln x \right]_0^1 = 2 \ln 1 - 2 \ln 0 = -\infty$$

Let $Z = \frac{1}{x^2}$, then

$$E[Z] = \int_0^1 \frac{1}{x^2} \cdot 2x \, dx = 2 \int_0^1 \frac{1}{x} \, dx = 2 \left[\ln x \right]_0^1 = -\infty$$

Now putting $z = \frac{1}{x^2}$ and evaluating this integral we get

$$E\left[\frac{1}{x^2}\right] = 2 \int_0^1 \frac{1}{x} \, dx = 2 \left[\ln x \right]_0^1 = -\infty$$

Similarly,

$$E\left[\frac{1}{x^4}\right] = \int_0^1 \frac{1}{x^4} \cdot 2x \, dx = 2 \int_0^1 \frac{1}{x^3} \, dx = 2 \left[-\frac{1}{2x^2} \right]_0^1 = -\infty$$

So (1), implies that

$$E\left[\log f(x)\right] = \int_0^1 \log f(x) \cdot 2x \, dx = 2 \int_0^1 \log f(x) \cdot x \, dx$$

So any unbiased estimator, say t_n of $\frac{1}{2}$ must satisfy

$$\text{Var}(t_n) = \frac{1}{n} E\left[\log f(x)\right]^2 = \frac{1}{n} \int_0^1 \left(\log f(x)\right)^2 \cdot 2x \, dx$$

An unbiased estimator of $\frac{1}{2}$ (when θ is known) is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, so it attains the Rao-Cramer lower bound.

Cramer lower bound.

But in case of θ is unknown, it does not attain the Rao-Cramer lower bound because

$$\text{Var}(S) = \frac{1}{n-1}$$

Lecture no.20 (108 to 111)

Application Q 6.2.2

Given $f(x) = \frac{1}{x^2}$, $x > 0$, zero elsewhere, with $f(0) = 0$, formally compute the reciprocal of

$E[\log L_n(x)]$. Compare this with the variance of $\frac{1}{n} \sum_{i=1}^n \log Y_i$, where Y_i is an observation of a random sample of size n from this distribution. Comment.

Solution:

The log of the function is

$$\log f(x) = -2 \log x$$

Now $E[\log L_n(x)] = \frac{1}{n} \sum_{i=1}^n E[\log Y_i]$

Hence $nE[\log L_n(x)] = \sum_{i=1}^n E[\log Y_i]$

$$I = \frac{1}{n} \sum_{i=1}^n \log Y_i$$

Now the distribution of the largest observation is

$$f_{Y_n}(y) = \frac{1}{n} \cdot \frac{1}{y^2}$$

$$E[Y_n] = \int_0^\infty y \cdot \frac{1}{n} \cdot \frac{1}{y^2} dy = \frac{1}{n} \int_0^\infty \frac{1}{y} dy$$

$$E[Y_n^2] = \int_0^\infty y^2 \cdot \frac{1}{n} \cdot \frac{1}{y^2} dy = \frac{1}{n} \int_0^\infty 1 dy$$

$$E[Y_n^0] = \int_0^\infty 1 \cdot \frac{1}{n} \cdot \frac{1}{y^2} dy = \frac{1}{n} \int_0^\infty \frac{1}{y^2} dy$$

$$n \cdot \frac{1}{n} \int_0^\infty \frac{1}{y^2} dy = \int_0^\infty \frac{1}{y^2} dy$$

Now $\text{Var} \left(\frac{1}{n} \sum_{i=1}^n \log Y_i \right) = \frac{1}{n} \text{var} Y$

So it is obvious that $\text{Var} \ln \bar{Y} = \frac{1}{n} \text{Var} \sum_{i=1}^n \ln Y_i = \frac{1}{n} \sum_{i=1}^n \text{Var} \ln Y_i = \frac{1}{n} \cdot \frac{2}{n} = \frac{2}{n^2}$, i.e. $\frac{2}{n^2}$.

Application Q 6.2.3

Given the pdf

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R},$$

Show that the Rao-Cramer lower bound is $2/n$, where n is the size of a random sample from the Cauchy distribution.

Solution:

The given pdf is

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}.$$

The log of the function is

$$\log f(x) = \log \frac{1}{\pi(1+x^2)} = -\log \pi - \log(1+x^2).$$

$$\frac{d}{dx} \log f(x) = \frac{d}{dx} [-\log \pi - \log(1+x^2)] = -\frac{2x}{1+x^2}.$$

$$I = E \left[\left(\frac{d}{dx} \log f(x) \right)^2 \right] = E \left[\left(\frac{2x}{1+x^2} \right)^2 \right] = 4 E \left[\frac{x^2}{(1+x^2)^2} \right].$$

$$= 4 \int_{-\infty}^{\infty} \frac{x^2}{(1+x^2)^2} \cdot \frac{1}{\pi(1+x^2)} dx = \frac{4}{\pi} \int_{-\infty}^{\infty} \frac{x^2}{(1+x^2)^3} dx.$$

$$\text{Let } x = \tan z, \quad dx = \sec^2 z dz, \quad z = \tan^{-1} x,$$

$$\text{if } x \in \mathbb{R}, \text{ then } z \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

$$I = \frac{4}{\pi} \int_{-\pi/2}^{\pi/2} \frac{\tan^2 z}{(1+\tan^2 z)^3} \sec^2 z dz = \frac{4}{\pi} \int_{-\pi/2}^{\pi/2} \frac{\tan^2 z}{\sec^4 z} \sec^2 z dz = \frac{4}{\pi} \int_{-\pi/2}^{\pi/2} \tan^2 z \cos^2 z dz.$$

$$\text{Now we evaluate } \int_{-\pi/2}^{\pi/2} \tan^2 z \cos^2 z dz = \int_{-\pi/2}^{\pi/2} \frac{\sin^2 z}{\cos^2 z} \cos^2 z dz = \int_{-\pi/2}^{\pi/2} \sin^2 z dz = \int_{-\pi/2}^{\pi/2} \frac{1 - \cos 2z}{2} dz = \frac{1}{2} \left[z - \frac{\sin 2z}{2} \right]_{-\pi/2}^{\pi/2} = \frac{1}{2} \left[\frac{\pi}{2} - \frac{\sin \pi}{2} - \left(-\frac{\pi}{2} - \frac{\sin(-\pi)}{2} \right) \right] = \frac{1}{2} \left[\frac{\pi}{2} - \frac{\pi}{2} \right] = \frac{\pi}{2}.$$

But also know that $\sin^2 z = \frac{1 - \cos 2z}{2}$ and $\cos^2 z = \frac{1 + \cos 2z}{2}$,

So where $\frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos 2z \, dz = 0$,

$$\frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos 2z \, dz = 0.$$

Hence

$$I = \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos^2 z \, dz = \frac{1}{2} \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2z}{2} \, dz = \frac{1}{4} \int_{-\pi/2}^{\pi/2} (1 + \cos 2z) \, dz.$$

$$= \frac{1}{4} \left[z + \frac{\sin 2z}{2} \right]_{-\pi/2}^{\pi/2} = \frac{1}{4} \left(\frac{\pi}{2} + \frac{\sin \pi}{2} - \left(-\frac{\pi}{2} + \frac{\sin -\pi}{2} \right) \right) = \frac{1}{4} \pi.$$

$$= \frac{\pi}{4}.$$

Hence the Rao-Cramer lower bound is $\frac{1}{nE} \frac{d^2 \log f(x; \theta)}{d\theta^2} = \frac{1}{n}.$

MLE are Asymptotically Efficient

Now let consider θ is the MLE of θ_0 and we assume the true value of θ_0 is denoted by θ_0 . We shall prove that as the sample size increases the distribution of MLE estimator, $\hat{\theta}_n$

approximately normal with mean 0 and variance $\frac{1}{nI}$. Since this is a limiting result, which

holds as the sample size tends to infinity, we say that the MLE is asymptotically unbiased and the variance of the limiting normal distribution. For this we have the following theorem, known as the asymptotic distribution of MLE:

Theorem: Let $X_1, \dots, X_n$ be a sample of size n from a distribution for which the pdf or pmf is $f(x; \theta)$, where θ is the unknown parameter. Assume that the true value of θ is θ_0 , and the MLE of θ is $\hat{\theta}_n$. Then the asymptotic distribution of $\hat{\theta}_n$ is $N(\theta_0, \frac{1}{nI})$, i.e. $\hat{\theta}_n \xrightarrow{d} N(\theta_0, \frac{1}{nI})$.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \frac{1}{I})$$

Proof: We know that the log-likelihood function is

Now applying Taylor expansion of $\log f(x)$ at the point x_0 , giving

implies that

giving that

or

,

(multiplying both sides by n)

Let first consider the numerator of right side and taking its expectation

as

Its variance is

So

Next, we consider the denominator

By the law of large number, the above expression converges to

Thus we have

☐ ☐ Applying ce




particularly a large data sample then lot of information are redundant as we know the sufficient statistic for that sample. Thus we get about (moving around) the big data set and focus on the summary statistic. So sufficient statistic is a function of your sample statistic that convey exactly same information about the data generating process as does the entire sample. For example, we consider an experiment of flipping a coin and we assume our random variable X follows the Bernoulli distribution with parameter θ , representing our data generating process i.e.

$X = \bigcup_{i=1}^n X_i$ and if we

So we notice here one thing state away essentially that our likelihood depends on $\sum_{i=1}^n x_i$. We know the summary statistic $\sum_{i=1}^n x_i$ then essentially we know everything is useful in sample for inference, so $\sum_{i=1}^n x_i$ is a sufficient statistic. If we go one step further that what is the inference we obtain, we need to take log likelihood of this function.

which is our sample mean. Thus in simple words if we

It and then we can draw inference about our parameter

know X without regarding the order of 0's and 1's.

Definition: Let $X_1, \dots, X_n$ be a random sample of size n from a distribution that has the pdf or pmf $f(x)$. A statistic Y whose pdf or pmf is $g(y)$ is called sufficient statistic for θ if and only the conditional distribution of the sample given Y does not depend on θ , i.e.

$$f(x_1, \dots, x_n | Y) = \frac{f(x_1) \dots f(x_n)}{g(Y)}$$

Lecture no.21 (112 to 116)

Example

Let $Y_1, \dots, Y_n$ denote the order statistics of a random sample of size n from the distribution

with pdf

$f(x) = e^{-x}$, $x > 0$. Show that Y_n is sufficient statistic for θ .

Solution:

Since $Y_1, \dots, Y_n$ are the order statistics of a random sample of size n , then the distribution

of

$Y_1, \dots, Y_n$ is given by $f_Y(y) = n! e^{-y_n} \prod_{i=1}^{n-1} e^{-y_i} = n! e^{-y_n} e^{-\sum_{i=1}^{n-1} y_i}$.

So

Now $f_Y(y) = n! e^{-y_n} e^{-\sum_{i=1}^{n-1} y_i}$.

So $f_Y(y) = n! e^{-y_n} e^{-\sum_{i=1}^{n-1} y_i}$.

$$f(x_1, \dots, x_n | y) = \frac{f(x_1) \dots f(x_n)}{f_Y(y)} = \frac{e^{-x_1} \dots e^{-x_n}}{n! e^{-y_n} e^{-\sum_{i=1}^{n-1} y_i}}$$

which is independent of θ . So Y_n is sufficient statistic for θ .

Remarks $\square \square$

1- Any statistic t_x represents a type of data reduction or data summary, i.e. the statistic summarizes the data instead of reporting the entire sample $X_1, \dots, X_n$. Thus if $\square \square \square$, then t_x does not report the actual sample values but only sum.

2- In most of the cases $X_1, \dots, X_n$ represent the observation of a random sample which means they are independently and identically distributed (iid). However in general these random variables are independent but they do not need to be identically distributed. So more specifically the definition of sufficiency of a statistic, y would be extended to

read that $\square \square$ $f(x_1, x_2, \dots, x_n)$ is the joint pdf or pmf of $X_1, \dots, X_n$ and $g(t_x)$ does not depend upon t_x . t_x is the subset of X .

3- Sometimes the terminology “ t_x is sufficient statistic for θ ” is a bit misleading. For example, suppose we are interested not in θ but in e^θ --do we need to consider of find a sufficient statistic for e^θ ? The answer is No. When we say that statistic t_x is sufficient we mean t_x contains all the information in the data related to θ . So if t_x is sufficient statistic for θ then it is also sufficient for e^θ or any other function of θ .

Example 1

In this example we use moment generating function to find the sufficient statistics. Let $X_1, \dots, X_n$ be a random sample from a gamma distribution with $\alpha = 2$ and $\beta = 0$. Because the mgf associated with this distribution is given by $M(t) = (1 - \beta t)^{-\alpha}$ Show that $\sum_{i=1}^n X_i$ is sufficient statistic for θ .

Solution: The mgf of $\sum_{i=1}^n X_i$ is

$$E[e^{tY}] = E[e^{t(X_1 + \dots + X_n)}] = E[e^{tX_1}] \dots E[e^{tX_n}] = (1 - \beta t)^{-\alpha} \dots (1 - \beta t)^{-\alpha} = (1 - \beta t)^{-n\alpha}$$

So Y has a gamma distribution with $2n$ and 0 and its pdf is

$$f(y) = \frac{1}{\Gamma(2n)} \left(\frac{y}{\beta} \right)^{2n-1} e^{-\frac{y}{\beta}}$$

Since the mgf of X is, $M_X(t) = 1 - \frac{1}{2}t^2$ so its pdf is

$$f_X(x) = \begin{cases} 1 - \frac{1}{2}x^2 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X_1, \dots, X_n | Y}(x_1, \dots, x_n | y) = \frac{f_{X_1, \dots, X_n, Y}(x_1, \dots, x_n, y)}{f_Y(y)}$$

$$\text{where } f_{X_1, \dots, X_n, Y}(x_1, \dots, x_n, y) = \frac{1}{2^n} e^{-\frac{1}{2}y} \prod_{i=1}^n \mathbb{I}_{\{0 \leq x_i \leq y\}}$$

for

Example 2

Let $X_1, \dots, X_n$ be independent random samples from the pdf

$$f_X(x) = \begin{cases} 1 - x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is a sufficient statistic for θ . Show that

is not a sufficient statistic but

Solution: The joint distribution of $X_1, \dots, X_n$ is

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n (1 - x_i) \mathbb{I}_{\{0 \leq x_i \leq 1\}}$$

The pdf of the largest observation Y_n is

$$f_{Y_n}(y) = n y^{n-1} (1 - y) \mathbb{I}_{\{0 \leq y \leq 1\}}$$

$$f_{X_1, \dots, X_n | Y_n}(x_1, \dots, x_n | y) = \frac{f_{X_1, \dots, X_n, Y_n}(x_1, \dots, x_n, y)}{f_{Y_n}(y)}$$

$$f(x_1, \dots, x_n | y) = \frac{f(x_1, \dots, x_n) f(y)}{f(y)}$$
 which is independent of y . So Y is sufficient statistic for θ .

Now the pdf of the smallest observation Y is

$$f_Y(y) = \frac{d}{dy} P(Y \leq y) = \frac{d}{dy} [1 - P(Y > y)]$$

$$= \frac{d}{dy} [1 - P(X_1 > y, \dots, X_n > y)]$$

$$= \frac{d}{dy} [1 - P(X_1 > y)^n]$$

$$= \frac{d}{dy} [1 - e^{-n\theta y}] = n\theta e^{-n\theta y}$$
 which is not independent of θ .

So Y is not a sufficient statistic for θ .

Application Q 7.2.2

Prove that the sum of the observations of a random sample of size n from a Poisson distribution having parameter $\theta, \theta > 0$ is a sufficient statistic for θ .

Solution: In this question we have to show that $\sum_{i=1}^n X_i$ is sufficient statistic for θ . Since X_i 's are random sample from the Poisson distribution therefore $\sum_{i=1}^n X_i$ is also Poisson distribution with parameter $n\theta$. So it's pdf is

$$f(s) = \frac{e^{-n\theta} (n\theta)^s}{s!}, s = 0, 1, \dots$$

Now conditional distribution of sample given $S = s$ is

$$f(x_1, \dots, x_n | s) = \frac{f(x_1, \dots, x_n) f(s)}{f(s)}$$

$$= \frac{\prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} \cdot \frac{e^{-n\theta} (n\theta)^s}{s!}}{\frac{e^{-n\theta} (n\theta)^s}{s!}}$$

$$= \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!}$$

$$\frac{e^{-s} s^n}{n!} \frac{x^n}{s^n} = \frac{e^{-s} s^n}{n!} \frac{x^n}{s^n}$$
 which is independent of s , so $\sum_{i=1}^n X_i$ is sufficient statistic for θ .

Lecture no.22 (117 to 121)

Application Q 7.2.5

Show that the sum of the observations of a random sample of size n from a gamma distribution that has the pdf $f(x) = \frac{1}{\Gamma(n)} e^{-x} x^{n-1}$, $x > 0$, zero elsewhere, is a sufficient statistic for θ .

Solution: The X_i 's are random sample from the gamma distribution and let $Y = \sum_{i=1}^n X_i$. We know

that the mgf of X is $M_X(t) = (1-t)^{-1}$. So the mgf of Y is

$$M_Y(t) = E[e^{tY}] = E[e^{t(X_1 + \dots + X_n)}] = E[e^{tX_1} \dots e^{tX_n}] = (1-t)^{-n}$$

So its pdf is

Since X_i 's are random sample of size n from the gamma distribution i.e $f(x) = \frac{1}{\Gamma(n)} e^{-x} x^{n-1}$ so its pdf is

$$f(x) = \frac{1}{\Gamma(n)} e^{-x} x^{n-1}, \quad x > 0$$

Now conditional distribution of sample given $Y = y$ is

$$f(x_1, \dots, x_n | y) = \frac{f(x_1, \dots, x_n)}{f_Y(y)} = \frac{\prod_{i=1}^n \frac{1}{\Gamma(n)} e^{-x_i} x_i^{n-1}}{\frac{1}{\Gamma(n)} e^{-y} y^{n-1}} = \frac{e^{-y} \prod_{i=1}^n x_i^{n-1}}{y^{n-1}}$$

where $e^{-\sum_{i=1}^n x_i}$, $YX_n = \sum_{i=1}^n x_i$
 $\frac{1}{n!} \left(\sum_{i=1}^n y_i \right)^n$
 which is independent of y , so YX_n is sufficient statistic for θ .

Application Q 7.2.7

Show that the product of the sample observations is a sufficient statistic for θ if the random sample is taken from a gamma distribution with parameters θ and 6.

Solution: The pdf of the gamma distribution with θ and 6 is

$$f(x) = \frac{1}{\Gamma(6)\theta^6} e^{-x/\theta} x^{5} \quad x > 0$$

so the pdf of Y is

$$f(y) = \frac{1}{\Gamma(6n)\theta^{6n}} e^{-\sum_{i=1}^n y_i/\theta} \left(\sum_{i=1}^n y_i \right)^{6n-1}$$

as $YX_n = \sum_{i=1}^n y_i$

Now conditional distribution of sample given YX_n is

$$f(x_1, \dots, x_n | y) = \frac{f(x_1, \dots, x_n)}{f(y)}$$

which is independent of θ , so

YX_n is sufficient statistic for θ .

Neyman-Fisher Factorization Theorem

By the definition of the sufficient statistic, that a statistic say y is or not a sufficient statistic for θ , we must know the pdf of y i.e. $f(y; \theta)$. There are some cases when it is difficult to find its pdf. This problem can be avoided by using the following theorem.

Theorem: Let $X_1, \dots, X_n$ denote a random sample from a distribution that has pdf or pmf $f(x; \theta)$. The statistic Y is a sufficient statistic for θ if and only if we can find

two non-negative functions h and g such that

$$f(x_1, \dots, x_n; \theta) = h(t(x_1, \dots, x_n); \theta) g(x_1, \dots, x_n),$$

Where $g(x_1, \dots, x_n)$ does not depend upon θ .

Proof: We prove this theorem in two parts, i.e. necessary and sufficient parts. First to prove the necessary part, suppose that the statistic S is sufficient then we have

$$\begin{aligned} f(x_1, \dots, x_n; \theta) &= f(x_1, \dots, x_n; \theta | S) f(S; \theta) \\ &= f(x_1, \dots, x_n | S; \theta) f(S; \theta) \\ &= f(x_1, \dots, x_n | S) h(S; \theta) \end{aligned}$$

This implies that

Since we partitioned the joint distribution of sample observation in two components and hence the proof is completed. To prove the sufficient part we assume that

$$f(x_1, \dots, x_n; \theta) = g(S; \theta) h(x_1, \dots, x_n)$$

holds. Now

$g(S; \theta) = \int f(x_1, \dots, x_n; \theta) dx_1 \dots dx_n$, where integration is done over all the values of $x_1, \dots, x_n$ that lead to S . Now (1)

is holds, therefore

$$\begin{aligned} f(S; \theta) &= \int g(S; \theta) h(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= g(S; \theta) \int h(x_1, \dots, x_n) dx_1 \dots dx_n \\ &= g(S; \theta) k(S) \end{aligned}$$

where $k(S)$ is some function of S . Now

Since the conditional distribution of sample given S , under the assumption (1) holds, is independent from parameter θ , therefore it satisfies the condition of being a sufficient statistic. Hence if (1) holds then S is sufficient statistic for θ .

Above theorem states that if we partitioned the joint distribution of sample in two components, one containing a statistic S and a parameter θ and other free from the parameter of θ then that statistic S is sufficient for parameter θ . So this theorem can also be used to find the sufficient statistic.

Example 1

—

$\begin{array}{c} \square \\ \text{fx} \end{array}$

$$\frac{2}{22}n$$

Now first we evaluate $\sum_{i=1}^n x_i$

[illegible]

Example 2

Let $\bar{X}_1, \dots, \bar{X}_n$ denote a random sample from a distribution with pdf is

By the factorization theorem, show that

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for } x \in \mathbb{R},$$

Solution:

The joint distribution of $\mathbf{X}_{1:n}, \dots, \mathbf{X}_n$ is

X is a sufficient
 $\square$
n
i

$f(x) = \frac{1}{n!} x^n e^{-x}$, $x \geq 0$. Thus X is a sufficient statistic for θ .
 Since kx does not depend on θ , thus the product $\prod_{i=1}^n kx_i$ is also a sufficient statistic.

Lecture no.23 (122 to 125)

Properties of the sufficient statistic

Here we discuss how sufficiency is used to determine the minimum variance unbiased estimator (MVUE). Before this, one thing must keep in mind that the sufficient statistic is never unique in any sense. If Y is a sufficient statistic and g is a one-to-one function, $g(Y)$ is also a sufficient statistic.

Let Y be an unbiased statistic of θ , then we have $EY = \theta$ and $EY^2 = \theta^2 + \text{var } Y$. Thus EY^2 defines a statistic Y_2 , where Y is an unbiased estimator of θ with a variance not greater than of Y_2 . In simple words when we are looking for MVUE, we can just use the function of a sufficient statistic. This can be further discussed in the upcoming theorem i.e. Rao-Blackwell theorem.

Example

Let $X_1, \dots, X_n$ denote a random sample from a distribution with pdf is $f(x) = e^{-x}$, $x \geq 0$. We are interested to find MVUE of θ .

Solution: The likelihood function is

$L(x_1, \dots, x_n) = e^{-\sum_{i=1}^n x_i}$, for $x_i \geq 0$, $i = 1, \dots, n$. By factorization theorem $\sum_{i=1}^n x_i$ is a sufficient statistic. Now log-likelihood of the above function is

$$\ln L = -\sum_{i=1}^n x_i = -Y.$$

It is observed that Y is a function of the sufficient statistic $\sum_{i=1}^n x_i$. Also Y is the MLE of θ so it

is asymptotically unbiased. The expected value of Y can be obtained as:

are iid having

1

therefore

Since X_i

$\gamma = 1$,

so Y_i

$X_i \sim \gamma = 1$,

,

$EY_i = E$

$E \frac{1}{n} \sum_{i=1}^n X_i = 1$

nt

nt

Let $z = \frac{1}{n} \sum_{i=1}^n X_i$ and simplifying we get

$EY_i = E$

$E \frac{1}{n} \sum_{i=1}^n X_i = 1$

nt

nt

Thus the statistic

$\frac{1}{n} \sum_{i=1}^n X_i$

is an MVUE of

1.

Application Q 7.3.3

If X_1, X_2 is a random sample of size 2 from a distribution having pdf

$f(x) = e^{-x}$,

$x > 0$,

zero elsewhere, find the joint pdf of the sufficient statistic

$Y = (X_1, X_2)$

and

show that

Y_1 is an unbiased estimator of

1 with variance $\frac{1}{2}$.

Find EY_1

and the variance of

Y_1 .

Y_2 .

Solution: First we find the joint distribution of Y_1 and Y_2 .

Let $Y = (Y_1, Y_2)$ implies that

$Y_1 = X_1$

$Y_2 = X_2$

$J_{xx} = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$

So

$f_{Y_1, Y_2}(y_1, y_2) = e^{-y_1} e^{-y_2}$

Since we know that $EY_1 = 1$,

$EY_2 = 1$

and $\text{var } Y_1 = \frac{1}{2}$

$\text{var } Y_2 = \frac{1}{2}$

Now the marginal distribution of Y_1 is

$f_{Y_1}(y_1) = \int_0^\infty e^{-y_1} e^{-y_2} dy_2 = e^{-y_1}$

$f_{Y_2}(y_2) = \int_0^\infty e^{-y_1} e^{-y_2} dy_1 = e^{-y_2}$

$f_{Y_1}(y_1) = e^{-y_1}$

$f_{Y_2}(y_2) = e^{-y_2}$

$$\begin{aligned}
 \text{Now } f(y) &= \frac{1}{\sigma^2 \sqrt{2\pi}} e^{-\frac{y^2}{2\sigma^2}} \\
 \text{So } E(Y) &= \int_{-\infty}^{\infty} y f(y) dy = 0 \\
 \text{var } Y &= E(Y^2) = \int_{-\infty}^{\infty} y^2 f(y) dy = \sigma^2
 \end{aligned}$$

Minimum variance unbiased estimator (MVUE)

In many estimation problems, where we have several estimators competing with one another, we have to choose for a suitable one. A rigorous way of choosing such an estimator is that the chosen estimator should have smaller variance. Thus MVUE is an unbiased estimator that has lower variance than any other unbiased estimator for all possible values of the parameter.

Definition: For a given positive integer n , Y is called an MVUE of the parameter θ if Y is unbiased, i.e. $EY = \theta$ and if the variance of Y is less than or equal to the variance of every other unbiased estimator of θ .

Conventionally it is not an easy task to compare the variances of all the other unbiased estimators. For example, let $X_1, \dots, X_n$ denote a random sample from a distribution that is $N(\theta, \sigma^2)$, where σ^2 is known. Because the statistic $\bar{X} = \frac{X_1 + \dots + X_n}{n}$ is $N(\theta, \frac{\sigma^2}{n})$, so $\bar{X}$ is an unbiased estimator of θ . The statistic X_1 is $N(\theta, \sigma^2)$, so X_1 is also unbiased estimator of θ .

Although the variance of $\bar{X}$ is less than the variance of X_1 , But we cannot say that with $n > 1$ that $\bar{X}$ is the minimum variance unbiased estimator of θ . The definition actually requires that the comparison is made with the every unbiased estimator of θ . Indeed, it is almost impossible to tabulate all the other unbiased estimators of this parameter, so there is a need to develop other methods for making comparisons of the variance. The CRLB is a useful inequality to identify MVUE e.g. if an unbiased estimator has the variance equal to the CRLB, it must have the minimum variance amongst all unbiased estimators. Sufficiency can also be used to find unbiased, minimum variance estimation. If TY is an unbiased estimator of θ and S is a statistic sufficient for θ , then there is a function of S that is also an unbiased estimator of θ and has no larger variance than the variance of TY .

Lecture no.24 (126 to 128)

Sufficiency and unbiasedness

When we develop an estimator, the precision of the estimator is an important factor. It means that the estimator gives the right estimation (in terms of expectation) as many times as it is used and has the small variance.

A statistic is a function of data which extracts the comprehensive information from the data. A statistic Y is unbiased if $EY = g$, and any function g .

The sufficient statistic Y is the conditional expectation of X given Y does not depend on θ for any

In the previous modules the concept of sufficiency and its applications have been discussed. Now we study the concept of sufficiency in unbiased estimates as a useful tool. Amongst the unbiased estimators the best one is that having the smallest variance. This smallest variance can be accessed by using Rao-Cramer lower bound which tells us that if

T is an unbiased estimator of θ , then its variance cannot be less than $\frac{1}{nI}$. But in some situation the lower bound is not attainable. Rao-Blackwell theorem guides us to find the useful technique to overcome this problem. Rao-Blackwell theorem combines unbiased estimator and sufficient statistic in order to get an improved unbiased estimator with smaller variance. This theorem improves any unbiased estimator by taking conditional expectation given a sufficient statistic. It has two important impacts:

1. The unbiased estimator is a function of sufficient statistic.
2. It provides a proper way to improve the unbiased estimator with least variance.

Rao-Blackwell Theorem

To prove this theorem, we need the following results:

If X and Y are two any random variables, provided that their expectations exist, then we have:

- 1- $E(Y|X) = Y$
- 2- $\text{Var } X \geq \text{Var } E(X|Y) = E \text{Var } X|Y$.

Statement: Let W be any unbiased estimator of g , and Let T be the sufficient statistic for θ . Then the

Define $T = E(W|T)$ and the $\text{var } T \leq \text{var } W$, i.e. T is a uniformly better unbiased estimator of g .

Proof: It is given that

$$EW = g$$

So if T is sufficient statistic for θ , then

which allows us to obtain the unbiasedness by using the first result, so

So T is an unbiased estimator of g .

To find the variance, we use the second result as:

$$\text{Var } W = \text{var } E[W|T] + E[\text{var } W|T]$$

$$\text{var } T + E[\text{var } W|T]$$

Since $\text{var } W|T = 0$

$$\text{Var } W = \text{var } T$$

Hence T is uniformly better than W if we show that T is indeed an estimator. That is, we have to show that T is a function of only sample and independent of θ . From the definition of sufficient statistic that W is only the function of sample therefore it is independent from θ . Hence T is a uniformly better unbiased estimator of g .

Hence conditioning any unbiased estimator on a sufficient statistic will result in a uniform improvement, so searching for best unbiased estimators we only need to consider statistic which are the function of sufficient statistic.

The results (1) and (2) which are used to prove this theorem, not saying anything about sufficiency and may give an impression that conditioning anything may lead an improvement. This is, indeed, true but the resulting quantity may contain θ and not be an estimator. This issue can be seen in the next example.

Example (Conditioning on an insufficient statistic)

Let X_1, X_2 be iid $N, \sigma^2 = 1$. The statistic $X = \frac{1}{2}(X_1 + X_2)$ has

$$E[X] = \frac{1}{2}(E[X_1] + E[X_2]) = 0 \quad \text{and} \quad \text{var } X = \frac{1}{4}(\text{var } X_1 + \text{var } X_2) = \frac{1}{2}$$

$$\text{var } X = \frac{1}{4}(\text{var } X_1 + \text{var } X_2) = \frac{1}{2}$$

Now conditioning on X_1 which is not sufficient. Let $Y = \frac{1}{2}(X_1 + X_2)$ So by using

$$E[Y|X_1] = \frac{1}{2}(X_1 + E[X_2|X_1]) = \frac{1}{2}(X_1 + 0) = \frac{1}{2}X_1$$

Now using the fact $\text{Var } X = \text{Var } E(X|X) + E \text{Var } X|X$,

$$\text{var } X = \text{var } E(X|X) + E \text{var } X|X$$

$$\text{var } X = \text{var } X + E \text{var } X|X$$
Hence $\text{var } X = \text{var } X + E \text{var } X|X$.
Since $\text{var } X = \text{var } X + E \text{var } X|X$, X is better than X . However

$$\text{var } X = \text{var } X + E \text{var } X|X$$

$$\text{var } X = \text{var } X + E \text{var } X|X$$
because of independence.

$$\text{var } X = \text{var } X + E \text{var } X|X$$
Hence X is not an estimator as it involves X .

Example (Hogg, McKean & Craig) P-391

Let $X_1, X_2, \dots, X_n$ be a random sample from an exponential distribution with mean θ , so that the joint pdf is

$$f(x_1, x_2, \dots, x_n) = \frac{1}{\theta^n} e^{-\sum_{i=1}^n x_i / \theta}, \quad x_i > 0, \theta > 0$$

Lecture no.25 (129 to 131)

Uniqueness of unbiased estimator

Theorem 7.3.19: If W is a best unbiased estimator of $g(\theta)$, then W is unique.

Proof: Let W' is another best unbiased estimator, and consider the estimator

$$W'' = W - W'$$

$$E W'' = E W - E W' = g(\theta) - g(\theta) = 0$$

$$\text{var } W'' = \text{var } W + \text{var } W' - 2 \text{cov } W, W'$$

Applying Cauchy-Schwarz inequality on covariance, that is

$$\text{cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\text{var}(W) = \frac{1}{n} \sum_{i=1}^n (w_i - \bar{w})^2$$

When $\text{var}(W) = 0$, we get

$$\text{var}(W) = 0$$

If the above inequality holds strictly, then the best unbiasedness of W is contradicted, so we must have equality for all a . As it is because of Cauchy-Schwarz, we can have equality only if $W = a$. Now by using the properties of covariance, we have

$$\text{cov}(W, W) = \text{cov}(W, a) = W - a$$

But $\text{cov}(W, W) = \text{var}(W)$ and $\text{var}(W) = 0$ as the above equality holds. Thus $W = a$ and $W = 0$ showing that W is unique.

$$a = \bar{y} \text{ and } E(W) = \bar{y}$$

Continue
Theorem 7.3.20: If $E(W) = \bar{y}$ is the best unbiased estimator of $\bar{y}$ if and only if W is uncorrelated with all unbiased estimators of 0.

Proof: Using the above argument, if W is the best unbiased then it must satisfy $\text{cov}(W, U) = 0$, for all U and for any U , satisfying $E(U) = 0$ so the necessary condition is satisfied.

Suppose we have an unbiased estimator W which is uncorrelated with all unbiased estimators of 0. Let W be any other estimator such that $E(W) = \bar{y}$ we will show that W is better than W .

so we write $W = W + (W - W)$

$$\text{Taking variance on both sides, we get}$$

$$\text{var}(W) = \text{var}(W) + \text{var}(W - W) + 2\text{cov}(W, W - W)$$

The last quantity vanishes as W is an unbiased estimator of $\bar{y}$ and is uncorrelated with $W - W$. Since $W - W$ is arbitrary, assumption. Since $\text{var}(W - W) = 0$ which implies that $\text{var}(W) = \text{var}(W)$ it follows that W is the best unbiased estimator of $\bar{y}$.

Note: An unbiased estimator of θ is nothing more than random noise, i.e. there is no information in an estimator of θ .

Uniformly minimum variance unbiased estimator (UMVUE) in multi-parameter case

The results of finding the UMVUE can be extended for distributions having several parameters. For this we have an extension of the Rao-Blackwell theorem, which is given as:

“Let $X_1, \dots, X_n$ be a random sample from the probability density function $f(x; \theta_1, \dots, \theta_r)$, and let $T_1, \dots, T_r$ be a set of jointly sufficient statistics for $\theta_1, \dots, \theta_r$.

Let $h_1, \dots, h_r$ be an unbiased estimator of $g_1, \dots, g_r$, where $g_1, \dots, g_r$ are unbiased for $\theta_1, \dots, \theta_r$, and has the

defined by T_j for $j=1, \dots, r$, T_j are unbiased for $g_1, \dots, g_r$, and has the smaller variance than the unbiased estimates $h_1, \dots, h_r$. In addition, the generalized variance of

$T_1, \dots, T_r$ is less than the generalized variance of $h_1, \dots, h_r$.”

Lecture no.26

Topics (132 to 134)

Example

Suppose we have a random sample from the distribution

$f(x) = \frac{1}{2} e^{-x} I_{(0, \infty)}(x)$. Let $Y = \min X, \dots, X$ and $Z = \max X, \dots, X$

are jointly sufficient statistic for θ_1 and θ_2 . Find the UMVUE's of $g_1 = \frac{1}{2} e^{-\theta_1}$ and $g_2 = \frac{1}{2} e^{-\theta_2}$.

$g_1 = \frac{1}{2} e^{-\theta_1}$

Solution:

Any function of sufficient statistic which is unbiased will be UMVUE's. Now we find the EY and EZ . First we find the distribution of minimum and maximum observation.

Let $F_Y(y) = P(Y \leq y) = 1 - P(Y > y) = 1 - P(X_1 > y, \dots, X_n > y)$

So $F_Y(y) = 1 - [P(X > y)]^n = 1 - [e^{-y}]^n = 1 - e^{-ny}$

Now $F_Z(z) = P(Z \leq z) = P(X_1 \leq z, \dots, X_n \leq z) = [P(X \leq z)]^n = [1 - e^{-z}]^n$

So $f_Y(y) = \frac{d}{dy} F_Y(y) = n e^{-ny}$ and $f_Z(z) = \frac{d}{dz} F_Z(z) = n [1 - e^{-z}]^{n-1} e^{-z}$

So $EY = \int_0^\infty y f_Y(y) dy = \int_0^\infty y n e^{-ny} dy = \frac{1}{n}$

So $EZ = \int_0^\infty z f_Z(z) dz = \int_0^\infty z n [1 - e^{-z}]^{n-1} e^{-z} dz = \frac{n}{n-1}$

So

Now
 gy, n ny
 and solve it, we get
 Let y t
 E Y.
 EY y₂ ny
 and solve it, we get
 Let y t
 E Y.
 En YY n
 Similarly
 g.

Since the statistic T_n and T_n are unbiased estimators for g and g so they are the functions of jointly sufficient statistics therefore they are UMVUE's.

Completeness

Within the concept of sufficiency there is another important concept of completeness which is used to find the UMVUEs. We will discuss the completeness of a family of a probability distribution.

Definition: The family of the probability density function f_X is called complete family if for every function $g(X)$, the identity

$E[g(X)] = 0$ for all θ , implies

TasX complete statistic if the family of its pdf is

We can also define a statistic

complete. Further if sufficient statistic is complete then it is called a complete sufficient statistic.

In simple words, first we find the sufficient statistic and its distribution and check whether

Eg $\sum_{i=1}^n X_i$ or not for all θ . If it is satisfied then we call it complete sufficient statistic.

Completeness means the collection of distributions for all possible values of θ provides sufficiently rich set of vectors.

Example 1

$\sum_{i=1}^n X_i$ is sufficient

Let $X_1, \dots, X_n$ be a random sample from Poisson distribution, show that $\sum_{i=1}^n X_i$

statistic and is complete.

Solution:

The pmf of Poisson distribution is

$$f(x; \theta) = \frac{e^{-\theta} \theta^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

The joint distribution is

$$f(x_1; \theta), \dots, f(x_n; \theta) = \frac{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}{x_1! \dots x_n!}$$

So joint distribution can be written into two parts one containing θ and other independent of θ .

So $\sum_{i=1}^n X_i$

Next we consider the family of probability mass functions and use the condition

of completeness i.e. $E g(\sum_{i=1}^n X_i) = 0$ This requires that

$g(\sum_{i=1}^n X_i) = 0$ for all θ .

Eg

$$E g(\sum_{i=1}^n X_i) = \sum_{x=0}^{\infty} g(x) \frac{e^{-n\theta} \theta^x}{x!} = 0$$

Since e^{-n} and $g(0) = 2^n$ is an infinite series converges to zero, then each coefficient must be equal to zero, i.e. $a_n = 0$ and this is what we want to show.

Lecture no.27

(135-137)

Example 2

Consider the family of pdfs $h_z : \mathbb{R} \rightarrow [0, \infty)$. Suppose Z has a pdf in this family given as

$h_z(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$, zero elsewhere. Show that it is complete statistic.

Solution:

To show the given family h_z is complete for any measurable function U such that $E[U(Z)] = 0$ for any z , which implies that $U = 0$ almost surely. Now

as $z \geq 0$, so

$E[U(Z)] = \int_0^\infty U(z) e^{-z} dz = 0$

$\Rightarrow \int_0^\infty U(z) e^{-z} dz = 0$

The above integral is of the form of Laplace transformation of Uz . This transformation is useful to convert integral and differential equations into the algebraic equations. Thus the function Uz transforms into a function of s only it is identically equal zero, i.e. $Uz = 0$, for all points the probability is zero for all h_z . So the family h_z is complete.

Application 7.4.3

Let $X_1, \dots, X_n$ represent a random sample from the discrete distribution having pmf

$f(x) = \begin{cases} 1 - e^{-x} & x = 1, 2, 3, \dots \\ 0 & x = 0 \end{cases}$

$\bar{X}$ is a complete sufficient statistic for θ .

Show that $\bar{X}$ is

the MVUE of θ .

Find the unique function of $\bar{X}$ that is

Solution:

The joint distribution is

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n p(x_i) = \prod_{i=1}^n \frac{1}{n} = \frac{1}{n^n}$$

$\mathbf{g}(\mathbf{y}) = h(\mathbf{x}, \dots, \mathbf{x})$ is sufficient statistic for θ .

So by factorization theorem

To show that it also a complete,

let $\mathbf{g}(\mathbf{y})$ be any statistic for which complete, is the function must of $\mathbf{y}$, for which

$E_{\theta}[\mathbf{g}(\mathbf{y})] = 0$, for $\theta \in \Theta$. Now the pmf of $\mathbf{YX}$ is

$$P_{\mathbf{YX}}(\mathbf{y}, \mathbf{x}) = \prod_{i=1}^n p(y_i) = \prod_{i=1}^n \frac{1}{n} = \frac{1}{n^n}$$

Now $E_{\theta}[\mathbf{g}(\mathbf{y})] = 0$, gives

$$\sum_{\mathbf{y}} \mathbf{g}(\mathbf{y}) \prod_{i=1}^n p(y_i) = 0 \text{ where } \prod_{i=1}^n p(y_i) = \frac{1}{n^n}$$

The above is a polynomial in θ . For this polynomial to be zero, all coefficients must be zero. So

$C(\mathbf{g}(\mathbf{y})) = 0$ for $\mathbf{y} = 0, 1, \dots, n$.

so $C(\mathbf{g}(\mathbf{y})) = 0$ for $\mathbf{y} = 0, 1, \dots, n$. But $\mathbf{g}(\mathbf{y})$ therefore $\mathbf{YX}$ is complete.

To find the MVUE of θ , we need to find the expectation of y_i . Since $E y_i = E \frac{X_i}{n}$, so

$\frac{1}{n} \sum_{i=1}^n X_i$ is unbiased minimum variance estimator of θ .

Application 7.4.4(a)

Consider the family of probability density functions h_z : $\frac{1}{2}e^{-|x-z|}$, where $z \in \mathbb{R}$, zero elsewhere.

Show that the family is complete provided that $z \in \mathbb{R}$.

Solution: To show the given family h_z is complete for any measurable function

U such that $E[U(z)] = 0$ for any z , which implies that $U = 0$ almost surely. Now

U implies that $\int_{-\infty}^{\infty} U(z) h_z(x) dx = 0$ for $z \in \mathbb{R}$.

$\int_{-\infty}^{\infty} U(z) e^{-|x-z|} dz = 0$, $\int_{-\infty}^{\infty} U(z) e^{-|x-z|} dz = 0$,

Differentiating both sides with respect to x , we get

$U(z) = 0$. Hence the family is complete.

. $U = 0$

Lecture no.28

(Topics 138 to 140)

Lehmann-Scheffe Theorem

Lehmann-Scheffe theorem plays an important role in statistics by combining the ideas of completeness, sufficiency, uniqueness and the best unbiased estimation.

The theorem states that any estimator which is unbiased for a given unknown quantity and that depends on the data only through a complete, sufficient statistic are unique best unbiased estimator of that quantity.

Statement: Suppose T is a complete sufficient statistic for θ .

Let $g(\theta)$ be a real valued parametric function. If we define a statistic

$U(T) = E[g(\theta) | T]$, then $U(T)$ is the unique UMVUE of $g(\theta)$.

Proof: According to the Rao-Blackwell theorem, finding UMVUE of $g(\theta)$, we need to focus on unbiased estimator of $g(\theta)$, which is a function of sufficient statistic T alone.

We know that $U(T)$ is a function of T and is unbiased for $g(\theta)$.

Suppose that $V(T)$ is another function of T which is also unbiased for $g(\theta)$.

Define $h(T) = U(T) - V(T)$, then

$E[h(T)] = E[U(T)] - E[V(T)] = g(\theta) - g(\theta) = 0$, $E[h(T)] = 0$

Since T is complete and from (1), we have $h(T) = 0$ with probability 1, which implies that T is unique UMVUE of g .

Note. To find the UMVUE of g , we need not always have to go through the conditioning with respect to a complete sufficient statistic T . In some situations the following but an alternative equivalent result can be used.

Theorem: Suppose that T is a complete sufficient statistic for θ and there exist an unbiased estimate T^* of g , then there exist a unique UMVUE T of g with probability one.

The proof is the same as above theorem.

Example 1 (Uniform Distribution)

Let $X_1, \dots, X_n$ be a random sample from the uniform distribution with pdf $f(x) = \frac{1}{b-a}$, $a \leq x \leq b$, and zero elsewhere. It is already shown in the previous module that the distribution of maximum $Y = \max\{X_1, \dots, X_n\}$ is sufficient statistic for θ . Now here we show that it is also a complete and find MVUE of θ .

Solution:

We know that the distribution of Y using the given pdf is

$$f_Y(y) = \frac{n}{b-a} \left(\frac{y-a}{b-a} \right)^{n-1}, \quad a \leq y \leq b$$

To show it is complete, suppose that for any function $U(y)$ and for any θ , $E[U(Y)] = 0$. Then

$$\int_a^b U(y) \frac{n}{b-a} \left(\frac{y-a}{b-a} \right)^{n-1} dy = 0$$

Since $\frac{n}{b-a} \left(\frac{y-a}{b-a} \right)^{n-1} > 0$, therefore

$$U(y) = 0 \quad \text{for all } y \in [a, b]$$

Differentiating both sides with respect to θ , we get

$$U(y) = 0 \quad \text{for all } y \in [a, b]$$

As $U(y) = 0$, $\frac{d}{d\theta} U(y) = 0$. Thus

Hence MVUE of θ is

$$E[Y] = \frac{a+b}{2}$$

Y is complete. To find MVUE of θ , we proceed

$$f_Y(y) = \frac{n}{b-a} \left(\frac{y-a}{b-a} \right)^{n-1}$$

Application Q7.4.5

Show that the first order statistic Y_1 of a random sample of size n from the distribution having pdf $f(x); e^{-x}, x > 0$, zero elsewhere, is a complete sufficient statistic for θ . Find the unique function of this statistic which is the MVUE of θ .

Solution:

The distribution of the first order is

$$f_{Y_1}(y) = n f(y) [F(y)]^{n-1}$$

$$= n e^{-y} (1 - e^{-y})^{n-1}$$

$$= n e^{-y} \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k e^{-ky}$$

$$= n \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k e^{-(k+1)y}$$

$$= n \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k e^{-y}$$

$$= n e^{-y} \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k$$

$$= n e^{-y} (1 - 1)^{n-1}$$

Now the joint distribution of $X_1, \dots, X_n$ is

$$f(x_1, \dots, x_n) = e^{-x_1} e^{-x_2} \dots e^{-x_n}$$

then we say that Y is sufficient

Now by using the factorization theorem if $P_X(x) = h(x)$,

$$P_Y(y) = g(y)$$

statistic for θ . Now which is independent of

$$P_X(x) = e^{-x}$$

$$P_Y(y) = n e^{-y} (1 - e^{-y})^{n-1}$$

so it means that Y is sufficient

statistic for θ . To show Y is complete for any measurable function g such that $E g(Y) = 0$ for any θ , implies that $g = 0$ almost surely. Now

$$E g(Y) = \int_0^\infty g(y) f_{Y_1}(y) dy = 0$$

$$\int_0^\infty g(y) n e^{-y} (1 - e^{-y})^{n-1} dy = 0$$

Differentiating both sides with respect to θ , we get

$$g(y) = 0$$

Since $g(y) = 0$, Y is complete. As Y is complete and sufficient statistic for θ , so the MVUE of θ can be obtained as

$$E g(Y) = \int_0^\infty g(y) f_{Y_1}(y) dy = \int_0^\infty g(y) n e^{-y} (1 - e^{-y})^{n-1} dy = 0$$

Now integrating by parts by taking y as a first function, we get

Therefore if $E \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ then $\bar{y}$ is an unbiased estimator of μ , and hence UMVUE.

Lecture no.29

(141-145)

Application 7.4.6 (a)

Let a random sample of size n be taken from a distribution of the discrete type with pmf $f(x)$, $x = 1, 2, \dots$, where k is an unknown positive integer.

Show that the largest observation, say Y , of the sample is complete sufficient statistic for k .

Solution: (a)

First we find the distribution of the largest observation, Y . In case of discrete random variable, the distribution of the largest observation is

$$P(Y=y) = P(Y_1 \leq y, Y_2 \leq y, \dots, Y_n \leq y) - P(Y_1 \leq y-1, Y_2 \leq y-1, \dots, Y_n \leq y-1)$$

$$= [P(X_1 \leq y)]^n - [P(X_1 \leq y-1)]^n$$

$$= \left(\frac{y}{k}\right)^n - \left(\frac{y-1}{k}\right)^n$$

The joint distribution of the sample pmf is $P(X_1=x_1, X_2=x_2, \dots, X_n=x_n) = \prod_{i=1}^n f(x_i)$ then we say that Y is sufficient

Now by using the factorization theorem if $P(X_1=x_1, X_2=x_2, \dots, X_n=x_n) = h(x)g(y)$ then Y is sufficient

statistic for k . Now

$P(X_1=x_1, X_2=x_2, \dots, X_n=x_n)$ which is independent of

$$P(X_1=x_1, X_2=x_2, \dots, X_n=x_n) = \prod_{i=1}^n \frac{1}{k^n} = \frac{1}{k^n}$$

so it means that Y is sufficient

sufficient statistic for k . To show Y is complete for any measurable function g such that

$Eg(Y) = 0$ for any g , implies that $g = 0$ almost surely. Now

implies that
 $E\left[\sum_{i=1}^n Y_i\right] = \sum_{i=1}^n E[Y_i] = 10n$,
 so Y is complete statistic.

Application 7.4.6 (b)

Prove that $\frac{\sum_{i=1}^n Y_i}{n}$ is the unique MVUE of θ .

Solution:

To find the MVUE, we need to find the expected value of $\frac{\sum_{i=1}^n Y_i}{n}$. So

$$E\left[\frac{\sum_{i=1}^n Y_i}{n}\right] = \frac{1}{n} E\left[\sum_{i=1}^n Y_i\right] = \frac{1}{n} \sum_{i=1}^n E[Y_i] = \frac{1}{n} \cdot 10n = 10$$

Since it is unbiased in term of sufficient statistic Y so it is MVUE.

Exponential Families

There is class of distribution which has complete and sufficient statistics, is called exponential class or family. Consider a family f_X of pdf or pmf, where

where η and θ are unknown constants or it may be η and θ ,

$$f_X(x) = \exp\left(\eta^T T(x) - \theta^T \eta\right) \cdot h(x) \cdot q(\eta)$$

elsewhere,

where S is the support of X . The (1) has the following characteristics:

- 1- S does not depend on η .
- 2- P is a nontrivial continuous function of η .

3- If X is continuous, then $K(x) = 0$ and $H(x)$ is a continuous function of x . And if X is a discrete, the $K(x)$ is a nontrivial function of $x \in S$.

In this course we focus only a particular class, called regular exponential class. For example, each member of the family $f(x; \theta)$; $\theta \in \Theta$, where $f(x; \theta)$ is $N(0, 1)$, represents a regular exponential class of continuous type because

$$f(x; \theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$= \exp\left\{-\frac{x^2}{2}\right\} \log \frac{1}{\sqrt{2\pi}}, \quad x \in \mathbb{R}.$$

Whereas the uniform density function is

$$f(x) = \exp\left\{-\log \frac{1}{b-a}\right\} \quad \text{elsewhere } 0,$$

This can be written in the form of (1) but the support is the interval $[0, b]$, which depends on θ . Hence it is not a regular exponential family.

Example

Let X be a Poisson distribution, verify that it belongs to the exponential family.

Solution:

The pdf of Poisson distribution is

$$f(x; \theta) = \frac{e^{-\theta} \theta^x}{x!}, \quad x = 0, 1, 2, \dots$$

The above pdf can be written as

$$\exp\left\{x \log \theta - \theta\right\} \log \frac{1}{x!}.$$

Hence the Poisson distribution is a member of the regular exponential class, with $K(x) = x$ and $q(\theta) = \theta$.

Further the support of X is $S = \{0, 1, 2, \dots\}$, which does not depend on θ .

Theorem

The subsequent theorem is useful to the sufficient statistic for the regular exponential class, i.e. if $T(X)$ is a sufficient statistic for θ .

Let

$$X_1, X_2, \dots, X_n$$

Theorem 7.5.1: Let $X_1, \dots, X_n$ denote a random sample from the regular exponential class with pdf or pmf is

$f_X(x) = \begin{cases} \exp\left(-\frac{1}{K}x\right) & 0 \leq x < \infty \\ 0 & \text{elsewhere,} \end{cases}$

Consider the statistic $Y = \sum_{i=1}^n X_i$, then

1. The pdf or pmf has the form $f(y) = \frac{1}{\Gamma(K)} \left(\frac{1}{K}\right)^K y^{K-1} e^{-y/K}$ for $y \geq 0$ and 0 elsewhere. Neither K and Y depend on θ .
2. $EY = nK$.
3. $\text{var } Y = nK^2$.

The above theorem not only helps us in finding that Y is sufficient statistic but also very useful to find its general form along with mean and variance.

Further we present another theorem which helps us in establishing the completeness of Y .

Theorem 7.5.2: Let $f_X(x)$ be a pdf or pmf of a random variable X whose distribution is a regular exponential class. Then if $X_1, \dots, X_n$ (n is fixed positive integer) is a random sample from the distribution of X , the statistic $Y = \sum_{i=1}^n X_i$ is a sufficient statistic for θ and the family of pdf's of Y is complete, i.e. Y is a complete sufficient statistic for θ .

We do not discuss the proof of these theorem, we just need to learn the use of these theorems.

According to these theorems, if we have regular class (1), we can easily find that Y is a sufficient statistic. And if we are able to find the function of Y , say $h(Y)$, so that $EY = h(K)$, then statistic Y is unique and is MVUE of K .

Lecture no.30

(146-149)

Example

Let $X_1, \dots, X_n$ denote a random sample from a normal distribution that has the pdf

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad x \in \mathbb{R}.$$

Investigate for sufficient and complete statistic for θ .

Solution: The above pdf can be written as

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \exp\left(\frac{x^2}{2}\right) \exp\left(-\frac{x^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \exp\left(\frac{x^2}{2}\right) \exp\left(-\frac{x^2}{2}\right).$$

Here θ is any fixed positive number. This is a regular exponential class with K and P , $Y = X^2$ which is a complete sufficient statistic for θ . According to

every fixed value of θ , $EY = \theta$. Since $EY = \theta$, Y is the only function of Y which is an unbiased estimator of θ and being a function of sufficient statistic Y , it has a minimum variance. Thus X^2 is the unique MVUE of θ and a complete statistic for θ as well.

Jointly Sufficient

Some time we have a family of distribution having a set of parameters, not all known, and we need to find a set of statistics which is jointly sufficient statistics for all the unknown parameters. The jointly sufficient statistic can be defined as:

Let x be a sample from the pdf $f(x; \theta_1, \theta_2, \dots, \theta_k)$, where $\theta = (\theta_1, \theta_2, \dots, \theta_k)$ is a vector of unknown parameters. A set of statistics $T_1, T_2, \dots, T_m$ is said to be jointly sufficient statistic for the parameters $\theta_1, \theta_2, \dots, \theta_k$ if and only if the conditional distribution of the sample, x given the set $T_1, \dots, T_m$ is free from the parameters $\theta_1, \theta_2, \dots, \theta_k$.

Note that it could be possible that the numbers of set of statistics in T are different from the number of parameters or even they equal, it may not be true that T is sufficient for θ_1 , T is sufficient for θ_2 , etc. In general, one needs to show that whole T summarize all the information in x about θ .

The definition of jointly sufficient statistics has a natural extension of Neyman-Fisher factorization theorem which is given as:

Let x be a sample from the pdf $f(x; \theta_1, \theta_2, \dots, \theta_k)$, where $\theta = (\theta_1, \theta_2, \dots, \theta_k)$ is a vector of unknown parameters. A set of statistics $T_1, T_2, \dots, T_m$ is said to be jointly sufficient statistic for the parameters $\theta_1, \theta_2, \dots, \theta_k$ if and only if there exists functions g and h such that

$f(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, x > 0, \theta > 0$
 where $x_1, \dots, x_n$ is free from the parameters $\theta_1, \dots, \theta_n$.

Example

Suppose $X_1, \dots, X_n$ be a random sample from the uniform distribution having pdf

$$f(x; \theta) = \frac{1}{\theta}, 0 \leq x \leq \theta, \theta > 0.$$

Show that $Y_1 = \min\{X_1, \dots, X_n\}$ and $Y_2 = \max\{X_1, \dots, X_n\}$ are jointly sufficient for θ and θ^2 .

Solution:

We know that the joint distribution of Y_1 and Y_2 is

$$f(y_1, y_2) = n(n-1) f(y_1) f(y_2) [F(y_2) - F(y_1)]^{n-2}, \quad y_1 < y_2.$$

Now $F(y) = \frac{y}{\theta}$

$$f(y_1, y_2) = n(n-1) \frac{1}{\theta} \frac{1}{\theta} \left(\frac{y_2}{\theta} - \frac{y_1}{\theta}\right)^{n-2} = \frac{n(n-1)}{\theta^n} (y_2 - y_1)^{n-2}, \quad 0 \leq y_1 < y_2 \leq \theta.$$

Now the conditional distribution of sample given Y_1 and Y_2 is

$$f(x_1, \dots, x_n | y_1, y_2) = \frac{f(x_1, \dots, x_n)}{f(y_1, y_2)} = \frac{\frac{1}{\theta^n}}{\frac{n(n-1)}{\theta^n} (y_2 - y_1)^{n-2}} = \frac{1}{(n-2)! (y_2 - y_1)^{n-2}}, \quad 0 \leq x_i < y_2, y_1 < x_i \leq y_2.$$

Since the conditional distribution is free from θ and θ^2 , therefore Y_1 and Y_2 are jointly sufficient for θ and θ^2 .

Application Q 7.7.6(a)

Let $X_1, \dots, X_n$ be a random sample from the uniform distribution with pdf

θ_1 and θ_2 the pdf is equal to θ_1^2 where $\theta_1 < x$ and $\theta_2 > 0$,
 $f(x) = \frac{1}{2} \theta_1^2$, $0 < x < \theta_1$,
 zero elsewhere.

Show that $Y = \max(X_1, \dots, X_n)$ and $X = \min(X_1, \dots, X_n)$ are jointly sufficient statistics for θ_1 and θ_2 , are complete.

Solution:

The joint distribution of Y and X is

$$f_{Y,X}(y,x) = f_Y(y) f_X(x) = \frac{1}{2} \theta_1^2 \theta_2^n, \quad y_2 < y < \theta_1, \quad 0 < x < y.$$

So $f_Y(y) = \frac{1}{2} \theta_1^2 \theta_2^n$ and $f_X(x) = \frac{1}{2} \theta_1^2 \theta_2^n$.

$$\text{Now } F_Y(y) = \int_0^y f_Y(t) dt = \frac{1}{2} \theta_1^2 \theta_2^n y.$$

Now the conditional distribution of sample given Y and X is

$$f_{X|Y,X}(x|y,y) = \frac{f_{Y,X}(y,x)}{f_Y(y)} = \frac{\frac{1}{2} \theta_1^2 \theta_2^n}{\frac{1}{2} \theta_1^2 \theta_2^n} = 1, \quad 0 < x < y.$$

Since the conditional distribution is free from θ_1 and θ_2 , therefore Y and X are jointly sufficient for θ_1 and θ_2 . Next we look for the completeness, for completeness we need to find

$$E[g(Y,X)] = 0 \quad \text{So} \quad \int_0^{\theta_1} \int_0^y g(y,x) \frac{1}{2} \theta_1^2 \theta_2^n dy dx = 0$$

Multiplying both sides by $\frac{1}{\theta_1^2 \theta_2^n}$, we get

$$\int_0^{\theta_1} \int_0^y g(y,x) dy dx = 0$$

Now differentiating first with respect to θ_1 and then θ_2 , we get

g, , for all $i = 1, 2, \dots, n$ and hence $g(y)$, so by the definition of the complete sufficient statistic, Y and T are joint complete sufficient statistic.

Lecture no.31 (150-156)

Minimal Sufficiency

Definition: A statistic T is said to be minimal sufficient statistic for θ if and only if it is sufficient for θ and it is a function of every other statistic.

To find the minimal sufficient statistic, first we have to check that the given statistic is sufficient or not.

Suppose $X_1, X_2, \dots, X_n$ is a random sample from the sample space $\mathcal{X}$. Consider a statistic T is a mapping from $\mathcal{X}$ to some space $\mathcal{T}$, then for each value t of T define

$\mathcal{X}_t$: $\mathcal{X}$ which are disjoint subsets and $\bigcup_{t \in \mathcal{T}} \mathcal{X}_t = \mathcal{X}$.

collection of subsets $\mathcal{X}_t$ where $t \in \mathcal{T}$ form a partitioned of space $\mathcal{X}$ induced by statistic T . The following theorem plays a vital role in finding the minimal sufficient statistic:

Theorem: Let f be the common pdf or pmf of each X_i , where θ is an unknown parameter. For x, y , consider the ratio

$$\frac{f(x)}{f(y)} = \frac{\prod_{i=1}^n f(x_i)}{\prod_{i=1}^n f(y_i)}$$

Suppose, we have a statistic T such that $\frac{f(x)}{f(y)}$ is independent of θ if and only if $T(x) = T(y)$. Then T is minimal sufficient for θ .

Example

Suppose $X_1, \dots, X_n$ iid from $N(\mu, \sigma^2)$. Find the minimal sufficient statistic for μ and σ^2 .

Solution:

According to the Neyman-Fisher factorization theorem that the statistic T is minimal

sufficient statistic if and only if $\frac{f(x)}{f(y)}$ is independent of θ if and only if $T(x) = T(y)$.

The pdf of normal distribution is

[illegible]

The above equation is constant with respect to θ if and only if $\sum_{i=1}^n x_i = y$,
 respect to θ^2 if and only if $\sum_{i=1}^n x_i = y$ and $\sum_{i=1}^n y x_i$
 and the minimal sufficient statistic for θ^2 is $\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2$
 for θ is $\sum_{i=1}^n x_i$

The concept of ancillary statistic was given by R.A Fisher in 1925. The term ancillary is taken from a Latin word Ancillary meaning handmaiden and it is used in the context of Maximum Likelihood Estimation. It plays an important role in the theory of statistics. Ancillary statistic is somewhat opposite of sufficient statistic. Sufficient statistic contains full information about the unknown parameter θ , contained in the data X whereas ancillary statistic contains no information about θ but it is useful in the statistical methodology.

Definition: A statistic U is called **ancillary** if its sampling distribution contains no information about unknown parameter θ .

Since the sampling distribution is a place where the information about θ is stored therefore the sampling distribution free from θ means no information about θ .

There are two primary applications of ancillary statistic, one is Basu's theorem and the other in conditional inference (beyond this course).

Example

Suppose $X_1, \dots, X_n$ be the random sample from the distribution whose pdf is

$$f(x) = \frac{1}{\sigma} e^{-\frac{x}{\sigma}}, \quad x > 0, \quad \sigma > 0.$$

Show that the sample variance for the above distribution is an ancillary statistic.

Solution:

We know that the sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n X_i^2 - \frac{(\sum_{i=1}^n X_i)^2}{n}$$

The above distribution is a location distribution and the location parameter is θ . We use the location transformation i.e.

$$Z_i = \frac{X_i - \theta}{\sigma}, \quad \text{or} \quad X_i = \theta + \sigma Z_i$$

Now the distribution of Z_i is

$$f(z) = e^{-z}, \quad z > 0$$

The distribution of Z_i does not depend on θ . Now

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (\theta + \sigma Z_i)^2 - \frac{(\sum_{i=1}^n (\theta + \sigma Z_i))^2}{n} = \frac{\sigma^2}{n-1} \sum_{i=1}^n Z_i^2 - \frac{\sigma^2}{n} \left(\sum_{i=1}^n Z_i \right)^2$$

, which does not contain θ so s^2 is an ancillary on θ .

Location Problems

Let $X_1, \dots, X_n$ be the random sample from the model

$$X_i \sim W(\theta, 1), \quad i = 1, \dots, n,$$

where θ is representing the location parameter and $W_1, \dots, W_n$ are the iid random variables with pdf $f(w) = e^{-w}$ when the common pdf of X_i is $f(x) = \frac{1}{\sigma} e^{-\frac{x-\theta}{\sigma}}$. The statistic $U = \sum_{i=1}^n X_i$ is ancillary if $U - n\theta$ is ancillary.

For example, we know that the sample variance s^2 of a random sample from $N(\mu, \sigma^2)$ has a distribution which is

free from θ , the ratio $Y = \frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2}$ where $X_1, X_2, \dots, X_n$ is a random sample from gamma distribution with known parameter θ and unknown parameter θ^2 , then Y has a beta distribution which does not depend on θ . are the examples of an ancillary statistic.

In the context of model (1), let $U = W_1, \dots, W_n$ is a function of $X_1, \dots, X_n$ alone and does not depend on θ . We also call U a location-invariant statistic.

Some more examples of location-invariant statistic are the sample range $= \max X - \min X$, the mean deviation of the sample median $= \frac{1}{n} \sum_{i=1}^n |X_i - X_m|$ and so on.

Scale Problems

Consider the random sample $X_1, \dots, X_n$ from the following scale model

$X_i = W_i, i = 1, 2, \dots, n$, where $W_1, \dots, W_n$ are the iid random variables with pdf $f(w)$ which does not depend on θ . Then the common pdf of X_i is $f(x/\theta)$ where θ is a scale parameter. The statistic $U = X_1, \dots, X_n$ is ancillary if

In the context of model (1), let $R = X_1, \dots, X_n$. Since neither the joint distribution of U nor the U contains θ . We say that U is a scale-invariant statistic. The following are some examples of scale-invariant:

$$\frac{\sum_{i=1}^n X_i^2}{\sum_{i=1}^n X_i}, \frac{\min X_i}{\sum_{i=1}^n X_i}$$

Location-Scale problems

Consider the random sample $X_1, \dots, X_n$ from the following location and scale model

$X_i = \mu + W_i, i = 1, 2, \dots, n$, Where $W_1, \dots, W_n$ are the iid random variables with pdf $f(w)$ which does not depend on μ and θ . Then the common pdf of X_i is $f(\frac{x-\mu}{\theta})$, where μ is a location and θ is a scale parameter. The statistic $U = X_1, \dots, X_n$ is ancillary if

In the context of model (1), let $R = X_1, \dots, X_n$. Since neither the joint distribution of U nor the U contains μ and θ . We say that U is a location-and-scale-

invariant statistic. The following are some examples of location-and-scale-invariant:
 $\frac{\max_i X_i - \min_i X_i}{S}$, $\frac{\sum_{i=1}^n X_i^2}{n \sum_{i=1}^n X_i}$, $\frac{\sum_{i=1}^n X_i^3}{(\sum_{i=1}^n X_i)^3}$ so on.

Note: In context of Group Theory in algebra, these three types of problems belong to the group transformation problems where the model presents a certain kind of invariance under a group of transformations on the sample space.

Lecture no.32 (157-159)

Bayesian Point Estimate

To define the Bayes point estimator, first we need to understand the loss function. It is the difference between the estimated and the true value, i.e. $L(\hat{\theta}, \theta)$. It measures the accuracy of an estimator, i.e. the larger value of loss function reflects a worse estimator. There are different types of loss functions but we use here squared error loss function, i.e.

function which increases quadratically with difference.

The Bayes estimator or Bayes action is an estimator or decision rule that minimizes the posterior expected value of a loss function. Let $\hat{\theta}$ be a decision function (based on data X) and let $L(\hat{\theta}, \theta)$ be a loss function. The Bayes risk of $\hat{\theta}$

is the expected value of the loss function, i.e. $R(\hat{\theta}) = E[L(\hat{\theta}, \theta)]$. An estimator of θ is said to be the Bayes estimator if it minimizes

the Bayes risk among all the other estimator. In other words, the estimator which minimizes the posterior expected loss for each X is called the Bayes estimator or Bayes decision. It is given as

$$\hat{\theta} = \frac{\int \theta L(\hat{\theta}, \theta) \pi(\theta|X) d\theta}{\int L(\hat{\theta}, \theta) \pi(\theta|X) d\theta}$$

If the loss function, $R(\hat{\theta}, X) = \frac{1}{2} (\hat{\theta} - X)^2$, then the Bayes point estimate is the mean of the posterior distribution. Its proof is given as:

Considering the squared error loss function

$$L(\hat{\theta}, X) = \frac{1}{2} (\hat{\theta} - X)^2$$

The first order condition is

$$\frac{dR(\hat{\theta})}{d\hat{\theta}} = 0$$

$$X \sim \text{Normal}(\mu, \sigma^2)$$

$$p(X) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{X^2}{2\sigma^2}\right)$$

Differentiate both sides wrt X , we get

$$\frac{d}{dX} \left(\frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{X^2}{2\sigma^2}\right) \right) = 0$$

= posterior mean.

Example

Consider the model

$X_i \sim \text{binomial}(1, \theta)$
 $\theta \sim \text{beta}(\alpha, \beta)$ and α, β are known:

That is, the prior pdf is

$$p(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

We are interested to find a decision function which is a Bayes estimate or solution.

Solution

θ is sufficient statistic which has a binomial,

We know that $Y = \sum_{i=1}^n X_i$ is $\text{bin}(n, \theta)$ distribution. So its

pdf given

$$p(y) = \binom{n}{y} \theta^y (1-\theta)^{n-y}, y = 0, 1, \dots, n.$$

Thus the conditional pdf of θ given $Y = y$ is

$$p(\theta|y) = \frac{p(\theta)p(y)}{p(y)} = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

That is,

$$p(\theta|y) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

Hence the posterior density is a beta density with $\alpha + y$ parameters

we take loss function as squared-error loss function i.e.

Let $y_1, \dots, y_n$ be a random sample from a binomial distribution with parameters n and θ . The likelihood function is given by

$$L(\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

and the Bayes estimate of θ is the mean of this beta distribution, which is

$$\frac{y + \alpha}{n + \alpha + \beta}$$

The Bayes estimator can also be written as

which is a weighted average of the MLE $\hat{\theta} = \frac{y}{n}$ and the mean of the prior pdf $\frac{\alpha}{\alpha + \beta}$. The weights are

$\frac{\alpha}{\alpha + \beta}$ and $\frac{y}{n}$. For large n , the Bayes estimate is close to the maximum likelihood estimate of θ and the sum $\alpha + \beta$ is a consistent estimator of n . Thus in selecting α and β , not only the prior mean $\frac{\alpha}{\alpha + \beta}$ but also the prior variance $\frac{\alpha\beta}{(\alpha + \beta)^2}$ play an important role relative to the sample size n .

Application Q 11.2.4

Let $X_1, \dots, X_n$ denote a random sample from a Poisson distribution with mean θ . Let the prior distribution is $\theta \sim \text{Gamma}(\alpha, \beta)$. Let the likelihood function is

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!}$$

and α, β are known numbers. Find the Bayes solution for a point estimate for θ .

Solution

The likelihood function of the Poisson distribution is

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} = \frac{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

Thus the conditional pdf of θ given $X_1, \dots, X_n$ is

$$k(\theta | x) = \frac{\frac{e^{-\theta} \theta^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}}{\int_0^\infty \frac{e^{-t} t^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t} dt}$$

Where c is a constant and is equal to

$$c = \frac{1}{\Gamma(n)} \int_0^\infty x^{n-1} e^{-x} dx = 1$$

is now a proper density and has a gamma distribution.

Hence, the posterior density is a gamma density with parameters

n and $\sum_{i=1}^n x_i$. The Bayes estimate of θ is the mean of this gamma distribution, which is

$$\frac{\sum_{i=1}^n x_i}{n+1} \text{ which is the weighted average of } \bar{y} \text{ and } \frac{1}{n+1}$$

the prior mean of θ .

Lecture no.33 (Topics 160 to 162)

Unbiasedness of Bayes Estimator

In Bayesian, the mean of the posterior distribution is generally taken as a point estimator. If X has a pdf $f(x)$, $E(X)$ and there is a prior $\pi(\theta)$ then the posterior mean, the Bayesian point estimator of θ , is given by

$E(\theta|X)$. Now the question is whether $E(\theta|X)$ can be an unbiased estimator of θ and satisfy

$E(E(\theta|X)) = E(\theta)$. The answer is no, i.e. the posterior means are never unbiased estimators, but in case of squared error loss function, then it can be unbiased if Bayes risk is zero. Now we will try to prove this.

Let $\pi(\theta)$ be a proper prior, the Bayes estimator $E(\theta|X)$, implies that Bayes risk is zero. The Bayes risk of $E(\theta|X)$ can be calculated as repeated expectation in two ways as

$$R(E(\theta|X)) = E[E(E(\theta|X) - \theta)^2] = E[E(E(\theta|X) - \theta)^2 | X]$$

Thus conveniently choosing either $E[E(X|Y)]$ or $E[X]$ and using the properties of the conditional expectation, we have

$$E[X] = E[E(X|Y)] = E\left[\sum_{j=1}^n p_j E(X|Y=j)\right] = \sum_{j=1}^n p_j E(X|Y=j)$$

Thus no Bayes estimator can be unbiased unless its Bayes risk is zero.

Testing of Hypothesis

In the previous modules, we studied a method of inference known as point estimation. Now we discuss another method of inference called as hypothesis testing. We start with the definition of the hypothesis as:

Definition: A hypothesis is a statement about the population parameter.

The basic objective of the hypothesis testing is to decide, on the basis of sample information from the population, whether to accept or reject a statement about the population parameter. Generally we accept the hypothesis as being true if the sample data supported it otherwise we reject it.

Often we see the word “significant” in the theory of hypothesis testing. The result is said to be significant if the decision leads to reject the null hypothesis.

Basics of Testing

Some important concepts of hypothesis testing are discussed here:

Null and alternative hypothesis: A null hypothesis is a statement about the population parameter which is tested for the possible rejection under the assumption that it is true. It is generally denoted by H_0 . On the complementary, an alternative hypothesis is accepted when the null hypothesis is rejected. It is denoted by H_1 . If θ represents the population parameter, then the general format of null and alternative hypotheses is $H_0: \theta \in \omega_0$ and $H_1: \theta \in \omega_1$, where ω_0 is any subset of the parameter space. For example, if μ denotes the average change in patient's blood pressure after taking a medicine then experimenter might be interested in testing hypothesis $H_0: \mu = 0$ vs $H_1: \mu \neq 0$. It means that, on the average the medicine has no effect on the blood pressure and alternative hypothesis states that there is some effect. The alternative hypothesis could also take the form $H_1: \mu > 0$ or $H_1: \mu < 0$, depending upon the situation. For example, a consumer might be interested in the proportion of the defective items produced by a supplier. If p represents the proportion of the defective items, the consumer might be interested to test the hypothesis $H_0: p \leq p_0$ vs $H_1: p > p_0$.

Acceptance and rejection regions: After observing sample, the experimenter decides either to accept H_0 as the true or to reject it as a false and decide H_1 is true, a decision rule is stated as:

- 1- For which sample values the decision is made to accept H_1 is true
- 2- For which sample values H_1 is rejected and H_0 is accepted as true.

The subset of the sample space for which H_0 will be rejected is called rejection region or critical region and the complement of the rejection region is called the acceptance region.

Types of errors: When we perform the hypothesis testing there is a possibility that the sample evidence may lead us to make wrong decision. We may reject a null hypothesis when actually it is true or we may accept a null hypothesis when it is, in fact, wrong. The first type of error is called type-I error and the other is called type-II error. This can be seen in the following table as well.

	<i>Decision</i>	
	Accept H_0	Reject H_0
H_0 (true)	Correct decision	Type I error (α error)
H_0 (false)	Type II error (β error)	Correct decision

The probability of committing type-I is generally denoted by α while the probability of committing type-II is denoted by β .

Test Statistic: A statistic which provides the basis for testing the null hypothesis is called the test statistic and it is the function of our sample data i.e. $T = T(X)$ where X is a variety of

test statistic depending upon the situation and the problem in hand.ⁿ In the coming modules we will discuss some methods of test statistic.

Generally there are five basic steps in testing of hypothesis:

- 1- State the null and alternative hypotheses
- 2- Specify the level of significance
- 3- Select and compute the appropriate test statistic
- 4- State the decision rule
- 5- Conclusion and its interpretation

Lecture no.34

(Topics 163 to 166)

P-value

P-value or the probability value is a measure of strength of the evidence against the null hypothesis.

P-value is the maximum probability of rejecting the true null hypothesis. It is also known as the observed p-value as we compute it on the basis of observed data.

The general rule for making the decision on the basis of p-value is "If the observed p-value is greater than the assumed level of significance, we do not reject H_0 otherwise we reject it". So the smaller the p-value, greater the evidence against the null hypothesis and vice versa. If we have a given significance level α , then reject H_0 if p-value $\leq \alpha$. A result is said to be statistically significant if it allows us to reject the null hypothesis.

Suppose we are testing a hypothesis H_0 against H_1 and we use Z test.

Suppose we get $Z=2.60$ then the p-value can be calculated as

$$P\text{-value} = P(Z > 2.60)$$

Now, we calculate the area over the observed Z value, $Z = 2.60$. By using the Table of Normal Distribution, we obtain

$$P(Z > 2.60) = (0.5 - 0.4953) = 0.0047$$

In case of two tailed test, the p-value is

$$P(Z < -2.60) + P(Z > 2.60) = 2(0.5 - 0.4953) = 0.0094$$

Best Critical Region

Let H_0 represents the null hypothesis which we want to test against the alternative hypothesis H_1 . Let $f(x)$ denote the pdf or pmf of a random variable X, where

Based on the data analysis and results, we know that there are two types of errors. Type-I error, reject the null hypothesis when it is actually true and it is usually denoted by α and called significance level. Type-II error, accept null hypothesis when it is, in fact, false. The best rejection region is that which minimizes the probability of making Type-I and Type-II errors. The probability that H_0 is rejected when it is false is called the power of the test and is denoted by $1 - \beta$. Now we define a best critical region for testing a simple null hypothesis H_0 versus simple alternative hypothesis H_1 .

Definition: Let C denote a subset of the sample space. Then C is called a best critical region of size α for testing H_0 against H_1 , if for every subset A of the sample space, the following are true:

1. $P(C|H_0) \leq \alpha$ and $P(C|H_1) \geq P(A|H_1)$.
- 2.

One thing is important to note that the power of the test associated with C is at least as greater as the power of test associated with every other A, then C is called a best critical region of size α for testing H_0 against H_1 .

Example

Consider a random variable X that has a binomial distribution with $n = 5$ and $p = \frac{1}{2}$. We wish to

test $H_0: p = \frac{1}{2}$ versus $H_1: p > \frac{1}{2}$

The following table shows the probability of the values

$f(x; \frac{1}{2})$, $f(x; \frac{3}{4})$ and the ratio

$f(x; \frac{1}{2}) / f(x; \frac{3}{4})$.

X	$f(x; \frac{1}{2})$	$f(x; \frac{3}{4})$	$f(x; \frac{1}{2}) / f(x; \frac{3}{4})$
0	1/32	1/1024	32
1	5/32	15/1024	32/3
2	10/32	90/1024	32/9
3	10/32	270/1024	32/27
4	5/32	405/1024	32/81
5	1/32	243/1024	32/243

To satisfy the significance level $\alpha = 1$

we select the critical region as either

the only possible critical region that has probability $\geq \alpha$ greater than

1/32 under the null hypothesis H_0 . So at least one of them must be a best critical region.

However, under the alternative hypothesis, we observe that P_A is the best critical region.

Thus P_A is the best critical region.

Neyman-Pearson Lemma

Neyman-Pearson developed this lemma in 1933 which is considered to the theoretical foundation of the testing of hypothesis theory. To find the best test procedure this lemma helps to find a test with the highest power. The power of the test is the probability that the test correctly reject H_0 when H_1 is true.

Statement: Let $X_1, \dots, X_n$ where n is a fixed positive integer, denote a random sample from a distribution with pdf $f(x; \theta)$. Then the joint pdf of $X_1, \dots, X_n$ is

$$L(x_1, \dots, x_n; \theta) = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta).$$

Let θ_0 and θ_1 be the distinct values of the parameter space Θ , and let k be a positive number.

Define C as the subset of the sample space such that

- 1- $L(x_1, \dots, x_n; \theta_0) \geq k$, for $x_1, \dots, x_n \in C$.
- 2- $L(x_1, \dots, x_n; \theta_1) \geq k$, for $x_1, \dots, x_n \in C$.
- 3- $P_{\theta_0}(C) = \alpha$.

Then C is a best critical region of size α for testing the simple null hypothesis H_0 against the simple alternative hypothesis H_1 .

Example

A random sample of size n from a normal population with $\mu = 1$ is to be used to test the $H_0: \mu = 1$ against $H_1: \mu = 0$ where $\sigma^2 = 1$. Use Neyman-Pearson lemma to find the most powerful critical region of size α .

Solution: The joint distribution of $X_1, \dots, X_n$ on the basis of null and alternative hypotheses are

$$L_0(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2} \sum_{i=1}^n x_i^2}$$

$$L_1(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2} \sum_{i=1}^n (x_i - 1)^2}$$

$$L_0 \geq L_1 \Leftrightarrow \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2} \sum_{i=1}^n x_i^2} \geq \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2} \sum_{i=1}^n (x_i - 1)^2}$$

$$\Leftrightarrow e^{-\frac{1}{2} \sum_{i=1}^n x_i^2} \geq e^{-\frac{1}{2} \sum_{i=1}^n (x_i - 1)^2}$$

$$\Leftrightarrow -\frac{1}{2} \sum_{i=1}^n x_i^2 \geq -\frac{1}{2} \sum_{i=1}^n (x_i - 1)^2$$

$$\Leftrightarrow \sum_{i=1}^n x_i^2 \leq \sum_{i=1}^n (x_i - 1)^2$$

By the Neyman-Pearson lemma the test is rejected H_0 when

$$\sum_{i=1}^n x_i^2 \leq c$$

$$e^{-\frac{1}{2} \sum_{i=1}^n x_i^2} \geq e^{-\frac{1}{2} \sum_{i=1}^n (x_i - 1)^2}$$

Taking log on both sides

$$\frac{1}{n} \sum_{i=1}^n \log k_i = \frac{1}{n} \sum_{i=1}^n \log k_i$$

□ Sam and Xig the distribution of

1

Thus when we consider X_{-n} , i.e. probability of

rejecting H_0

☐ then

PX_C

X_n when $n \rightarrow \infty$

His true. Now

$$\text{PX}-\text{C}$$

PK

So the test is

Reject H_0 if $Z \geq Z_{1-\alpha}$

Do not reject H_0 if $|Z| \leq Z_{\alpha/2}$. This is the most powerful test of size α .

Now we use this for a practical situation. Suppose $\mu_1 = 0$, $\mu_2 = 1$, $\sigma_1^2 = 0.2$ and $\sigma_2^2 = 0.05$

Now we calculate

Reject H_0 if $|Z| > z_{\alpha/2}$

and $0.5045 > 1$ so we can not reject H_0 .
Now $Z = 1.96$

H. But if the level of significance is changed we may accept H_0 .

Also notice that if we consider $\frac{1}{10}$, the rejection rejoin becomes $\frac{1}{10}S_6$.

PX-C

PK

☐ 100 ✓

So the most powerful test is

Reject H_0 if $Z \geq Z_{\alpha}$ accept otherwise.

Lecture no.35 (Topics 167 to 170)

Application Q 8.1.5

Let $X_1, \dots, X_n$ is a random sample from a distribution having pdf of the form

$$f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Show that a best critical region for testing $H_0: \theta = 1$ against $H_1: \theta = 2$ is

$$C = \{X_1, \dots, X_n : \sum_{i=1}^n X_i \geq c\}$$

Solution

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \prod_{i=1}^n x_i^{\theta-1}$$

$$L(1) = \prod_{i=1}^n 1 \cdot x_i^{1-1} = 1$$

$$L(2) = \prod_{i=1}^n 2 \cdot x_i^{2-1} = 2^n \prod_{i=1}^n x_i$$

By the Neyman-Pearson lemma the test is rejected H_0 when

$$\frac{L(2)}{L(1)} \geq k \implies 2^n \prod_{i=1}^n x_i \geq k$$

Uniformly Most Powerful Tests

It is already explained that the probability of rejecting H_0 when it is false is called the power of the test. A test is uniformly most powerful (UMP) test of size α if it has a size α and if among all the tests of size less than or equal to α , it has the largest power for all alternative values of θ . The term “uniformly” refers to “all” alternative values. It is important to note that the UMP test do not exist for all testing problems. But if it exists, it has the greatest chance of rejecting H_0 .

Although we use Neyman-Pearson lemma for deriving the best hypothesis tests for testing a simple null hypothesis against the simple alternative hypothesis but in fact we are interested in testing a simple hypothesis against a composite alternative hypothesis. The Neyman-Pearson lemma can be used for the composite alternative hypothesis as well.

Definition: The critical region C is a uniformly most powerful (UMP) critical region of size α for testing the simple hypothesis H_0 against an alternative composite hypothesis H_1 if the set C is a best critical region of size α for testing H_0 against each simple hypothesis in H_1 . A test defined by this critical region C is called a uniformly most powerful (UMP) test, with significance level α , for testing the simple hypothesis H_0 against the alternative composite hypothesis H_1 .

Example 1

Let $X_1, \dots, X_n$ be the random sample from $N(\mu, \sigma^2)$. We are interested to test the hypothesis

$H_0: \mu = 0$ vs $H_1: \mu > 0$. As σ^2 is unknown, so H_1 is composite. Find the most power (MP) test for size α .

Solution:

$$L_0 = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}}$$

$$L_1 = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}}$$

and x_i for $i = 1, \dots, n$.

L_1 , where k is determined by the

According to the Neyman-Pearson lemma, rejected H_0 if $\frac{L_1}{L_0} > k$ size condition.

$$\frac{L_1}{L_0} = \frac{\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}}}{\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}}} = e^{\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2}$$

Taking log on both sides and adjusting constant terms in C , we get

$$\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2$$

To find the C , we need to apply size condition, i.e.

P Type I error

P Reject H_0 when it is true

χ^2 , under the null hypothesis. For this we need to find the distribution of χ^2 .
 Considering original X 's and Y 's, $Y_1, \dots, Y_n$ are assumed to be independent, then
 Hence test is rejected if $\chi^2 \geq \chi^2_{n, \alpha}$ otherwise accept H_0

Lecture no.36 (Topics 171 to 174)

Example 2

Let X has the observation from the density $f(x)$ Test the hypothesis

$H_0: f(x) = \frac{1}{2}(1-x)$
 $H_1: f(x) = \frac{1}{2}x$ (triangular distribution) and

where $f(x) = \begin{cases} 1-x & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$ Find the MP test with size α .

Solution:

According to the Neyman-Pearson lemma by considering densities at a point

$f_1(x) = \frac{1}{2}x$
 $f_0(x) = \frac{1}{2}(1-x)$
 This implies that $\frac{f_1(x)}{f_0(x)} = \frac{x}{1-x}$

So, $\frac{x}{1-x} \geq K$ under H_0 so
 $x \geq \frac{K}{1+K}$ under H_0 .
 And $\frac{x}{1-x} \geq K$ under H_1 so
 $x \geq \frac{K}{1+K}$ under H_1 .

Next we write the size condition for α as

$$E_0 \left[\int_0^1 \int_0^1 \frac{1}{2} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \right)^2 dx \right] = \alpha$$

Hence MP test of size α is:

$$\text{Reject } H_0 \text{ if } \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \geq \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad \text{or} \quad \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \leq \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

For instance, if $\alpha = 0.01$, the test will reject H_0 if $|x| \geq 1.96$. So any value of X within the range $[-1.96, 1.96]$ we accept H_0 otherwise reject it.

Important Remarks

The Neyman-Pearson lemma makes no claims on the dimension of data or parameter

θ . So, the result holds for vector or even function parameters without changing a word.

The only limitation of the result is that it holds only for simple-vs-simple hypotheses.

In general, the most powerful test will depend on θ

However, in many cases, it can

be shown that the test actually does not depend on θ . In those cases, the optimality of the test can be extended to a sort of uniform optimality for a more general kind of alternative hypothesis.

Let $X_1, \dots, X_n$ denote a random sample from a distribution that has pdf $f(x; \theta)$. Y is a sufficient statistic for θ .

Suppose that Y is a sufficient statistic for θ . Then by the factorization

theorem, the joint pdf of $X_1, \dots, X_n$ can be written as

$$L(x_1, \dots, x_n; \theta) = h(x_1, \dots, x_n) k(u; \theta), \quad \text{where } u = u(x_1, \dots, x_n) \text{ does not contain } \theta.$$

This results, the ratio

$$\frac{L(x_1, \dots, x_n; \theta_1)}{L(x_1, \dots, x_n; \theta_0)} = \frac{h(x_1, \dots, x_n) k(u; \theta_1)}{h(x_1, \dots, x_n) k(u; \theta_0)} = \frac{k(u; \theta_1)}{k(u; \theta_0)}$$

depends only on $X_1, \dots, X_n$ through $u = u(x_1, \dots, x_n)$. Thus

if there is a sufficient statistic Y for θ and if a best test or UMP test is required, no test is needed which based on any statistic other than the sufficient statistic.

Likelihood Ratio Tests

In this module we try to develop test that can be used to test the general hypotheses, called likelihood ratio test (LRT). These tests are based on a ratio of the likelihood functions. Let X be a

random variable with pdf or pmf $f(x; \theta)$, $\theta \in \Omega$, we want to test

$$H_0: \theta \in \Omega_0 \quad \text{vs} \quad H_1: \theta \in \Omega_1$$

Let us write down their likelihood functions L ;

and $\hat{\theta}_1$, then consider the ratio

$$\frac{\sup_{\theta \in \Theta_0} L(\theta; \mathbf{x})}{\sup_{\theta \in \Theta_1} L(\theta; \mathbf{x})}$$

The possible rejection region. The criteria is if $\hat{\theta}_1$ has a higher value then $\hat{\theta}_0$ is more likely and if H_0 is more likely to happen then $\hat{\theta}_0$ has higher value. Therefore the rejection region is for the larger value of this ratio and acceptance region for the lower value of this ratio. This ratio requires actually two maximizations and at least one of the maximization could be more complicated, e.g in one sided hypothesis.

An equivalent procedure can be adopted by considering full parameter space

$$\frac{L(\hat{\theta}_0; \mathbf{x})}{L(\hat{\theta}_1; \mathbf{x})}$$

The likelihood ratio is to reject H_0 if $\frac{L(\hat{\theta}_0; \mathbf{x})}{L(\hat{\theta}_1; \mathbf{x})} > C$. Since the denominator is the maximization of the full parameter space, therefore naturally

$$\sup_{\theta \in \Theta} L(\theta; \mathbf{x})$$

If the distribution of $\mathbf{x}$ is continuous then size is attainable. In case $\mathbf{x}$ has a discrete distribution, it may not be possible to derive a LRT whose size is exactly α . This happens because LRT is a non-randomized test.

Example 1

Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, 1)$ population. Consider testing $H_0: \mu \leq \mu_0$ versus $H_1: \mu > \mu_0$. Here μ_0 is a number fixed by the experimenter prior to the experiment. Write down the likelihood ratio statistic.

Solution:

According to the LR test

$$\frac{L(\hat{\theta}_0; \mathbf{x})}{L(\hat{\theta}_1; \mathbf{x})} = \frac{\exp\left(-\frac{n(\bar{x} - \mu_0)^2}{2}\right)}{\exp\left(-\frac{n(\bar{x} - \hat{\mu})^2}{2}\right)}$$

$$\hat{L} = \frac{1}{n} \sum_{i=1}^n \exp\left(-\frac{1}{2} \left(\frac{x_i - \mu_0}{\sigma_0}\right)^2\right)$$

(by adding and subtracting

Now where $\bar{x}$ is the sample mean and μ_0 is the hypothesized mean. The likelihood ratio test is $\frac{\hat{L}_1}{\hat{L}_0}$ where k is determined by the size condition. So

$$\exp\left(-\frac{n}{2} \left(\frac{\bar{x} - \mu_0}{\sigma_0}\right)^2\right) \leq k$$

or

$$n \left(\frac{\bar{x} - \mu_0}{\sigma_0}\right)^2 \geq \frac{2}{\ln k}$$

$$n \left(\frac{\bar{x} - \mu_0}{\sigma_0}\right)^2 \geq \frac{2}{\ln k}$$

$$\text{Let } g = \frac{2}{\ln k}$$

To find the k we need to apply size condition, i.e.

P Type I error

P Reject H_0 when it is true

$$P_{nX} = \int_{\bar{x} \geq \mu_0 + \frac{\sigma_0}{\sqrt{n}} \sqrt{g}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu_0}{\sigma_0}\right)^2\right) dx$$

, under the null hypothesis. For this we need to find the distribution of

Considering original

's, then we know that

$X \sim N(\mu_0, \sigma_0^2)$ In the given case

$nX \sim N(n\mu_0, n\sigma_0^2)$ then

Since the alternative hypothesis is $H_1: \mu > \mu_0$ so the decision rule is, reject H_0 at level α test if

$$n \left(\frac{\bar{x} - \mu_0}{\sigma_0}\right)^2 \geq \frac{2}{\ln k} \quad \text{or} \quad n \left(\frac{\bar{x} - \mu_0}{\sigma_0}\right)^2 \geq \frac{2}{\ln k}$$

Lecture no.37

(Topics 175 to 177)

Example 2

Let $X_1, \dots, X_n$ be iid random sample from with pdf

$$f(x) = \begin{cases} \exp(-x) & x > 0 \\ 0 & x \leq 0 \end{cases}$$

We want to test $H_0: \lambda = \lambda_0$ vs $H_1: \lambda < \lambda_0$. Find the likelihood ratio test.

Solution:

Actually this is a negative exponential distribution. To find the maximum likelihood, we need to choose λ as large as we can in order to maximize the likelihood function. But we have the constrain that $x_i \geq 0$, so we pick the smallest λ i.e. $\lambda = 1/\bar{x}$ so the likelihood function will be

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

So $\hat{\lambda}$ is our maximum likelihood estimator of λ and is an increasing function of $\sum_{i=1}^n x_i$. So the denominator of the LR test is as (1). But we have an additional constrain in the numerator. Thus we have two possibilities

- 1- If $\lambda_0 \leq \hat{\lambda}$, then this constrain has no effect so our unrestricted MLE is the same as the restricted MLE, so the likelihoods are the same and thus the LR is one.
- 2- If $\lambda_0 > \hat{\lambda}$, then we have to choose $\lambda = \lambda_0$ without violating $x_i \geq 0$ and we take the ratio of two likelihoods. In this case the numerator is $L(\lambda_0)$.

Now the likelihood ratio is

$$\lambda = \lambda_0 \quad \lambda = \hat{\lambda} = 1/\bar{x}$$

We reject H_0 if $\lambda_0 \bar{x} < k$, which implies a rejection region $x : x > k/\lambda_0$. Note that the

rejection region depends on the sample only through the sufficient statistic $\sum_{i=1}^n x_i$.

Likelihood Ratio Test for the exponential distribution Suppose we are interested to test

Let $X_1, \dots, X_n$ are iid with pdf $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$.

the hypothesis $H_0: \lambda = \lambda_0$ versus $H_1: \lambda < \lambda_0$. Develop the likelihood ratio test.

Solution:

According to the LR test

where

$$\hat{L} = \frac{L(\lambda_0)}{L(\hat{\lambda})} = \frac{\lambda_0^n e^{-\lambda_0 \sum_{i=1}^n x_i}}{\left(\frac{1}{\bar{x}}\right)^n e^{-\frac{1}{\bar{x}} \sum_{i=1}^n x_i}}$$

$$\hat{L} = \frac{\prod_{i=1}^n \exp(-x_i)}{\prod_{i=1}^n \exp(-k)} = \exp\left(-\sum_{i=1}^n x_i + nk\right)$$

The likelihood ratio is to reject H_0 if $\sum_{i=1}^n x_i > k$, where k is determined by the size conditions.

Now

$$\frac{\prod_{i=1}^n \exp(-x_i)}{\prod_{i=1}^n \exp(-k)} = \exp\left(-\sum_{i=1}^n x_i + nk\right)$$

$$= \exp\left(-\sum_{i=1}^n x_i + nk\right)$$

$$= \exp\left(-\sum_{i=1}^n x_i + nk\right)$$

$$= \exp\left(-\sum_{i=1}^n x_i + nk\right)$$

where

$$k = \frac{\sum_{i=1}^n x_i}{n}$$

or

where

$$k = \frac{\sum_{i=1}^n x_i}{n}$$

We know by the property of exponential distribution, under the null hypothesis $\sum_{i=1}^n x_i \sim \chi^2_{2n}$, so

Since the alternative hypothesis H_1 is $\lambda > \lambda_0$, so the decision rule is, reject H_0 at level α if $\sum_{i=1}^n x_i > k$, where k is determined by the size conditions. k and k_2 are the upper and lower quantile of χ^2_{2n} distribution with $2n$ degree of freedom.

Application 6.3.5

Let $X_1, \dots, X_n$ are the random samples from a $N(\mu, \sigma^2)$ distribution, where σ^2 is known. Show that the likelihood ratio test of $H_0: \mu = 0$ versus $H_1: \mu \neq 0$ can be based

upon the statistic $W = \frac{\sum_{i=1}^n X_i^2}{n}$. Determine the null hypothesis of W and give explicitly, the rejection rule for a level α test.

Solution

We have

$$\hat{L}_0 = \frac{1}{(2\pi)^{n/2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n X_i^2\right\}$$

$$\hat{L}_e = \frac{1}{(2\pi)^{n/2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n X_i^2\right\}$$

But we know that the MLE of μ is $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, so

$$\hat{L}_1 = \frac{1}{(2\pi)^{n/2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2\right\}$$

$$= \frac{1}{(2\pi)^{n/2}} \exp\left\{-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2\right)\right\}$$

which implies that

$$\frac{\hat{L}_1}{\hat{L}_0} = \exp\left\{\frac{n\bar{X}^2}{2\sigma^2}\right\}$$

So this test depends on W . The likelihood ratio is to reject H_0 if $W > k$, where k is determined by the size conditions. Now

or

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To find the k , we need to apply size condition, i.e.

Reject H_0 when it is true, under the null hypothesis. For this we need to find the distribution

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Since the alternative hypothesis is H_{10}

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Lecture no.38

(Topics 178 to 180)

Application 6.3.10

Let $X_1, \dots, X_n$ are the random samples from a gamma

(a) Show that the likelihood ratio test of H_0 versus H_{10} is based upon the

statistic, W

(b) For $n=5$, find c and c_0 so that the test that reject H_0 when

significance level 0.05.

Solution: (a)

The pdf of gamma

$f(x)$

$f(x)$

$f(x)$

$f(x)$

According to the LR test

where

$\hat{\theta}$

$\hat{\theta}$

e

e

$$\hat{L}_X = \prod_{i=1}^n \frac{1}{2} \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma}\right)^2\right\}$$

The likelihood ratio is to reject H_0 if $\sum_{i=1}^n x_i^2 > k$, where k is determined by the size conditions.

Now

$$\frac{\hat{L}_X}{\hat{L}_{H_0}} = \prod_{i=1}^n \frac{1}{2} \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma}\right)^2\right\} \cdot \frac{1}{\prod_{i=1}^n \frac{1}{2} \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma_0}\right)^2\right\}}$$

To obtain the null distribution we need to multiply both sides by $\frac{2}{\sigma_0^2}$, so

$$\frac{2}{\sigma_0^2} \prod_{i=1}^n \frac{1}{2} \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma_0}\right)^2\right\} = \prod_{i=1}^n \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma_0}\right)^2\right\}$$

Since X_i under the null hypothesis, then the m.g.f is

$$M_{X_i} = \int_{-\infty}^{\infty} \exp\left\{t x_i\right\} \frac{1}{\sigma_0} \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma_0}\right)^2\right\} dx_i$$

$$M_{X_i} = \int_{-\infty}^{\infty} \exp\left\{t x_i\right\} \frac{1}{\sigma_0} \exp\left\{-\frac{1}{2} \left(\frac{x_i}{\sigma_0}\right)^2\right\} dx_i = \exp\left\{\frac{1}{2} \sigma_0^2 t^2\right\}$$

$$M_W = \left(\exp\left\{\frac{1}{2} \sigma_0^2 t^2\right\}\right)^n = \exp\left\{\frac{1}{2} n \sigma_0^2 t^2\right\}$$

(b)

It is given that $\theta_0 = 3$ and $n=5$, then

$$W_c = \frac{2 \cdot 0.025 \cdot 30}{0.2} = 7.5 \quad W_c = \frac{2 \cdot 0.025 \cdot 30}{0.2} = 7.5$$

Theorem 6.3.1

Statement: Assuming the regularity conditions, under the null hypothesis, H_0 ,

$$2 \log \frac{L(\hat{\theta})}{L(\theta_0)}$$

This theorem is also called Wilks theorem.

L Proof: Let $L(\theta)$ and expand the function $L(\theta)$ into Taylor series about θ_0 and evaluate at the value $\hat{\theta}$,

$$L(\hat{\theta}) = L(\theta_0) + \frac{1}{n} \frac{dL(\theta_0)}{d\theta} (\hat{\theta} - \theta_0) + \frac{1}{2n} \frac{d^2 L(\theta_0)}{d\theta^2} (\hat{\theta} - \theta_0)^2 + \dots$$

Define the likelihood ratio as

$$\lambda = \frac{L(\hat{\theta})}{L(\theta_0)} \quad \text{or} \quad \log \lambda = \log L(\hat{\theta}) - \log L(\theta_0)$$

$$2 \log \lambda = 2 \log L(\hat{\theta}) - 2 \log L(\theta_0)$$

Now using (1) in the above, we get

$$2 \log \lambda = \frac{1}{n} \frac{d^2 L(\theta_0)}{d\theta^2} (\hat{\theta} - \theta_0)^2 + \dots$$

We know that by the LLN $\frac{1}{n} \frac{d^2 L(\theta_0)}{d\theta^2} \rightarrow I(\theta_0)$, and $\frac{1}{n} \frac{dL(\theta_0)}{d\theta} \rightarrow 0$, where

$$Z = \frac{\frac{1}{n} \frac{dL(\theta_0)}{d\theta}}{\sqrt{\frac{1}{n} \frac{d^2 L(\theta_0)}{d\theta^2}}} \rightarrow 0$$

Since $\frac{n \ln I}{N}$ and its square converges to $\frac{2}{N}$ in distribution.

From the above theorem we can construct an approximate size- α test based on likelihood ratio without considering exact sampling distribution. By this theorem if n is large $2 \log$

when H_0 is true. So we just need to calculate λ and find χ^2 distribution.

Example

Let $X_1, \dots, X_n$ be a random sample having the common beta $(\alpha, 1)$ pdf. We use this pdf to develop the above test statistic for the hypotheses

$H_0: \alpha = 1$ versus $H_1: \alpha < 1$

Solution:

The pdf of beta distribution is

$$f(x) = \frac{1}{B(\alpha, 1)} x^{\alpha-1}, \quad 0 < x < 1.$$

Now $L(x)$

We know that its ML estimate is $\hat{\alpha} = \frac{n}{\sum_{i=1}^n \log x_i}$

Now $L(x)$

Under H_0

$$\begin{aligned} 2 \log \lambda &= \log \frac{L(\hat{\alpha})}{L(1)} \\ &= \log \frac{\left(\frac{1}{B(\hat{\alpha}, 1)} \right)^n \prod_{i=1}^n x_i^{\hat{\alpha}-1}}{\left(\frac{1}{B(1, 1)} \right)^n \prod_{i=1}^n x_i^{1-1}} \\ &= \log \frac{\left(\frac{1}{B(\hat{\alpha}, 1)} \right)^n \prod_{i=1}^n x_i^{\hat{\alpha}-1}}{\left(\frac{1}{B(1, 1)} \right)^n} \\ &= \log \frac{\left(\frac{1}{B(\hat{\alpha}, 1)} \right)^n \prod_{i=1}^n x_i^{\hat{\alpha}-1}}{\left(\frac{1}{B(1, 1)} \right)^n} \\ &= \log \frac{\left(\frac{1}{B(\hat{\alpha}, 1)} \right)^n \prod_{i=1}^n x_i^{\hat{\alpha}-1}}{\left(\frac{1}{B(1, 1)} \right)^n} \\ &= \log \frac{\left(\frac{1}{B(\hat{\alpha}, 1)} \right)^n \prod_{i=1}^n x_i^{\hat{\alpha}-1}}{\left(\frac{1}{B(1, 1)} \right)^n} \end{aligned}$$

Lecture no.39

(Topics 181 to 183)

Application Q6.3.12

Let $X_1, \dots, X_n$ be a random sample from the beta distribution with parameters α and β . Show that the likelihood ratio test statistic for testing $H_0: \alpha = 1$ versus $H_1: \alpha > 1$ is a function of the statistic $W = \sum_{i=1}^n \log X_i$.

H: $\alpha > 1$ is a function of the statistic

$$W = \sum_{i=1}^n \log X_i$$

Solution:

The desired pdf is

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1.$$

Now $L(x)$

The log-likelihood function is

$$\begin{aligned} \ln L(x) &= \sum_{i=1}^n \ln f(x_i) \\ &= \sum_{i=1}^n \left[\ln \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} + (\alpha - 1) \ln x_i + (\beta - 1) \ln (1 - x_i) \right] \\ &= n \ln \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} + (\alpha - 1) \sum_{i=1}^n \ln x_i + (\beta - 1) \sum_{i=1}^n \ln (1 - x_i) \end{aligned}$$

To find the MLE of α we need to equate above equation to zero and solve it for α . Since this is a non-linear equation and involves digamma function so numerical solution could only be possible by using some iterative methods. So we can write it as

$$\ln L(x) = n \ln \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} + (\alpha - 1) \sum_{i=1}^n \ln x_i + (\beta - 1) \sum_{i=1}^n \ln (1 - x_i)$$

Also under H_0

where test statistic

$$f = W$$

is a function of W , which we only need to show.

Application Q6.3.15 (a)

Let $X_1, \dots, X_n$ be a random sample from a distribution with pmf

$$p_X(x) = \begin{cases} 1-x & \text{if } x=0 \\ x & \text{if } x=1 \end{cases} \quad \text{where } 0 < x < 1$$

We wish to test $H_0: p = 1/3$ versus $H_1: p > 1/3$. Find the MLE of p .

Solution:

The likelihood function is

$$L(p) = \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} = p^{\sum_{i=1}^n X_i} (1-p)^{n - \sum_{i=1}^n X_i}$$

The MLE of p for this function is $\hat{p} = \bar{X}$. So

$$\hat{L}(\hat{p}) = \left(\frac{\sum_{i=1}^n X_i}{n} \right)^{\sum_{i=1}^n X_i} \left(1 - \frac{\sum_{i=1}^n X_i}{n} \right)^{n - \sum_{i=1}^n X_i}$$

Also under the null hypothesis, $L(p_0)$ is

$$L(p_0) = \left(\frac{1}{3} \right)^{\sum_{i=1}^n X_i} \left(\frac{2}{3} \right)^{n - \sum_{i=1}^n X_i}$$

LR test for Binomial Proportion

Let X be a binomial random variable. We are considering the hypothesis $H_0: p = p_0$ versus $H_1: p > p_0$. The full parameter space is $\{p \in (0, 1)\}$ whereas parameter space under null hypothesis is $\{p = p_0\}$.

Now $C = \{p \in (0, 1) : p > p_0\}$

$$1 - \log L = \log C - \sum_{i=1}^n X_i \log p - (n - \sum_{i=1}^n X_i) \log (1-p)$$

So, therefore

$$\hat{L}(\hat{p}) = \left(\frac{\sum_{i=1}^n X_i}{n} \right)^{\sum_{i=1}^n X_i} \left(1 - \frac{\sum_{i=1}^n X_i}{n} \right)^{n - \sum_{i=1}^n X_i}$$

Now under the null hypothesis $p = p_0$ we have the two following cases:

n then 0

If $\hat{p} > \hat{p}_0$ then $\hat{p} > \hat{p}_0$

If $\hat{p} < \hat{p}_0$ then $\hat{p} < \hat{p}_0$

., p Hence $\hat{p} > \hat{p}_0$

f L H_0 C_p p 1 f i R x n

So x n , then n 1 p implies that always accept H_1

When x p 0

xxx pp 1 nx

nn 1 nx

x log x $xlog$ p n x log 11 p $xlog$ x n x log x

So reject H_1 if x c . We need to analysis the behaviour of x so

dx x log p log 11 p log n x n log n n x n

dx x log p log 11 p log n x n log n n x n

np x 1 n p , so

Now we choose x p n x n 1 p dx x 0 for p , i.e. is

decreasing function for x and also x is decreasing for x as well. Therefore the LR test can be written as reject H_1 if x c where c is determined on the size condition i.e.

Sup P x c , X C_{nx} p p nx

This probability on the right tail of the binomial distribution corresponds to the higher number of success, so it is an increasing in p , sp p p . Thus c is determined for

P_{xc} $Hence$ X Bin $binomial$ table can be used to find the probabilities.

It may be possible that no value of c_j gives an exact χ^2 . so we choose c_j such that χ^2 is minimum and χ^2 is minimum.

Lecture no.40 (Topics 184 to 187)

Minimum Chi-square Method

Let $X_1, \dots, X_n$ be a random sample from the density given by $f(x)$; and let $s_1, \dots, s_k$ be the partition of the range of X . The probability that an observation falls in any cell s_j is denoted by p_j . If $f(x)$ is that pdf of a random variable, then $\sum_{j=1}^k p_j = 1$.

Let the random variable X denote the number of X in the sample which falls in the cell s_j , then n_j is the number of X in the sample which falls in the cell s_j . Then the statistic χ^2 is defined as follows:

$$\chi^2 = \sum_{j=1}^k \frac{(n_j - np_j)^2}{np_j}$$

is known as the minimum Chi-square statistic, where n_j is a value of N . In hypothesis testing the null hypothesis is rejected if the under used statistic provide a large value. In minimum chi-square (MCS), we find the values of the parameter that make this test statistic as small as possible. In other words the MCS estimator of θ is that estimator of θ which minimizes the χ^2 . It is that θ among all possible θ 's which makes the expected number of observations in cells nearest the observed number.

Example

Let $X_1, \dots, X_n$ be a random sample from a Bernulli distribution

$$f(x) = p^x (1-p)^{1-x}, \quad x = 0, 1.$$

Take N . The range of X is partitioned in to two number of observation equal to j for $j=0,1$.

sets consisting of the number 0 and 1 respectively. Find the MCS estimator of p .

Solution

We know that

The $\hat{\theta}_n$ will be zero when $\hat{\theta}_n = 0$, $\hat{\theta}_n = 1$. Hence the MCS estimator of θ is $\hat{\theta}_n$.

Sequential Probability Ratio Test (SPRT)

So far we used the random sample having fixed sample size to estimate the parameter or test the hypothesis using full sample or full data. But there is another type of experimentation, where we may not take a decision on a fixed sample. For example, when we test the hypothesis under some certain conditions we need to draw more samples instead using the fixed sample size. This type of procedure is called sequential type procedure or sequential testing hypothesis procedure. This is given by Wald and that is why it is called Wald's Sequential Probability Ratio Test.

Suppose that the sample size n is not fixed initially and it is a random variable N with sample space, $1, 2, 3, \dots$. Let $X_1, X_2, \dots$ be a sequence of observations (not necessarily iid) and consider the hypothesis testing problem H_0 versus H_1 . Let the likelihood ratio at the m -th stage is

$$L_m = \frac{L_1 L_2 \dots L_m}{L_0^m}$$

In NP-lemma the test based on L_m for a fixed n . However in SPRT, we propose a modified procedure which is as follows:

First choose two positive constants L_0 and L_1 such that $L_0 < L_1$. Consider the critical region in m -dimensional space, the hypothesis H_0 is rejected if and only if there exists a positive integer m , so that x belongs to the set

$$C = \{x \in R^m : L_0 \leq \prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} \leq L_1\}.$$

On the other hand, the hypothesis H_0 is accepted if and only if there exists a positive integer m , so that x belongs to the set

$$B = \{x \in R^m : L_0 \leq \prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} \leq L_1 \text{ and } \prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} < L_0\}.$$

We continue to study one more observation as long as $L_0 \leq \prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} \leq L_1$ and repeat above procedure

based on $m+1$ observation. Here L_0 and L_1 are the boundary points of SPRT. We stop taking observation in one of two ways:

Reject H_0 as soon as $\prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} \geq L_1$, or accept H_0 as soon as $\prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} \leq L_0$. This type of test is

called Wald's Sequential Probability Ratio Test.

The inequality $L_0 \leq \prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)} \leq L_1$ can be expressed in an equivalent form

where $u = \prod_{i=1}^m \frac{f_1(x_i)}{f_0(x_i)}$ is the statistic and c_0 and c_1 depend on the constants L_0, L_1 , and n .

We stop to take the observation and reach the decision as soon as $u \geq c_1$ or $u \leq c_0$.

Further to observe the power of the test we define PH_{reject} when it is true

The PH_{accept} when it is false c_0, c_1 is called the strength of the test. We state two

results in term of theorems relating L_0, L_1 and c_0, c_1 .

Theorem 1: If the SPRT of strength c_0, c_1 and the boundary points L_0, L_1 terminates with probability one, then

$$(i) \quad L_0 \leq \frac{1}{1 - PH_{\text{reject}}} \leq L_1$$

$$(ii) \quad L_0 \leq \frac{1}{1 - PH_{\text{accept}}} \leq L_1.$$

Theorem 2: If for the choice L_0, L_1 the SPRT terminates with probability

one and is of strength c_0, c_1 , then $c_0 \leq L_0 \leq 1$ and $1 \leq c_1 \leq L_1$.

$$(i) \quad c_0 \leq L_0 \leq 1$$

$$(ii) \quad 1 \leq c_1 \leq L_1$$

(iii) 01 .

Example 1

Let X has a pmf

$$f(x) = \begin{cases} 1 - \frac{1}{2^n} & x = 0, 1, \dots, n-1 \\ \frac{1}{2^n} & x = n \end{cases}$$

Using sequential probability ratio test, let $H_0: H_{12}$ and H_{33} determine the constants c_0 and c_1

Solution:

The likelihood ratio at the m -th stage is

$$L_m = \frac{L_{12}^m}{L_{33}^m} = \frac{\prod_{i=1}^m \frac{f_{12}(x_i)}{f_{33}(x_i)}}{\prod_{i=1}^m \frac{f_{12}(x_i)}{f_{33}(x_i)}}$$

Where $L_{12} = \frac{f_{12}(x)}{f_{33}(x)}$ and $L_{33} = \frac{f_{33}(x)}{f_{33}(x)}$. Then

$$L_m = \frac{2^{n \sum_{i=1}^m x_i}}{2^{n \sum_{i=1}^m x_i}}$$

Now the inequality $L_m \geq c_1$ becomes $2^{n \sum_{i=1}^m x_i} \geq c_1$. Taking log of base 2, the inequality becomes

$$\log L_m = n \sum_{i=1}^m x_i \geq \log c_1$$

$$\sum_{i=1}^m x_i \geq \frac{\log c_1}{n}$$

The observation of the X^n

We continue to observe outcomes as long as

outcomes is discontinued. If $\sum_{i=1}^m x_i \geq \frac{\log c_1}{n}$ leads to acceptance H_0 and $\sum_{i=1}^m x_i < \frac{\log c_1}{n}$ leads to rejection H_0 . The inequality $\sum_{i=1}^m x_i \geq \frac{\log c_1}{n}$ leads to acceptance H_0 and $\sum_{i=1}^m x_i < \frac{\log c_1}{n}$ leads to rejection H_0 .

Lecture no.41

(Topics 188 to 190)

Example 2

Let X be $N, \sigma^2 = 100$. To find the sequential probability ratio test for testing

$H_0: \mu = 75$ against $H_1: \mu = 78$ such that each of α and β is approximately equal to 0.10.

⁰¹
Solution:

The likelihood ratio at the m -th stage is

$$L_m = \frac{\prod_{i=1}^m \frac{1}{\sqrt{2\pi \cdot 100}} \exp\left\{-\frac{(X_i - 75)^2}{2 \cdot 100}\right\}}{\prod_{i=1}^m \frac{1}{\sqrt{2\pi \cdot 100}} \exp\left\{-\frac{(X_i - 78)^2}{2 \cdot 100}\right\}}$$

After simplification we get

$$L_m = \frac{6^n \exp\left\{-\frac{X_n - 459n}{200}\right\}}{9^n \exp\left\{-\frac{X_n - 459n}{200}\right\}}$$

Now the inequality $L_m \geq 1$ becomes $L_m \geq 1$ or $\frac{6^n \exp\left\{-\frac{X_n - 459n}{200}\right\}}{9^n \exp\left\{-\frac{X_n - 459n}{200}\right\}} \geq 1$.

And $L_m \leq 1$ and $L_m \leq 1$ or $\frac{6^n \exp\left\{-\frac{X_n - 459n}{200}\right\}}{9^n \exp\left\{-\frac{X_n - 459n}{200}\right\}} \leq 1$. Hence

$$\frac{6^n \exp\left\{-\frac{X_n - 459n}{200}\right\}}{9^n \exp\left\{-\frac{X_n - 459n}{200}\right\}} \geq 1$$

Taking log on both sides, the inequality can be written as

$$\log 99 \frac{6^n \exp\left\{-\frac{X_n - 459n}{200}\right\}}{9^n \exp\left\{-\frac{X_n - 459n}{200}\right\}} \geq \log 1, \text{ or}$$

$$\frac{153}{23} \frac{100}{23} \frac{153}{23} \frac{100}{23}$$

The observation of the
 We continue to observe outcomes as long as $X_n \leq c_n$
 outcomes is discontinued. If $X_n > c_n$ or $X_n < c_n$ the inequality $X_n \leq c_n$ leads to acceptance and the inequality $X_n < c_n$ leads to rejection. The power of test is 0.10 when H_0 is true and 0.90 when H_0 is false.

Application Q 8.4.2

Let X have a Poisson distribution with mean λ . Find the sequential probability ratio test for testing $H_0: \lambda = 1$ against $H_1: \lambda = 2$. Show that this test can be based upon the statistic

$\sum_{i=1}^n X_i$, 0.20 and 0.10 find c_1 and c_2 .

Solution:

The likelihood ratio at the m -th stage is

$$\frac{L_m}{L_0} = \frac{2^{\sum_{i=1}^m X_i} e^{-2}}{1^{\sum_{i=1}^m X_i} e^{-1}} = 2^{\sum_{i=1}^m X_i} e^{-1}$$

Where $L_m = \frac{2^{\sum_{i=1}^m X_i} e^{-2}}{m!}$ and $L_0 = \frac{1^{\sum_{i=1}^m X_i} e^{-1}}{m!}$

Now the inequality $L_m \geq L_0$ becomes $L_m \geq L_0$ and

And $L_m \leq L_0$ or $L_m \leq L_0$. Hence

$$2^{\sum_{i=1}^n X_i} e^{-1} \geq 1 \quad \text{or} \quad 2^{\sum_{i=1}^n X_i} \geq e$$

$$2^{\sum_{i=1}^n X_i} \geq e \quad \text{or} \quad \sum_{i=1}^n X_i \geq \frac{1}{\ln 2}$$

Taking log on both sides

$$\log \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \log \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \quad \text{and} \quad \log \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \log \left(\frac{1}{n} \sum_{i=1}^n X_i \right)$$

Hence the test is based on $\sum_{i=1}^n X_i$

The observation of the outcomes is discontinued with the first value of n

leads to $\sum_{i=1}^n X_i$ or X The inequality $\sum_{i=1}^n X_i$ leads to acceptance. and the inequality $\sum_{i=1}^n X_i$ leads to rejection.

Bootstrap Testing Procedures

The bootstrap method was introduced by Efron in 1979. It is a computer-intensive procedure for assessing the accuracy and properties of an estimator and it works by resampling the original n -data n -times with replacement and recalculating the point estimates such as mean. This method works as:

1. This method has an equal probability of drawing each original data point for inclusion in the resampled data.
2. The procedure can select a data point more than once for resampled data.
3. Generally procedure creates resampled datasets that has the same sample size as the original dataset.

For initial understanding, let's take a simple example. A study consists five data points and creates five bootstrap samples as:

Original	Bootstrap1	Bootstrap2	Bootstrap3	Bootstrap4	Bootstrap5
2	2	4	4	8	4
4	2	6	4	8	2
6	6	10	4	2	6
8	10	2	6	10	8
10	10	2	10	4	8

Bootstrap method can also be used effectively in testing of hypotheses. To illustrate the technique, consider the two sample independent case:

- Observed sample 1 of size n: $x_1, \dots, x_n$
- Observed sample 2 of size m: $y_1, \dots, y_m$
- Observed the sample mean difference: $\bar{x} - \bar{y}$

H_0 : Both samples are from the same population

H_a : Both samples are not from the same population

$\alpha = 0.05$

Bootstrap procedure

- Combine the samples into one sample of $(n+m)$ observations
- Draw a bootstrap sample of $(n+m)$ observations with replacement from the combined sample
- Calculate the mean of the first n observations and call it $\bar{x}^*$ and calculate the mean of the remaining m observation and call it $\bar{y}^*$ and find the test statistic: $\bar{x}^* - \bar{y}^*$
- Repeat step 2 and 3 for B (e.g. 3000) times and obtain B values of the test statistic
- Calculate the p-value as $\frac{\text{No. of times } \bar{x}^* - \bar{y}^* \geq \bar{x} - \bar{y}}{B}$

Reject H_0 if p value $\geq \alpha$.

Important Note: we do not need to assume the normality assumption and the sampling distribution of all possible $\bar{x} - \bar{y}$ generally does not follow the t-distribution.

Lecture no.42

(Topics 191 to 193)

Example using R

In this example the data is generated from a contaminated normal distribution. Thirty independent observations were generated from this distribution. The first 15 observations are represented by X-variate and the last 15 observations are denoted by Y-variate. The data is given as

X variates							
94.2	111.3	90.0	99.7	116.8	92.2	166.0	95.7
109.3	106.0	111.7	111.9	111.6	146.4	103.9	
Y variates							
125.5	107.1	67.9	98.2	128.6	123.5	116.5	143.2
120.3	118.6	105.0	111.8	129.3	130.8	139.8	

Using bootstrap method to test whether the two variates come from the same distribution.

Solution:

The R-code is given as:

```
d <- read.table("data1.csv",header=T,sep="," ,dec=".")
boxplot(d$data~d$variate,las=1,xlab="variate",ylab="data")
mean(d$data[d$variate=="x"])
mean(d$data[d$variate=="y"])
test1 <- abs(mean(d$data[d$variate=="x"])-
mean(d$data[d$variate=="y"]))
test1
set.seed(192348)
n <- length(d$variate)
B <- 3000      # number of bootstrap samples
variable <- d$data  #variable use for resample
bootstrapsamples <-
matrix(sample(variable,size=n*B,replace=T),nrow=n,ncol=B)
# we get 30*3000 matrix where each column is a bootstrap
resamples
dim(bootstrapsamples)
#now we calculate the test statistic for bootstrap samples. so
we need to create an empty vector to store these test
statistics
boot.test1 <- rep(0,B) #a vector of 3000 zeros
for (i in 1:B){
  boot.test1[i] <- abs(mean(bootstrapsamples[1:15,i])-
mean(bootstrapsamples[16:30,i]))
}
# just to recall the observed test statistic for test1
test1
```

```
boottest1 <- round(boot.test1,1)
pvalue1 <- mean(boottest1>=test1)
pvalue1
```

The following are the outputs:

```
> mean(d$data[d$variate=="x"])
[1] 111.1133
> mean(d$data[d$variate=="y"])
[1] 117.74
> test1
```

```
[1] 6.626667
```

```
dim(bootstrapsamples)
```

```
[1] 30 3000
```

```
boottest1 <- round(boot.test1,1)
```

```
pvalue <- mean(boottest1>=test1)
```

```
> pvalue
```

```
[1] 0.3453333
```

Do not reject H_0

Bayesian Tests

In Bayesian scenario we need to decide either we accept H_0 or H_1 . Suppose X be the random variable with pdf or pmf $f(x)$. Let our hypotheses are

H_0 versus H_1

H_0 : $\theta = \theta_0$ H_1 : $\theta = \theta_1$

where θ is a parameter. Let $\pi(\theta)$ denote the prior distribution of the random variable θ . X represents a random variable and the posterior pdf or pmf is denoted by $\pi(\theta|X)$.

Let $X_1, X_2, \dots, X_n$

We use the posterior distribution under the H_0 and H_1 to compute the following conditional probabilities:

$P(X_1, X_2, \dots, X_n | H_0)$ under H_0 and $P(X_1, X_2, \dots, X_n | H_1)$ under H_1

A simple rule is to compare the posterior probabilities $P(H_0 | X)$ and $P(H_1 | X)$ and accept the hypothesis with the highest posterior probability. In other words we choose H_0 if $P(H_0 | X) \geq P(H_1 | X)$ otherwise accept H_1 .

and only if $P(H_0 | X) < P(H_1 | X)$ H_1 . $X_1, X_2, \dots, X_n$ is a random sample from the Poisson distribution with mean θ . Suppose we are interested to test the hypotheses

$H_0: \mu = 10$ versus $H_1: \mu = 12$ We want to test H_0 against H_1 using a gamma prior distribution. The data for this problem are

11 7 11 6 5 9 14 10 9 5
 8 10 8 10 12 9 3 12 14 4

Although the mean of these 20 observations is 8 but in practice we did not know it. Now the likelihood of our sampling model is

and the pdf of the prior distribution is

$$f(x) = \frac{\gamma^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\gamma x}$$

Thus the posterior distribution is

$$h(x) = \frac{L(x) \pi(x)}{\int L(x) \pi(x) dx}$$

2 $p(x) = \frac{\gamma^{n+\alpha}}{\Gamma(n+\alpha)} x^{n+\alpha-1} e^{-\gamma x}$

Now we compute the conditional probabilities under the null and alternative hypotheses as:

$P_0(x) = \frac{\gamma^{n+\alpha_0}}{\Gamma(n+\alpha_0)} x^{n+\alpha_0-1} e^{-\gamma x} = 0.9368$ under the H_0

$P_1(x) = \frac{\gamma^{n+\alpha_1}}{\Gamma(n+\alpha_1)} x^{n+\alpha_1-1} e^{-\gamma x} = 0.0632$ under the H_1

Thus by using the rule, we accept H_0

Interval Estimation

In the previous modules we discussed different methods of point estimation for θ . The point estimator alone does not give much information about θ , e.g. we do not know how close it is to the real θ . So instead of estimating one value of θ , we will determine an interval which likely includes the true value of the parameter θ . In interval estimation we find two estimates for θ , a high estimate $\hat{\theta}_u$, also called upper limit and a low estimate $\hat{\theta}_l$, also called lower limit. There are two important concepts in interval estimation; one is the length of interval, $\hat{\theta}_u - \hat{\theta}_l$, which reflects the precision. The smaller the interval, the higher the precision for which we estimate θ . The other one is the confidence level, that is how much are we confident that it contains the true value of the population parameter. It is in fact the probability and we therefore desire high confidence levels.

General Framework of the Interval Estimation: Let $X_1, \dots, X_n$ be a random sample from a distribution with parameter θ . Our objective is to find the two estimators for θ : the lower estimator, $\hat{\theta}_l$ and the upper estimator, $\hat{\theta}_u$. The interval estimators

are given by the interval $[\hat{\theta}_l, \hat{\theta}_u]$. These estimators are chosen in such a way that the probability that the interval $[\hat{\theta}_l, \hat{\theta}_u]$ includes θ is large or equal to 1. The term 1 is known as

confidence level. We would like $1 - \alpha$ to be small so that we get high confidence level. Common levels for $\alpha = 0.01, 0.05$ and 0.10 represent 99%, 95% and 90% confidence level respectively. Thus when we construct 95% confidence interval for θ , we need to find $\hat{\theta}_l$ and $\hat{\theta}_u$ such that $P(\hat{\theta}_l \leq \theta \leq \hat{\theta}_u) = 0.95$

Definition: Let $X_1, \dots, X_n$ be a random sample from a distribution having parameter θ . An interval estimator with confidence level $1 - \alpha$ is a function of $X_1, \dots, X_n$ such that $P(\hat{\theta}_l \leq \theta \leq \hat{\theta}_u) = 1 - \alpha$ for every possible value of θ . Therefore $1 - \alpha$ is a confidence interval for θ .

The inequality $P(\hat{\theta}_l \leq \theta \leq \hat{\theta}_u) = 1 - \alpha$ can also be expressed as $P(\hat{\theta}_l - \theta \leq 0 \leq \hat{\theta}_u - \theta) = 1 - \alpha$. It is important to note that the interval is a

random quantity and θ is a non-random unknown quantity. The quantities $\hat{\theta}_l$ and $\hat{\theta}_u$ are the random variables because they are the functions of the observed variables $X_1, \dots, X_n$. The term interval estimator is often used as confidence intervals.

Lecture no.43 (Topics 194 to 197)

Procedure of Finding Interval Estimator

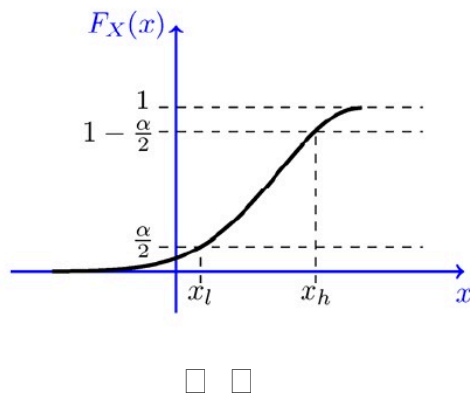
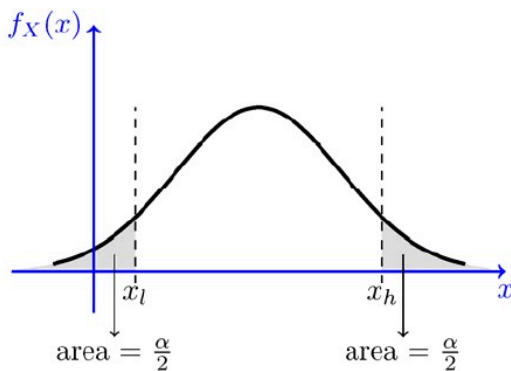
Let X be a continuous random variable with CDF $F_X(x)$.

Suppose we are interested in finding the values of x_l and x_u such that $P(x_l \leq X \leq x_u) = 1 - \alpha$. One way is to choose x_l and x_u such that

$$P(X \leq x_l) = \frac{\alpha}{2} \text{ and } P(X \leq x_u) = 1 - \frac{\alpha}{2}.$$

$$F_X(x_l) = \frac{\alpha}{2} \text{ and } F_X(x_u) = 1 - \frac{\alpha}{2}.$$

$$x_l = F_X^{-1}\left(\frac{\alpha}{2}\right) \text{ and } x_u = F_X^{-1}\left(1 - \frac{\alpha}{2}\right).$$



We explain it with the help of the following example. Find x_l and x_u such that

$$P(x_l \leq Z \leq x_u) = 0.95$$

and the CDF of Z is

$$F_Z(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

$$F_Z(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Thus for standard normal random variable Z , we have

$$P(Z \leq z) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Generally one can find z_p interval for standard normal random variable Z by Assuming

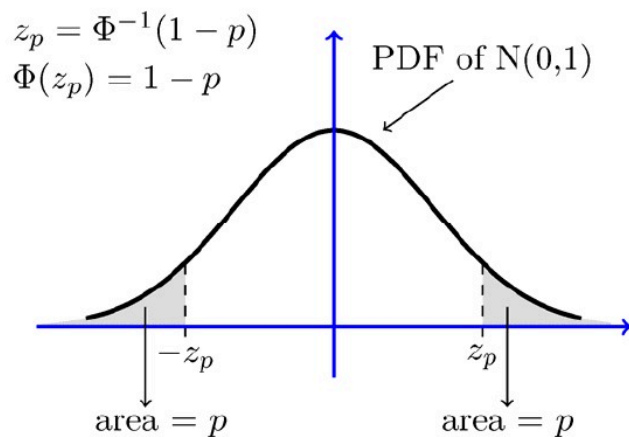
We define z_p as a real value such that

$$P(Z \leq z_p) = p$$

Since normal is a symmetrical distribution, so

Now using z notation, we can state 1 interval for standard normal random variable Z as

$$P(Z \leq z_p) = 1 - p$$



Confidence interval for mean when population variance is known

Assuming $X_1, \dots, X_n$ is a random sample from normal distribution that has point estimator which represents the mean of the distribution and is given as

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Since $X_1, X_2, \dots, X_n$ are independent variables with known variance σ^2 , hence we conclude that

By normalizing $\bar{X}$, we get standard normal distribution. The confidence interval for standard normal distribution is

Therefore

$$P\left(-z_p \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_p\right) = 1 - 2p$$

$$P\left(\bar{X} - z_p \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_p \frac{\sigma}{\sqrt{n}}\right) = 1 - 2p$$

Hence the confidence interval for μ is $\bar{X} \pm z_p \frac{\sigma}{\sqrt{n}}$. It is important

to note that here we do not require n to be large, so it could be any positive integer.

Confidence interval for mean when population variance is unknown

Another interesting and more realistic case is when the population variance is unknown. Let $X_1, \dots, X_n$ be random sample drawn from the normal population $N(\mu, \sigma^2)$. We are interested to find the confidence interval for μ but σ^2 is not known. Assume s is the standard deviation for this random sample, then the random variable T is defined as

which has a t -distribution with $n - 1$ degree of freedom.

It is already known that the ratio of the standard normal variate and the Chi-square distribution with $n - 1$ degree of freedom has a t -distribution i.e. if $Z \sim N(0, 1)$ and $Y \sim \chi^2_{n-1}$ then

$$T = \frac{Z}{\sqrt{Y/n}} \sim t_{n-1}$$

The random variable T is our pivot variable and using the definition of confidence interval for T is as follows

$$P\left(-t_{\alpha/2, n-1} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{\alpha/2, n-1}\right) = 1 - \alpha$$

Confidence interval for population variance when population mean is unknown

Let $X_1, \dots, X_n$ be random sample drawn from the normal population $N(\mu, \sigma^2)$, where μ is not known and n could be any positive integer. We know that if $X_1, \dots, X_n$ are the iid from $N(\mu, \sigma^2)$ and S^2 is the variance of this random sample, then the random variable Y is defined as $Y = \frac{(n-1)S^2}{\sigma^2}$ where $Y \sim \chi^2_{n-1}$ and $\bar{X}$ and S^2 are independent random variables. The random variable Y is our pivot variable and using the definition of confidence interval, the confidence interval for Y is

given as

$$P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}}\right) = 1 - \alpha$$

Lecture no.44 (Topics 198 to 201)

Confidence interval for population proportion

Suppose $X_1, X_2, \dots, X_n$ is a random sample from the Bernoulli distribution with probability of success p , where X is 1 or 0 if the outcome is success or failure respectively. Let $\hat{p} = \frac{\sum_{i=1}^n X_i}{n}$ be the sample proportion and $\hat{p}(1-\hat{p})$ be the sample variance. If the sample size is sufficiently large and p is not too close to zero or 1 then the random variable

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

will be approximately $N(0, 1)$. Thus a $100(1-\alpha)\%$ confidence interval $\sqrt{\frac{p(1-p)}{n}}$ is

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

After simplification, we get

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Application Q4.2.12

Let the observed value of Y is $y=75$, find an approximate 90% confidence interval for p .

Solution:

Here $n=300$, $\hat{p} = \frac{75}{300} = \frac{1}{4}$, then

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$pp = \frac{11}{4} \pm 1.645 \sqrt{\frac{44}{300}} = 4 \pm 1.645 \sqrt{\frac{44}{300}} = 10.$$

After simplification, we get 0.029% confidence interval proportion.

Bayesian Interval

Bayesian interval or credible interval is an interval in the domain of the posterior distribution used for the interval estimation. In Bayesian we assume that the true value is a random variable. Credible interval captures the uncertainty about the true value of the parameter by imposing a prior distribution on the true parameter vector say P.

Definition: The interval $L(X) \leq \theta \leq U(X)$ based on the observed data X is said to be a credible interval for the parameter θ if the posterior probability of θ being in the interval $L(X) \leq \theta \leq U(X)$ is equal to 1. In other words
$$P(L(X) \leq \theta \leq U(X) | x) = 1$$

It is important that not to confuse credible probability and coverage probability. Credible probability comes from the posterior distribution, which gets its probability from the prior distribution. Thus credible probability reflects the experimenter's belief which is expressed in the prior distribution and updated with the data to the posterior distribution. So in Bayesian, 95% coverage means that by combining the prior knowledge with the data the experimenter is 95% sure of coverage.

On the other hand, the coverage probability reflects the uncertainty in the sampling procedure of the repeated experimental trials. A classical 95% coverage means that in repeated identical trials there is a 95% chance that it will cover the true value of the parameter.

Example Using R

Suppose that we are interested in the proportion of the population of American college students that sleep at least eight hours each night. A random sample of 27 students is taken from this population and 11 were found that they slept at least eight hours each night. Assuming that the prior distribution is beta(3.3, 7.2). Find 90% credible interval for the parameter θ (say).

Solution: Let the parameter of interest is θ which represents the students who sleep at least eight hours each night, so our random variable $X \sim \text{Bin}(n, \theta)$. Since $n = 27$, it is given that the prior distribution on θ is beta(3.3, 7.2), then the posterior distribution is the Beta distribution i.e.

$$\text{beta}(\alpha + \sum_{i=1}^n X_i, \beta + n - \sum_{i=1}^n X_i) = \text{beta}(3.3 + 11, 7.2 + 27 - 11) = \text{beta}(14.3, 23.2)$$

In fact we need to solve

$$P(c \leq \beta \leq d) = 0.95$$

and

$$P(d \leq \beta \leq 1) = 0.05$$

The above integral is not in close form so we use R to find the intervals.

`c=qbeta(0.05,14.3,23.2)`

`>0.256`

`d= c=qbeta(1-0.05,14.3,23.2)`

`>0.514`

or

`>qbeta(c(0.05,0.95),14.3,23.2)`

`>0.256, 0.514`

So it means that given the observed data there is 90% probability that the proportion of the American students who sleep at least eight hours is between 0.256 and 0.514.

Lecture no.45 (Topics 202 to 205)

Highest Posterior Density

In Bayesian when we construct credible intervals there is a chance that some values fall outside the interval with higher probability than those inside the interval. This suggests to create more restrictive type of interval, called highest posterior density (HPD) interval.

Definition: C_{HPD} is called HPD interval for P if it is the shortest interval (has minimum length) among all intervals C and C : $P(C) = 1 - \alpha$, where k is the largest number such that $P(X \leq k) = 1 - \alpha$.

The value k can be thought of as a horizontal line placed over the posterior density whose intersection(s) with the posterior define regions with probability $1 - \alpha$.

All the points in an HPD region have higher posterior density than those outside the region. In case of symmetrical posterior density CC . HPD intervals are preferable to quantile or equal tail credible intervals when posterior density is highly skewed or multi-modeled.

Bootstrap Confidence Interval

We use another approach, known as bootstrap method, to build the confidence interval for the difference between two means. In bootstrap method we take random sample with replacement from each sample separately. Bootstrap method is useful when the sample size is small, the under study test statistic is difficult to use or confidence interval does not have a close form etc. Suppose $X_1, \dots, X_n$ represents the first sample and $Y_1, \dots, Y_n$ represents the second sample. Draw bootstrap samples individually for each sample with replacement and combine them. Repeat this process B-times (say 3000) and calculate the test statistic. In practice, 95% confidence interval uses 2.5th and 97.5th percentile of the bootstrap estimates considered as the lower and the upper bounds of the interval.

There are some existing packages in R for bootstrap confidence interval e.g. “boot” but here we develop our code to implement it so that one could understand it easily.

Example Using R

We use same dataset which we used for the bootstrap hypothesis testing. Thirty independent observations were generated from this distribution. The first 15 observations are represented by X-variate and the last 15 observations are denoted by Y-variate. The data is given as

X variates							
94.2	111.3	90.0	99.7	116.8	92.2	166.0	95.7
109.3	106.0	111.7	111.9	111.6	146.4	103.9	
Y variates							
125.5	107.1	67.9	98.2	128.6	123.5	116.5	143.2
120.3	118.6	105.0	111.8	129.3	130.8	139.8	

Using bootstrap method to construct 95% C.I for the difference between two means.

Solution:

The R-code is given as:

```
d <- read.table("data1.csv",header=T,sep=";",dec=".")
```

```
set.seed(192348)
```

#use to reproducibility means provide the same random data every time when we run this code (it is not necessary)

```
n1 <- 15
```

```
n2 <- 15
```

```
B <- 3000
```

```

# number of bootstrap samples
variable <- d$data
#variable use for resample
boot.samples.x1
<-matrix(sample(variable,size=n1*B,replace=T),nrow=n1,ncol=B)
# we get 15*3000 matrix where each column is a bootstrap resamples
boot.samples.x2
<- matrix(sample(variable,size=n2*B,replace=T),nrow=n2,ncol=B)
dim(boot.samples.x1);dim(boot.samples.x2)
[1] 15 3000
[1] 15 3000
#check to make sure that they are not empty
boot.samples.x1[1:10,1:10]
      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,] 143.2 129.3 120.3 116.5 116.8 105.0 111.8 105.0 105.0 99.7
[2,] 116.5 146.4 90.0 123.5 116.5 105.0 146.4 95.7 118.6 129.3
[3,] 90.0 111.6 118.6 95.7 128.6 118.6 95.7 116.5 94.2 111.7
[4,] 111.7 143.2 123.5 166.0 120.3 105.0 111.7 111.8 130.8 128.6
[5,] 118.6 106.0 130.8 90.0 146.4 146.4 98.2 106.0 111.7 90.0
[6,] 125.5 107.1 99.7 128.6 111.8 146.4 95.7 146.4 111.7 118.6
[7,] 116.5 90.0 128.6 125.5 99.7 111.9 116.5 143.2 116.5 92.2
[8,] 111.7 94.2 92.2 105.0 111.3 98.2 109.3 103.9 146.4 107.1
[9,] 129.3 103.9 111.3 111.9 67.9 90.0 120.3 99.7 125.5 130.8
[10,] 139.8 105.0 166.0 116.8 143.2 139.8 143.2 166.0 103.9 111.8
boot.samples.x2[1:10,1:10]
      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,] 67.9 111.6 123.5 94.2 146.4 106.0 111.9 123.5 67.9 118.6
[2,] 98.2 107.1 94.2 99.7 95.7 111.7 130.8 125.5 111.9 90.0
[3,] 111.8 99.7 94.2 129.3 139.8 123.5 111.8 107.1 111.7 94.2
[4,] 118.6 139.8 103.9 116.5 94.2 105.0 94.2 111.7 111.6 103.9
[5,] 129.3 120.3 166.0 116.5 128.6 128.6 116.8 92.2 166.0 146.4
[6,] 111.3 111.9 109.3 103.9 116.8 139.8 123.5 116.5 111.6 116.5
[7,] 90.0 92.2 130.8 123.5 146.4 98.2 67.9 109.3 90.0 128.6
[8,] 129.3 111.6 139.8 95.7 98.2 128.6 128.6 94.2 116.8 129.3

```

```

[9,] 90.0 92.2 99.7 111.7 103.9 107.1 109.3 111.6 139.8 146.4
[10,] 67.9 111.7 111.3 90.0 146.4 166.0 106.0 120.3 111.7 67.9
#now we calculate bootstrap estimate of difference between means.

boot.diff.means <- colMeans(boot.samples.x1)-
colMeans(boot.samples.x2)

#you will obtain 3000 difference between two means

length(boot.diff.means)

[1] 3000

#confidence intervals using 95% confidence
# there are different methods for building bootstrap C.I, here we use "Percentile Method"

quantile(boot.diff.means,prob=0.025)

2.5%
-13.62

quantile(boot.diff.means,prob=0.975)

97.5%
13.86233

Robustness

```

The term robustness means the stability of the tests, procedures and statistical models under the specific assumptions of the statistical analysis. In simple words robust statistics is a measure which does not affect by the extreme values. Let's start with a simple example to understand this concept. Suppose we have two data sets; 1,2,3,4,5,6 and 1,2,3,4,5,1000. The mean and the median of the first dataset are 3.5 and 3.5 while the mean and the median of the second dataset are 169 and 3.5 respectively. We observe that that the extreme value does not affect the median so we say that the median is robust. The reason is that mean depends on all the values of the dataset whereas the median depends only on the middle value of the dataset. Similarly IQR is more robust than the standard deviation because IQR depends on the median. Robust statistic is more useful for the skewed distribution and long tailed distribution while non-robust statistic is useful for symmetric distribution.

Generally we use two methods to measure the robustness of an estimator i.e. influence function (IF) of the local stability and the breakdown point (BDP) of global reliability. IF measures the effect of a single observation whereas BDP measures the fraction of contamination. Harrel-Davis quantile estimator is an example of IF while Huber Location M-estimator is an example of BDP.
